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17 QUALITY ASSURANCE

17.1 Quality Assurance During the Design and Construction Phases

This section is not applicable since these phases are completed. See Appendix D of Amendment 43 to FSAR for the Quality Assurance Program applicable during design and construction.

17.2 Quality Assurance During the Operations Phase

The licensee is responsible for the operation and maintenance of Salem/Hope Creek Generating Stations. These activities are presently conducted in accordance with the NRC approved Quality Assurance Topical Report (QATR) which describes the formal and comprehensive control methods established to assure compliance with 10CFR50, Appendix B.

The QATR describes how the Quality Assurance Program is to be functionally implemented with due regard to the safety and health of the public and the personnel on site. The Salem/Hope Creek Quality Assurance Program as described by the QATR is incorporated herein by reference.

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INTRODUCTION

The information in Appendix A addresses lessons learned from the accident at Three Mile Island (TMI), Unit 2. The material is organized specifically in response to NUREG-0578, "TMI-2 Lessons Learned Task Force Status Report and Short-Term Recommendations." Two items (Instrumentation to monitor containment conditions and reactor coolant system vents) were not included in NUREG-0578, but were later incorporated as lessons learned by the Director of Nuclear Reactor Regulation.

Item numbering in this Appendix corresponds with that of NUREG-0578. For each item, the NRC recommendation is first stated, followed by the Public Service Electric & Gas (PSE&G) response.

2.1.1 Pressurizer Power Sources

Emergency Power Supply Requirements for the Pressurizer Heaters, Power-Operated Relief Valves and Block Valves, and Pressurizer Level Indicators in PWRs NRC Position

Consistent with satisfying the requirements of General Design Criteria 10, 14, 15, 17, and 20 of Appendix A to 10CFR50 for the event of loss of offsite power, the following positions shall be implemented:

Pressurizer Heater Power Supply

1. The pressurizer heater power supply design shall provide the capability to supply, from either the offsite power source or the emergency power source (when offsite power is not available), a predetermined number of pressurizer heaters and associated controls necessary to establish and maintain natural circulation at hot standby conditions. The required heaters and their controls shall be connected to the emergency buses in a manner that will provide redundant power supply capability.

2. Procedures and training shall be established to make the operator aware of when and how the required pressurizer heaters shall be connected to the emergency buses. If required, the procedures shall identify under what conditions selected emergency loads can be shed from the emergency power source to provide sufficient capacity for the connection of the pressurizer heaters.

3. The time required to accomplish the connection of the preselected pressurizer heater to the emergency buses shall be consistent with the timely initiation and maintenance of natural circulation condition.

4. Pressurizer heater motive and control power interfaces with the emergency buses shall be accomplished through devices that have been qualified in accordance with safety grade requirements.

Response

The Salem design is such that it has the capability to manually connect approximately 800 kW of pressurizer heaters from two backup groups to the emergency power source. These connections are accomplished by an installed manually operated interlocked transfer scheme between the pressurizer heaters and the "A" diesel generator and the "C" diesel generator.

An analysis performed by Westinghouse indicates that 150 kW of pressurizer heaters is needed to assure maintenance of natural circulation. These backup heater groups will be manually set up such that only 150 kW can be supplied from each vital bus. Each redundant heater group has access to only one Class 1E division power supply. Motive and control interfaces with the vital buses will be through safety grade circuit breakers.

Emergency Operating Procedures address the transfer of the heaters to the vital buses.

The diesel generators are capable of supplying the 150 kW of pressurizer heaters concurrent with the equipment loads required for a loss-of-coolant accident (LOCA). The diesel loads during the injection phase (with the inclusion of the pressurizer heater load) would be slightly above the 2000-hour rating but well below the 30-minute rating. Under blackout conditions, the diesel generators have sufficient capacity to supply the required equipment loads including the pressurizer heaters and meet the continuous diesel rating.

The connection of the pressurizer heaters to a vital bus is through a normally open Class 1E circuit breaker. This circuit breaker is mechanically key interlocked with the heaters' normal, non-vital power feed circuit breaker. A manually operated disconnect switch must also be closed to make the connection. In addition, the backup group heaters are set up to supply only 150 kW. Once the connection pathway is established manually, the final connection of the pressurizer heaters to the vital bus (open/close the Class 1E circuit breaker) can be accomplished in the Control Room. The setup of the vital bus feed to the pressurizer heaters can be completed in a time frame consistent with maintenance of natural circulation.

The manual action of opening the non-vital pressurizer heater supply circuit breaker prior to closing the vital power supply circuit breaker by mechanical key interlocks is necessary to eliminate any possibility of feeding other non-vital loads from the vital power supply.

It is not necessary that certain equipment loads be shed in order to connect the pressurizer heaters to the vital buses. As a precaution, statements will be added to the operating procedures alerting the operator to remain within the appropriate diesel ratings.

The diesel generator ratings are included in the Emergency Operating Procedures. Connection of the pressurizer heaters to the vital buses does not require the reset of an SI signal.

The pressurizer heaters are not automatically tripped from the vital buses upon a safety injection actuation signal. An event where the pressurizer heaters are on the vital buses when a LOCA occurs is a highly improbable occurrence. Normal operation of the heaters does not and will not require their supply from the vital buses. They will only be needed when normal power is lost (blackout). A LOCA would have to occur following a blackout during which time it became necessary to operate the heaters to maintain natural circulation.

<u>Power Supply for Pressurizer Relief and Block Valves and Pressurizer Level Indicators</u>

1. Motive and control components of the power-operated relief valves (PORVs) shall be capable of being supplied from either the offsite power source or the emergency power source when the offsite power is not available.

2. Motive and control components associated with the PORV block valves shall be capable of being supplied from either the offsite power source or the emergency power source when the offsite power is not available.

3. Motive and control power connections to the emergency buses for the PORVs and their associated block valves shall be through devices that have been qualified in accordance with safety grade requirements.

4. The pressurizer level indication instrument channels shall be powered from the vital instrument buses. These buses shall have the capability of being supplied from either the offsite power source or the emergency power source when offsite power is not available.

Response

The pressurizer PORVs and their associated block valves are powered from the emergency power source. Motive and control power interfaces with the emergency power source satisfy safety grade requirements.

The design of the pressurizer relief and block valve arrangement for both Salem units was predicated on ensuring the ability to relieve. This concept resulted in providing two parallel relief paths which are completely independent and redundant. Such a design concept is necessary to provide protection for anticipated transient without SCRAM (ATWS) conditions, for pressurizer water-solid conditions due to inadvertent operation of the safety injection system events, and low temperature overpressure transients.

Incorporation of complete independence between the relief valve and block valve would negate the system's ability to meet the single failure criterion for the events identified above. The existing design, however, does incorporate the use of diverse power supplies for the PORVs and their associated block valves. The relief valves are supplied by Class 1E, 125 V dc systems while the block valves use 230 V and 115 V vital ac.

Pressurizer level indication instrument channels are powered from the vital instrument buses.

2.1.2 Relief and Safety Valve Testing

Performance Testing for BWR and PWR Relief and Safety Valves

NRC Position

Pressurized water reactor (PWR) and boiling water reactor licensees and applicants shall conduct testing to qualify the Reactor Coolant System relief and safety valves under expected operating conditions for design basis transients and accidents. The licensees and applicants shall determine the expected valve operating conditions through the use of analyses of accidents and anticipated operational occurrences referenced in Regulatory Guide 1.70, Revision 2. The single failures applied to these analyses shall be chosen so that the dynamic forces on the safety and relief valves are maximized. Test pressures shall be the highest predicted by conventional safety analysis procedures. Reactor Coolant System relief and safety valve qualification shall include qualification of associated control circuitry piping and supports as well as the valves themselves.

Response

1. Safety and Relief Valves

Public Service Electric & Gas (PSE&G) is a participant in the Generic PWR Safety and Relief Valve Test Program implemented by the Electric Power Research Institute (EPRI) at the request of participating PWR utilities in response to the USNRC recommendations for safety and relief valve testing.

The primary objective of the Test Program was to provide full scale test data confirming the functionability of primary system PORVs and safety valves for expected operating and accident conditions. The second objective of the program was to obtain sufficient piping thermal hydraulic load data to permit confirmation of models which may be utilized for plant unique analysis of safety and relief valve discharge piping systems. Relief valve tests were completed in August 1981 and safety valve tests were completed in December 1981. The reports prepared by EPRI documenting the Test Program results are as follows:

"Valve Selection/Justification Report"

This report documents that the selected test valves represent all participating PWR plant safety and relief valves. Salem PORVs are 2-inch NPS Copes Vulcan valves with 17-4PH plug and cage. Three-inch Copes Vulcan valves similar to the Salem valves were tested in a configuration similar to that of the Salem Station. Salem safety valves are 6M6 Crosby valves which were tested in a configuration similar to that at the Salem Station.

"Test Condition Justification Report" and the "Westinghouse Plant Condition Justification Report"

These reports document the basis and justification of the valve test conditions for all participating PWR plants. The PORV fluid conditions, safety valve fluid conditions and cold overpressurization conditions at Salem are enveloped by the test conditions.

"Safety and Relief Valve Test Report"

This report provides evidence demonstrating the functionability of the selected test valves under the selected test conditions for all participating PWR plants. Tests conducted on the relief valve have confirmed that the valve opened and closed on demand and that the valve suffered no damage that would preclude future operation. Requirements for the pressurizer power-operated relief valves are addressed in UFSAR Chapter 15.2.14, "Spurious Operation of the Safety Injection System at Power".

Although the tests indicated acceptable valve performance, test valve disassembly showed galling of the cage and plug guiding surfaces. In view of this, the internals of Salem Units 1 and 2 PORVs have been changed to 316 stellited plugs instead of

17-4PH. The Salem valves with this combination of internals have shown no indication of leakage or galling.

The safety valves were shown to open and close. The functionability of the valves to provide overpressure protection for the Final Safety Analysis Report (FSAR) events was found to be adequate (1). However, depending upon the test conditions (steam - transition - water), the valves were shown to flutter and/or chatter during loop seal discharge. The observed instability of the safety valves during loop seal discharge has been evaluated by Westinghouse and PSE&G and is not considered to be a safety concern.

Upstream piping and valve ring adjustments have been shown to have a substantial effect on valve performance. Performance of the Salem safety valves during steam discharge conditions has been judged stable through the use of a valve dynamic model developed by Continuum Dynamics, Inc. (2). The model also predicted blowdown of less than 10 percent which is considered acceptable. Liquid discharge challenges to safety valves are predicted to occur significantly less frequently than a LOCA and the consequences of such liquid discharge are much less severe; liquid discharge from the Salem safety valves has been shown to be an unlikely event (3). Thus, safety valve liquid discharge is an insignificant safety concern when compared with FSAR transient events.

"Application of Relap 5/MOD 1 for Calculations of Safety and Relief Valves Discharge Piping Hydrodynamic Loads"

This report presents an analytical model benchmarked against test data that may be used for plant unique analysis of safety and relief valve discharge piping systems.

During a portion of the tests performed on safety valves, it was noted that the pressure upstream of the valve tended to oscillate, especially for loop seal configurations. These oscillations are thought to be related to valve instability which, in turn, is related to the valve and the inlet piping parameters. The tests that exhibited pressure oscillations are summarized in Volume 10, Section 4, of the EPRI Safety and Relief Valve Test Report.

For the test configuration that most nearly approximates the Salem configuration, it was determined that a maximum peak transient pressure of 6300 psi was experienced. An analysis of the expected behavior of the Salem 1 and 2 safety valve inlet piping assuming unstable valve behavior has been performed. The analysis indicates that the peak transient pressure for Salem 2 will be less than the test value (6300 psi) whereas the expected peak transient pressure for Salem 1 will be below 7100 psi. When compared with ASME Code allowables, it was determined that the pressure pulses are within acceptable limits. (The maximum permissible pressure for the inlet piping is calculated to be 7130 psi for Westinghouse.) Necessary inspection of the inlet piping subsequent to a safety valve discharge will be instituted to comply with the ASME requirement. Hence, the review of inlet pressure oscillations indicates no safety concern.

EDS-Impell Corporation was retained by PSE&G to perform Salem plant specific analysis based on the EPRI test results. The thermal hydraulic analysis performed by Impell indicated that to make the pressurizer piping code compliant, the piping supports would have to be strengthened. To minimize the magnitude of discharge piping forces on a safety valve actuation, also required was encapsulation of the loop seals in insulation boxes so that loop seal water will be at a higher temperature.

The following modifications were completed at Salem to ensure the discharge piping integrity and in turn the safety valve operability.

At Salem 1, 18 new supports were added and 33 supports were modified. At Salem 2, 42 new supports were added and 27 supports were modified.

Subsequently, further analysis was performed for both Salem Units 1 and 2 as a result of concerns raised by the NRC in their Safety Evaluation Report of PSE&G's response to NUREG-0737, ItemII.D.1, Sections 1.2.3 and 1.2.8. Engineering analysis of the Pressurizer Safety Valve piping and Power Operated Relief Valve piping for Salem Units 1 and 2 indicated that the Pressurizer Safety Valves (SVs) and the Power Operated Relief Valves (PORVs) discharge piping and welded attachments exceeded code allowable stress levels after actuation. The current PORV and SV inlet piping configuration allows steam from the pressurizer to fill the inlet piping and then condense to form a water seal. These water loop seals have been identified as a major contributor to exceeding the allowable stress levels.

Analysis was performed for Salem Units 1 and 2 to verify that the discharge pipe stresses are within code allowable limits with no water loop seals present. This was accomplished by a modification to remove the water loop seals of the SVs and the PORVs, by continuously draining the condensation on the upstream side of the PORVs and SVs back to the pressurizer liquid space. This required the installation of drain lines from the upstream side of the PORVs connecting to the existing drain lines on the SVs loop seals. A new line was added to connect the drain lines to the Pressurizer liquid space.

As a result of eliminating the water loop seals for the PORVs and SVs, the valve internals are subjected to a saturated steam environment. The safety valve internals were modified to ensure against leakage. The PORV internals and materials were changed. Also, it is no longer required to maintain the SV loop seals in a heated condition, and therefore the heat boxes and associated temperature instrumentation were permanently removed.

In addition, the modification for Salem 2 includes the installation of 8 new supports on the drain line piping system, modification to 4 supports and modification to a pressurizer support column. Unit 1 modification includes the installation of 6 new supports on the drain line piping system, modification to 5 supports and modification to a pressurizer support column.

2. Block Valves

Salem block valves are 3-inch Velan valves with SMB-00-15 motor operators which were tested by EPRI at the Marshall Facility. The Velan valve fully opened and closed on demand for each of the 21 test cycles (4). No operability concern has been identified.

3. ATWS Testing

Results from the current program are likely to provide most of the information necessary to address ATWS events (i.e., relief capability at high pressures).

References for Section 2.1.2

1. "Review of Pressurizer Safety Valve Performance as Observed in the EPRI Safety and Relief Valve Test Program," Westinghouse Report WCAP-10105.

2. "Valve Dynamic Model Performance Simulation for the Salem PWR Plant Unit 1," Report No. 82-19 from Continuum Dynamics, Inc.

3. "Results of Probabilistic Evaluation of Liquid Discharge Challenges to Salem Units 1 and 2 Safety Valves," Report 1-231-01-052-00 from Science Applications, Inc.

4. "EPRI PWR Safety and Relief Valve Test Program - PORV Block Valve Information Package."

2.1.3.a Relief and Safety Valve Position Indications

Direct Indication of Power-Operated Relief Valve and Safety Valve Position for PWRs and BWRs

NRC Position

Reactor system relief and safety valves shall be provided with a positive indication in the control room derived from a reliable valve position detection device or a reliable indication of flow in the discharge pipe.

Response

Each PORV is equipped with a limit switch to provide an alarm in the Control Room if the PORV is not fully closed. These switches are seismically and environmentally qualified.

To provide positive indication of safety valve position, an environmentally qualified limit switch is mounted in each safety valve bonnet which will indicate open and closed position in the Control Room.

Both of the above schemes utilize a single switch on each valve powered from a vital bus. Several reliable backup methods are available to detect an open valve. These methods, which are addressed in Emergency Procedure EI 4.24, "Malfunction of Pressurizer Relief Valve," are the following:

1. Pressurizer pressure
2. Valve discharge piping temperature
3. PRT level, pressure and temperature
4. PORV open/close indication in conjunction with PORV block valve open/close position indication
5. Control Room alarms for all of the above indicators

2.1.3.b Instrumentation for Inadequate Core Cooling

<u>Instrumentation for Detection of Inadequate Core Cooling in PWRs BWRs</u>

<u>NRC Position</u>

1. Licensees shall develop procedures to be used by the operator to recognize inadequate core cooling with currently available instrumentation. The licensee shall provide a description of the existing instrumentation for the operators to use to recognize these conditions. A detailed description of the analyses needed to form the basis for operator training and procedure development shall be provided pursuant to another short-term requirement, "Analysis of Off-Normal Conditions, Including Natural Circulation" (see Section 2.1.9).

   In addition, each PWR shall install a primary coolant saturation meter to provide online indication of coolant saturation and condition. Operator instruction as to the use of this meter shall include consideration that it is not to be used exclusive of other related plant parameters.

2. Licensees shall provide a description of any additional instrumentation or controls (primary or backup) proposed for the plant to supplement those devices cited in the preceding section giving an unambiguous, easy-to-interpret indication of inadequate core cooling. A description of the functional design requirements for the system shall also be included. A description of the procedures to be used with the proposed equipment, the analysis used in developing these procedures, and a schedule for installing the equipment shall be provided.

Response

The existing instrumentation in the Control Room is sufficient to recognize inadequate core cooling. The indications available for verification of core heat removal are:

1. Reactor Coolant System (RCS) ΔT less than full load ΔT.

2. RCS or core exit thermocouple temperatures constant or decreasing.

3. Steam generator pressure constant or decreasing at a rate equivalent to the rate of decrease of RCS temperatures while maintaining steam generator level with continuous auxiliary feedwater.

A further guide for recognition of inadequate core cooling is a computerized subcooling monitor. The margin to saturation calculation is performed in a micro-processor supplied by Combustion Engineering. RCS pressure and core exit temperatures are inputted and compared to a standard steam table to calculate the actual margin to saturation. The margin to saturation is then adjusted automatically by the subcooling margin monitor computer to compensate for instrument channel inaccuracies. This provides the operator with reasonable confidence that the RCS fluid will be subcooled at the core exit whenever a positive margin to saturation is indicated by the subcooling monitor. Continuous digital and recorder displays of margin in °F are located in the main control room. Control room annunciators provide alarm at 10°F margin decreasing. Emergency Operating Procedures and Operations Procedures have been revised to address the use of this processor output.

Public Service Electric & Gas is a member of the Westinghouse Operating Plant Owners' Group. Westinghouse, under the direction of the Westinghouse Owners' Group, is performing further analyses to aid in selection of more direct indicators of inadequate core cooling, and to serve as a basis for augmented emergency procedures.

A preliminary report on adequate core cooling was submitted to the NRC on October 30, 1979 by the Westinghouse Owners' Group. The Salem Emergency Procedures were revised on an interim basis to specify precautions and operator actions to recover from a condition in which the core has experienced inadequate cooling. The station procedures will be further updated after completion of the final Owners' Group report.

A device to indicate reactor vessel water level has been installed in Units 1 and 2. This device is similar to the proposed Westinghouse design for Virginia Power's North Anna plant.

The instrumentation for monitoring inadequate core cooling conforms to NUREG-0578, NUREG-0737 and the letter from H. R. Denton to all operating plant licensees, dated October 30, 1979. If the devices were to fail, backup parameters, such as, RCS pressure and temperature, steam generator level, and auxiliary feedwater flow are available to enable determination of adequate core cooling. The existing procedure and availability of alternate indications is sufficient to ensure proper determination of core cooling adequacy.

The following is an evaluation of conformance of the ICC Instrument System to NUREG-0737, Item II.F.2, Attachment 1 entitled, "Design and Qualification Criteria for Pressurized-Water Reactor Incore Thermocouples." Paragraph numbering corresponds to that of Attachment 1:

1. There are 58 in-core thermocouples, with a minimum of ten in each quadrant and an additional seven thermocouples shared with two adjoining quadrants to provide indication of radial distribution of temperature rise across representative regions of the core. See Section 7.

2. Primary Operator Displays

   The plant computer and SPDS are the primary displays. Additional description of SPDS is found in Section 7.10.

   a. A core map for each quarter of the core is available to the operator on demand on the computer output CRT. The core map gives the temperature at each core exit thermocouple location in that quarter of the core.

   b. The core map will give the location of the hottest in-core thermocouple. This hottest in-core thermocouple reading is the basis for the subcooling calculations and procedures.

   c. There is direct readout and hard copy capability for all thermocouple temperatures. The range extends from 30°F to 2200°F.

   d. Trend capability for the thermocouples is available on demand on the trend printer located on a vertical panel in the control room.

   e. Alarm capability is provided consistent with operator procedure requirements: when reactor power <0.25 percent, alarm at 1200°F; when reactor power >0.025 percent, alarm at 630°F.

   f. The operator display device is human factor designed.

3. A backup display is available which meets the stated requirements as follows:

   All thermocouples may be read, but only one at a time. The range is 35°F to 2300°F. The readout meter on the Core Exit Thermocouple Processing Cabinet is located in a room adjacent to the Control Room and will allow the operator to read 16 thermocouples within a time interval of less than 6 minutes. The backup display is referenced in the emergency procedures.

4. The primary display CRT is located on the control console. The hard copy and trend displays are located on the console or on a vertical panel in the control room.

5. The instrumentation meets the stated design and qualification criteria for accident monitoring instrumentation. See Items 6 through 9 below.

6. The primary display channel utilizes the station computer which is energized from an uninterruptible power supply. The primary CRT display and printers which supply direct digital readout and trend capability are also energized from the same power supply. The primary display channel and associated hardware are Class 1E.

   The backup display is supplied from redundant station power sources. The backup display and hardware are not Class 1E.

   The reference junction heater power is derived from the same source as the backup display.

7. The in-core thermocouples are environmentally qualified as required in Appendix B (to NUREG-0737), Item 1. The computer and display have not been environmentally qualified or seismically qualified.

8. The primary display channel uses a computer with an estimated reliability of 98 to 99 percent. The overall primary display channel reliability is 90 to 95 percent. The SPDS primary display and the backup display are designed to achieve 99% availability.

9. The in-core thermocouples are purchased to 10CFR50 Appendix B Quality Assurance requirements.

2.1.4 Containment Isolation

<u>Containment Isolation Provisions for PWRs and BWRs</u>

<u>NRC Position</u>

1. All containment isolation system designs shall comply with the recommendations of Standard Review Plan (SRP) 6.2.4; i.e., that there be diversity in the parameters sensed for the initiation of containment isolation.

2. All plants shall give careful reconsideration to the definition of essential and non-essential systems, shall identify each system determined to be essential, shall identify each system determined to be non-essential, shall describe the basis for selection of each essential system, shall modify their containment isolation designs accordingly, and shall report the results of there-evaluation to the NRC.

3. All non-essential systems shall be automatically isolated by the containment isolation signal.

4. The design of control systems for automatic containment isolation valves shall be such that resetting the isolation signal will not result in the automatic reopening of containment isolation valves. Reopening of containment isolation valves shall require deliberate operator action.

Response

The containment isolation system complies with the requirement for initiation by diverse parameters as described in Section 6.2. A number of isolation signals are provided for valve closure. Each signal is indicative of certain operating conditions and is generated by diverse input parameters. The isolation signals and their input parameters are as follows:

Containment Isolation - Phase A

1. Manual actuation
2. High containment pressure
3. Low pressurizer pressure
4. High differential pressure between steam lines
5. High steam line flow coincident with low steam line pressure or Low-Low T_{avg}.

Containment Isolation - Phase B

1. Manual Actuation
2. High-High containment pressure

Containment Ventilation Isolation

1. Manual actuation
2. High containment pressure
3. Low pressurizer pressure
4. High differential pressure between steam lines
5. High steam line flow coincident with low steam line pressure or Low-Low T_{avg}.
6. High containment radiation - particulate (Mode 6)
7. High containment radiation – iodine
8. High containment radiation – gaseous

Main Steam Line Isolation

1. Manual actuation
2. High-High containment pressure

3. High steam line flow coincident with low steam line pressure or Low-Low T_{avg}.

Feedwater Isolation

1. Manual actuation
2. High containment pressure
3. Low pressurizer pressure
4. High differential pressure between steam lines
5. High steam line flow coincident with low steam line pressure or Low-Low T_{avg}.
6. High-High steam generator water level
7. Reactor trip coincident with low T_{avg}.

The Containment Isolation System isolates those systems which are not required for the mitigation of accidents specified in Section 15 of the FSAR. A review of Salem design has demonstrated conformance with these requirements.

The valves and systems isolated by the various isolation signals are indicated in Section 6.2. Isolation provision for all lines penetrating the containment are described in Section 6.2. All non-essential systems are either automatically isolated upon a containment isolation signal, or provided with non-return check valves, or closed during power operation and under administrative control. Essential systems are not isolated since they are required to perform functions needed to maintain the plant in a safe condition following an accident. These essential systems are as follows:

1. Residual Heat Removal – Part of Safety Injection
2. Safety Injection
3. Containment Fan Coolers – Service Water
4. Steam Supply to Auxiliary Feedwater Pump Turbine
5. Main Steam Atmospheric Relief
6. Auxiliary Feedwater
7. Charging – Portion for Safety Injection

The Westinghouse Owners' Group has prepared a report entitled "Classification of Lines Penetrating Containment and a Review of Containment Isolation Logic and Philosophy." Salem

conforms with the established essential/non-essential categories and recommended isolation provisions.

A review of the containment isolation valve Control Systems has been performed to verify that the valves remain closed upon resetting of the isolation signal until the operator takes deliberate action to reposition them.

Two instances, reactor coolant drain tank pump discharge line and pressurizer relief tank gas analyzer line, were discovered which could result in inadvertent transfer of radioactive material from the containment.

The reactor coolant drain tank line is provided with two isolation valves, WL12 (inside containment) and WL13 (outside containment). Both of these valves are closed upon a containment phase "A" isolation signal. Upon a reset condition, WL13 will remain closed unless the operator takes action to change its position. Valve WL12 is provided with a number of automatic control signals which, if present, will cause the valve to reopen.

The pressurizer relief tank line is provided with two isolation valves, PR17 (inside containment) and PR18 (outside containment). Both of these valves are closed upon a containment phase "A" isolation signal. The valves can reopen following a protection system reset when in the auto-control mode. An open pathway would be established if both valves were in the auto-control mode.

Design changes have been implemented (for valves WL12, PR17, and PR18) to revise the control circuitry to prevent these occurrences.

2.1.5.a Dedicated Containment Penetrations

Dedicated Penetrations for External Recombiners or Post-Accident Purge Systems

NRC Position

Plants using external recombiners or purge systems for post-accident combustible gas control of the containment atmosphere should provide Containment Isolation Systems for external recombiner or Purge Systems that are dedicated to that service only, that meet the redundancy and single failure requirements of General Design Criteria 54 and 56 of Appendix A to 10CFR50, and that are sized to satisfy the flow requirements of the recombiner of Purge System.

Response

Each Salem unit incorporated two redundant, physically separated, permanently installed electric hydrogen recombiners, located inside the reactor containment, as previously described in Section 6.2. Each recombiner was capable of maintaining post-accident hydrogen concentration in the containment below limit of flammability in air of 4 percent as previously noted in Section 6.2.

Requirements for the hydrogen recombiners have been deleted based on Technical Specification Amendment numbers 281 and 264 to Facility Operating License numbers DPR-70 and DPR-75.

2.1.5.c Hydrogen Recombiners

Capability to Install Hydrogen Recombiner at Each Light Water Nuclear Power Plant

NRC Position

1. All licensees of light water reactor plants shall have the capability to obtain and install recombiners in their plants within a few days following an accident if containment access is impaired and if such a system is needed for long-term post-accident combustible gas control.

2. The procedures and bases upon which the recombiners would be used on all plants should be the subject of a review by the licensees in considering shielding requirements and personnel exposure limitations as demonstrated to be necessary in the case of TMI-2.

Response

Post-accident hydrogen control capability is described in the response to Item 2.1.5.a.

The procedure for use of the hydrogen recombiners, 0I II – 15.3.1, "Hydrogen Recombiner – Normal Operation," had been reviewed and revised as required.

Requirements for the hydrogen recombiners have been deleted based on Technical Specification Amendment numbers 281 and 264 to Facility Operating License numbers DPR-70 and DPR-75.

2.1.6.a Leakage Outside Containment

Integrity of Systems Outside Containment Likely to Contain Radioactive Materials for PWRs and BWRs

NRC Position

Applicants and licensees shall immediately implement a program to reduce leakage from systems outside containment that would or could contain highly radioactive fluids during a serious transient or accident to as-low-as-practical levels. This program shall include the following:

1. Immediate Leak Reduction

    a. Implement all practical leak reduction measures for all systems that could carry radioactive fluid outside of containment.

b. Measure actual leakage rates with system in operation and report them to the NRC.

2. Continuing Leak Reduction

   Establish and implement a program of preventive maintenance to reduce leakage to as-low-as-practical levels. This program shall include periodic integrated leak tests at a frequency not to exceed refueling cycle intervals.

Clarification

Licensees shall, by January 1980, provide a summary description of their program to reduce leakage from systems outside containment that would or could contain highly radioactive fluids during a serious transient or accident. Examples of such systems are given on Page A-26 of NUREG-0578. Other examples include the Reactor Core Isolation Cooling and Reactor Cleanup (Letdown function) Systems for BWRs.

Include a list of systems which are excluded from this program. Testing of gaseous systems should include helium leak detection or equivalent testing.

Response

Public Service Electric & Gas has completed a comprehensive leak reduction program, which is intended to maintain the leakage rates as-low-as-practical for the following systems outside containment:

1. The Residual Heat Removal (RHR) System, in its entirety.

2. The Safety Injection System (SIS) – all those portions which have direct contact with the containment building.

3. Containment Spray System – only the portion which would have direct contact with the RHR System from the isolation valve to the containment building.

4. Chemical and Volume Control System (CVCS) – the operational portion of the system which includes the letdown heat exchanger lines, seal water heat exchangers, centrifugal charging pumps, and lines to and from the volume control tank.

5. Waste Gas System – the entirety of the system except for the lines to and from the nitrogen supply portion of the system.

6. Liquid Radwaste System – the waste evaporator and the waste holdup tanks.

7. Sampling System – all sample lines which have direct contact with the primary systems.

The systems and portions of systems excluded from the program are: the boric acid recovery portion of the CVCS, because boric acid recovery will not be used during an accident; the primary water recovery portion of the CVCS, because it is not anticipated that this portion of the CVCS will be utilized during an accident; the Liquid Radwaste and Gaseous Radwaste Systems contain portions which will not be utilized during an accident. The entire Liquid Radwaste System except for the waste evaporator and the waste holdup tanks is excluded from the program. The Gas System nitrogen supply portion is excluded since this portion of the system will not be radioactive.

Most of the Containment Spray System will not be directly in contact with the radioactive fluid immediately after an accident. During an accident the Containment Spray System utilizes water from the refueling water storage tank. Once this source is depleted the pumps are shut down and spray water is supplied from the RHR System. Thus, only the section of the system directly in contact with the RHR System will contain radioactive fluid (and be included in the program).

The program consists of integrated leak tests of these systems. The baseline study and the actual leak rates with the systems in operation will be submitted as soon as practical, but no later than prior to entering Mode 4 following the next refueling on each unit.

At intervals not to exceed each refueling outage, an operating pressure leak test will be performed on portions of the SIS, RHR, CVCS, Reactor Coolant Sampling, Liquid Radwaste, Gaseous Radwaste, and Containment Spray Systems. The pressurized systems will be visually inspected for leakage into the building environment. Observed leakage will be corrected to the extent reasonably practical. Where feasible, leakage from liquid containing systems will be determined by counting the number of drops from each system. Where a system is not normally operating (such as the RHR and the SIS leakage will be determined by pressurizing applicable portions of each system to operating pressure and conducting a walk-through of the system to determine the amount of leakage. Gaseous Radwaste System leakage will be determined by operating the waste gas compressors in the recirculation mode while maintaining the system pressure using a regulated nitrogen source. The entire Gaseous Radwaste System, including the gas decay tanks and their relief valves, will be pressurized and leak tested. The system makeup rate will be determined with a gas flow meter at the supply regulator. All nonwelded connections in the system will be checked with a soap-and-water solution to locate any leaks, and appropriate corrective action will be taken.

The operating leak reduction program consists of a daily review of radioactive liquid waste processing, calculated containment leakage, and leakage observations. The daily leakage observations include those portions of the previously identified systems which are not shielded, enclosed, or otherwise controlled. Using this information, unidentified leakage baseline data is accumulated. Leakage quantities which exceed the baseline by a factor of two or more will be investigated to determine the source of leakage. Another feature of the program is the use of strategically located air monitors in the Auxiliary Building. These monitors have the capability to detect low leakage volumes of primary coolant at normal activity levels. Automatic monitoring is supplemented with smear surveys in controlled areas which contain major components of the listed systems.

Utilizing these monitoring programs, investigations and corrective actions will be implemented to reduce system leakage outside of the containment. When sources of leakage are identified, appropriate corrective measures to reduce the leakage will be initiated. Should correction of the leakage require reduction in operating mode, the correction will be made during the next scheduled shutdown.

Public Service Electric & Gas is taking other steps to eliminate unnecessary leakage into the Auxiliary Building. For example, currently capped valve leakoff connections are being hardpiped to the Liquid Radwaste System.

A further review was performed in accordance with IE Circular 79-21, "Prevention of Unplanned Releases of Radioactivity." The results of this review indicate that the system designs are acceptable.

2.1.6.b Shielding Review

Design Review of Plant Shielding and Environmental Qualification of Equipment for Spaces/Systems Which May be Used in Post Accident Operations

NRC Position

With the assumption of a post accident release of radioactivity equivalent to that described in Regulatory Guides 1.3 and 1.4, each licensee shall perform a radiation and shielding design review of the spaces around systems that may, as a result of an accident, contain highly radioactive materials. The design review should identify the location of vital areas and equipment, such as the Control Room, radwaste control stations, emergency power supplies, motor control centers, and instrument areas, in which personnel occupancy may be unduly limited or safety equipment may be duly degraded by the radiation fields during post accident operations of these systems.

Each licensee shall provide for adequate access to vital areas and protection of safety equipment by design changes, increased permanent or temporary shielding, or post accident procedural controls. The design review shall determine which types of corrective actions are needed for vital areas throughout the facility.

Clarification

Any area which will or may require occupancy to permit an operator to aid in the mitigation of or recovery from an accident is designated as a vital area. In order to assure that personnel can perform necessary post accident operations in the vital areas, we are providing the following guidance to be used by licensees to evaluate the adequacy of radiation protection to the operators:

1. Source Term

   The minimum radioactive source term should be equivalent to the source terms recommended in Regulatory Guides 1.3, 1.4, 1.7 and SRP 15.6.5. with appropriate decay times based on plant design.

   a. Liquid Containing Systems: 100 percent of the core equilibrium noble gas inventory, 50 percent of the core equilibrium halogen inventory, and 1 percent of all others are assumed to be mixed in the reactor coolant and liquids injected by Safety Injection.

   b. Gas Containing Systems: 100 percent of the core equilibrium noble gas inventory and 25 percent of the core equilibrium halogen activity are assumed to be mixed in the containment atmosphere. For gas containing lines connected to the primary system (e.g., BWR steam lines) the concentration of radioactivity shall be determined assuming the activity is contained in the gas space in the Primary Coolant System.

2. Dose Rate Criteria

   The dose rate for personnel in a vital area should be such that the guidelines of General Design Criterion (GDC) 19 should not be exceeded during the course of the accident. GDC 19 limits the dose to an operator to 5 rem whole body or its equivalent to any part of the body. When determining the dose to an operator, care must be taken to determine the necessary occupancy time in a specific area. For example, areas requiring continuous occupancy will require much lower dose rates than areas where minimal occupancy is required. Therefore, allowable dose rates will be based upon expected occupancy, as well as the radioactive source terms and shielding. However, in order to provide a general design objective, we are providing the following dose rate criteria with alternatives to be documented on a case-by-case basis. The recommended dose rates are average rates in the area. Local hot spots may exceed the dose rate guidelines provided occupancy is not required at the location of the hot spot. These doses are design objectives and are not to be used to limit access in the event of an accident.

   a. <u>Areas Requiring Continuous Occupancy:</u>

      <15mR/hr. These areas will require full time occupancy during the course of the accident. The Control Room and onsite technical support center are areas where continuous occupancy will be required. The dose rate for these areas is based on the Control Room occupancy factors contained in SRP 6.4.

   b. <u>Area Requiring Infrequent Access:</u>

      GDC 19. These areas may require access on a regular basis, but not continuous occupancy. Shielding should be provided to allow access at a frequency and duration estimated by the licensee. The plant radiochemical/chemical analysis

laboratory, radwaste panel, motor control center, instrumentation locations, and reactor coolant and containment gas sample stations are examples where occupancy may be needed often but not continuously.

Response

The Salem station radiation shielding design basis includes TID-14844 and Reg. Guide 1.183 assumptions for certain areas of the plant.

Considerable shielding is installed throughout the plant that would allow access to many other areas of the plant for shorter than normal periods of time after an accident even though the shielding in these areas was not designed specifically for post-LOCA sources.

The airborne source terms inside containment used for this review are based on a release of 100 percent of the noble gases and 40 percent of the iodine in the core to the containment atmosphere. Credit is taken for removal of iodine by plateout and by containment sprays. The recirculating sump water source terms used for this review are based on a release of 40 percent of the Halogens and 1 percent of all remaining fission products to a water volume equal to the sum of the primary coolant system, the RWST, the accumulators and the Boron Injection Tank (Unit 2 Only). These release fractions are considered representative of Regulatory Guide 1.4 and 1.183. Calculations are performed at one hour, one day and seven days (one week) following the reactor shutdown caused by the accident to allow adjustment for radioactive decay.

Calculations are focused on areas in the Auxiliary Building and penetration areas. Dose rates in the containment were calculated for locations where necessary (post-LOCA) equipment and instruments are located.

The systems and areas reviewed include the following:

1. RHR System
2. Safety Injection
3. CVCS Demineralizer Area
4. Charging Pump Compartments
5. Reactor Coolant Filter
6. Seal Water Filter Area
7. Chemistry Lab
8. Primary Sample Lab
9. Fuel Handling Building

10. Spent Fuel Pool Heat Exchanger Area
11. Liquid Radwaste
12. Control Room
13. Technical Support Center
14. Diesel Generator Compartments
15. Diesel Oil Supply Tank Compartments
16. Electrical Relay and Switchgear Rooms
17. Gaseous Radwaste Valve Stations
18. Liquid Radwaste Valve Stations
19. Component Cooling

Accessibility to systems and areas:

The effect of the design basis LOCA on the radiation environment in the plant was determined by:

1. Identifying the equipment expected to be handling recirculating sump water by reviewing ESF systems and associated P&ID's.
2. Locating this equipment in the plant by the use of mechanical drawings, and
3. Calculating the dose rates in each area of the plant affected by the equipment considering intervening shielding material.

The areas of the plant affected by post LOCA sources are shown in Tables A-4 and A 5. These zones are identified by TRIS zone number and are assigned a radiation zone based on the following key:

Zone	Dose Rate
I	≤ 15 mrem/hr
II	≤ 100 mrem/hr
III	≤ 1 rem/hr
IV	≤ 5 rem/hr

V	≤ 50 rem/hr
VI	≤ 500 rem/hr
VII	≤ 5000 rem/hr
VIII	> 5000 rem/hr

Separate zone ratings are presented for one hour, one day and one week following the accident. The following is a discussion of the accessibility of specific areas of the plant.

Residual Heat Removal System - Elevation 45 Feet and 55 Feet Auxiliary Building

1. The RHR pump compartments on elevation 45 feet (Location Code # 01045002, 12045002, 01045006, 12045006) in the Auxiliary Building would be a zone VIII during pump operation one hour after the accident and would not be accessible.

2. The dose rate in the adjacent RHR compartment will be zone III (See Note 5 on Tables A-4 and A-5). This compartment is accessible for limited periods of time while the other RHR system is operating.

3. The radiation zone on elevation 55 feet from the operating RHR System below (Location Code # 01055002, 12055002, 01055005, 12055005) is zone V at one hour post accident. This drops to zone III after 1 week of decay. Six inches of lead was installed to shield an exposed portion of RHR suction pipe.

Safety Injection System

1. The safety injection pump compartment (Location Code # 01084005, 12084005, 01084006, 12084006) is inaccessible while operating.

2. Radiation zones in adjacent areas, such as the spent fuel pool heat exchanger area (Location Code # 01084004, 12084004) and component cooling heat exchanger compartments (Location Code # 01084009, 12084009) are as high as zone V at contact with the pipe chase and pump compartment shield wall surfaces. This dose rate drops off substantially several feet from the walls. Limited access is afforded to these areas and no additional permanent shielding is planned.

Charging Pump Compartments

1. The radiation zone in the vicinity of these pumps (Location Code # 01084035, 12084035, 01084036, 12084036, 01084037, 12084037) may be as high as zone VIII, thus precluding access while the pumps are operating.

2. The dose rate through the wall separating the pump compartments produces a radiation zone V (Location Code # 01084041, 12084041).

3. The radiation zone outside the charging pump compartments is zone IV (e.g., Location Code # 01084025, 12084025); therefore, access to the components in the general area is available.

Chemical and Volume Control - Demineralizer Area

1. Dose rates from the demineralizers would not have a significant effect on access.

2. The dose rates from piping and valves located behind valve aisle shield walls would be the major source of radiation and result in radiation zone IV in the operating aisles (Location Code # 01084024, 12084024). This would be reduced by decay and will afford sufficient access to the area for limited valve operations.

Reactor Coolant and Seal Water Filters

1. The dose rates from these filters do not present a problem since the elements are replaced at predetermined radiation levels rather than high pressure drop. Post accident radiation levels in this area will not preclude access to this area. Each filter is located in an individual shielded compartment.

Primary Sample Lab

1. Use of the Primary Sample Lab may be required for post accident sample analyses. The Primary Sample Room may be required for post accident sampling of reactor coolant and containment sump (via RHR).

Counting Room

1. Direct dose rates in the Counting Room are not significantly affected by accident radiation source terms due to the location of the Counting Room. If high background dose rates preclude use of this area, alternate facilities are available.

Fuel Handling Building

1. Dose rates in the Fuel Handling Building due to direct radiation from the containment will not be significantly affected. The only exception to this is streaming from the elevation 130 feet containment personnel hatch and through the doorway into the Fuel Handling Building at elevation 130 feet.

2. The dose rates at the spent fuel pool heat exchanger and pump area in the Auxiliary Building (Location Code # 01084004, 12084004) produce a radiation zone V at one hour, thus affording limited access to this area.

Areas to Which Access May be Required Following an Accident

The areas discussed below are considered vital areas, i.e., areas to which access may be required following an accident. Accessibility is based on direct radiation levels due to contained radiation sources.

Control Room

The Control Room is located on elevation 122 feet and is sufficiently shielded from systems containing highly radioactive fluids. The radiation levels in the Control Room due to direct dose rates from the systems that may be required to operate after an accident are in the millirem per hour range.

Technical Support Center

The Technical Support Center is located in the Clean Facilities Building, and the doses due to the systems that will be operating in the Auxiliary Building are negligible. The doses to individuals in this building over the course of an accident are due to the cloud and airborne dose from plant releases. With installed shielding, the TEDE dose would be less than five rem. The Emergency Plan Implementing Procedures identify alternate facilities that are available if access to the TSC is limited.

Areas in the Auxiliary Building That do not Contain Highly Radioactive Sources of Radiation but May Require Access

These areas include:

Diesel generator compartments

Diesel oil supply tank compartments

Electrical relay and switchgear rooms

Analysis shows that sufficient shielding exists between these areas and adjacent compartments that contain radiation sources such that access to these areas is not precluded.

Access to Areas in the Auxiliary Building Which May Contain Highly Radioactive Sources

The hydrogen purge controls and containment isolation valve reset controls are operated from the Control Room. Access to other areas of the Auxiliary Building related to this equipment is unnecessary.

Chemistry Lab

The Chemistry Lab (Location Code # 01100005) is located on elevation 100 ft in the Auxiliary Building. If there is a LOCA in Unit 1, there will be a localized high dose rate area (zone V at one hour) in the south end of the room. Otherwise it is sufficiently shielded such that the major

contribution to the dose rate in the lab is due to streaming from the containment personnel hatch which produces a zone II. Alternate chemistry facilities are available if access to the Chemistry Lab is limited.

Gaseous Radwaste Control Center

The valve operating station for the gas decay tanks is accessible.

Liquid Radwaste Control Station (Valve Areas)

Liquid radwaste is processed by the Portable Liquid Radwaste System located on elevation 103 ft of the Truck Bay of the Auxiliary Building. Before processing post accident radwaste, appropriate radiological controls will be put in place to reduce potential exposures. After processing, the liquid waste is stored and sampled in the Waste Hold-up Tanks or Waste Monitor Hold-up Tank. The valves used to divert flow are remotely operated at the 104 panel located on elevation 64 ft in the Auxiliary Building. Remaining manual valves are located on elevation 84 ft of the Auxiliary Building in accessible areas.

Component Cooling Pump and Auxiliary Feedwater Pump and Valve Areas

These areas are located on elevation 84 feet (Location Code # 01084016, 12084016). The dose rates from shielded sources adjacent to this area produce a radiation zone III. This does not preclude access to this area.

Boric Acid Evaporator Room

These rooms (Location Code # 01100008, 12100008) are located on elevation 100 ft and contain the Post Accident Sample System (PASS) and Sampling System (SS) sample lines and PASS coolers (Unit 2 only). The post accident radiation zone in the east side of the room due to the safety injection pumps below is zone IV, which would allow limited access in the BAE room for the unit in which the LOCA occurs. The dose rate in the west part of the room near the door will be much lower (zone I). Once sampling is initiated, these areas become radiation zone VII due to the presence of the PASS and SS sample lines. If the LOCA is in Unit 1 and the PASS is used, both rooms will be affected by the PASS sample lines. If the SS is used, only the Unit 1 room is affected. For a LOCA in Unit 2, only the Unit 2 room is affected by the PASS sample lines. If the SS is used, both rooms are affected.

Electrical Penetration Areas

The areas adjacent to the containment on elevation 78 (Location Code # 02078001, 02078012, 13078001, 13078012) contain electrical busses that may require access for long term recovery. The radiation zone in these areas is zone V at one hour after the accident due to activity in containment, which drops to zone III by one day. If PASS sampling is initiated, a localized high dose rate area will exist on the west end of the penetration in the vicinity of the PASS valves. The dose rate due to these valves also produces a zone V in one hour, which drops to a zone IV at one day.

Equipment Qualification - Capability to Withstand Possible Radiation Doses Following an Accident

Vital components in the RHR, High Pressure Safety Injection, Low Pressure Safety Injection, and CVCS Systems have been identified. Integrated doses after 120 days were calculated for the following pieces of equipment: RHR motor-driven pumps, safety injection motor-driven pumps and centrifugal charging motor-driven pumps. The direct dose to each of these pieces of equipment is 2.31×10^7 rad, or less, after a period of 120 days. These doses were calculated using the TID source terms and reassessed consistent with Reg. Guide 1.183 Source Terms over a 30-day elapsed period of time.

The capability of the electrical equipment to withstand the radiation exposures has been evaluated and documented in the Environmental Equipment Qualification (EQ) Program.

The containment spray pumps are also considered vital components, but these pumps are located in a mild environment and will not be exposed to any radioactive fluid. During the injection mode, the pumps take suction from the refueling water storage tank (RWST). During the recirculation mode, the RHR pumps supply the spray headers and the containment spray pumps are not run.

Radiation exposures for Class 1E electrical equipment in the EQ Program have been qualified to the guidelines of 10CFR50.49.

Other Considerations

Local and Portable Shielding

Small quantities of temporary shielding such as lead bricks, lead blankets, and lead sheet are available at the station to shield local hot spots.

Source Terms

The source terms used for this study were based on 1-hour decay. The calculated dose rates would be reduced by a factor of 10 at one day and approximately 50 at one week after the start of an accident.

The radiation source terms used for the shielding design review were obtained by using the ORIGEN code. Shielding and dose rate calculations were performed using point kernel transport methods, primarily consisting of fixed source/shield configurations. These methods were utilized because the source geometries for the analysis were primarily pipes and tanks.

Original requirements concerning environmental qualification for post accident operation were addressed in a letter dated November 26, 1980 to the Director of NRR. This has been superseded by the Environmental Design Criteria (EDC) program.

2.1.7.a Preliminary Feedwater Initiation

Automatic Initiation of the Auxiliary Feedwater System for PWRs (Section

NRC Position

Consistent with satisfying the requirements of General Design Criterion 20 of Appendix A to 10CFR50 with respect to the timely initiation of the Auxiliary Feedwater System (AFWS), the following requirements shall be implemented in the short term:

1. The design shall provide for the automatic initiation of the AFWS.

2. The automatic initiation signals and circuits shall be designed so that a single failure will not result in the loss of AFWS function.

3. Testability of the initiating signals and circuits shall be a feature of the design.

4. The initiating signals and circuits shall be powered from the emergency buses.

5. Manual capability to initiate the AFWS from the Control Room shall be retained and shall be implemented so that a single failure in the manual circuits will not result in the loss-of-system function.

6. The ac motor-driven pumps and valves in the AFWS shall be included in the automatic actuation (simultaneous and/or sequential) of the loads to the emergency buses.

7. The automatic initiating signals and circuits shall be designed so that their failure will not result in the loss of manual capability to initiate the AFWS from the Control Room.

In the long term, the automatic initiation signals and circuits shall be upgraded in accordance with safety grade requirements.

Responses

The AFWS is described in Section 10.4. The system is designed to Class 1E criteria and is powered from the emergency power source.

Automatic initiation of the AFWS is provided by the following signals.

Motor-Driven Pumps

1. Loss of Offsite Power

2. Loss of Main Feed

3. Low-Low level in One Steam Generator

4. Safeguards Sequence Signal

Turbine-Driven Pump

1. Low-Low Level in Two Steam Generators

2. 4 kV Bus Undervoltage

Manual initiation of the systems may be accomplished either from the Control Room or locally at the pumps. The system and its components are designed for single failure considerations and are testable.

In a letter from R. L. Mittl to S. A. Vargo, dated December 31, 1980, PSE&G summarized the various detailed design data which has been furnished during the Staff's review of the Auxiliary Feedwater System.

2.1.7.b Auxiliary Feedwater Flow Indication

Auxiliary Feedwater Flow Indication to Steam Generators for PWRs

NRC Position

Consistent with satisfying the requirements set forth in GDC 13 to provide the capability in the Control Room to ascertain the actual performance of the AFWS when it is called to perform its intended function, the following requirement shall be implemented:

1. Safety grade indication of auxiliary feedwater flow to each steam generator shall be provided in the Control Room.

2. The auxiliary feedwater flow instrument channels shall be powered from the emergency buses consistent with satisfying the emergency power diversity requirements of the AFWS set forth in Auxiliary Systems BTP 10-1 of SRP Section 10.4.9.

Response

Safety grade indication of auxiliary feedwater flow to each steam generator is provided in the Control Room. These indicating channels are designed to the same criteria as the Protection System indicators. One testable flow instrument with an accuracy on the order of ± 2 percent is provided for each steam generator. In addition, three level instruments are provided for each steam generator. The instruments are all powered from the vital buses and are seismically qualified, with environmental qualification for the level instruments which are located inside the containment.

Assurance of sufficient water being provided to the steam generators is of primary concern. This is accomplished by control valve demand with steam generator level indication. Present indication of pump operation, valve demand/position, auxiliary feedwater flow (one/steam generator), auxiliary feedwater discharge pressure and steam generator level (three/steam generator) is adequate to meet the information requirements necessary to assure appropriate

operator action. All of the above equipment is powered from vital buses, and is adequate to meet short- and long-term requirements.

2.1.8.a Post-Accident Sampling

Improved Post-Accident Sampling Capability

NRC Position

A design and operational review of the Reactor Coolant and Containment Atmosphere Sampling Systems shall be performed to determine the capability of personnel to promptly obtain (less than 1 hour) a sample under accident conditions without incurring a radiation exposure to any individual in excess of 3 and 18 3/4 rems to the whole body of extremities, respectively. Accident conditions should assume a Regulatory Guide 1.3 or 1.4 release of fission products. If the review indicates that personnel could not promptly and safely obtain the samples, additional design features or shielding should be provided to meet the criteria.

A design and operational review of the radiological spectrum analysis facilities shall be performed to determine the capability to promptly quantify (less than 2 hours) certain radioisotopes that are indicators of the degree of core damage. Such radionuclides are noble gases (which indicate cladding failure), iodines and cesiums (which indicate high fuel temperatures), and non-volatile isotopes (which indicate fuel melting). The initial reactor coolant spectrum should correspond to a Regulatory Guide 1.3 or 1.4 release. The review should also consider the effects of direct radiation from piping and components in the Auxiliary Building, and possible contamination and direct radiation from airborne effluents. If the review indicates that the analyses required cannot be performed in a prompt manner with existing equipment, then design modifications or equipment procurement shall be undertaken to meet the criteria.

In addition to the radiological analyses, certain chemical analyses are necessary for monitoring reactor conditions. Procedures shall be provided to perform boron and chloride chemical analyses assuming a highly radioactive initial sample (Regulatory Guide 1.3 or 1.4 source term). Both analyses shall be capable of being completed promptly; i.e., the boron sample analysis within 1 hour and the chloride sample analysis within a shift.

Response

The Post-Accident Sampling System is described in Section 9.3. The PASS was removed from the licensing base as a result of License Amendments 254/235, dated November 5, 2002.

The eight PASS sample and return lines for Salem Units 1 and 2 have been cut and capped on both sides of the applicable containment penetration. The RCS sample lines were also cut and capped at the connection point to the sample system tubing.

2.1.8.b Increased Radiation Monitoring Range

Increased Range of Radiation Monitors

NRC Position

The requirements associated with this recommendation should be considered as advanced implementation of certain requirements to be included in a revision to Regulatory Guide 1.97, "Instrumentation to Follow the Course of an Accident," which has already been initiated, and in other Regulatory Guides, which will be promulgated in the near-term.

1. Noble gas effluent monitors shall be installed with an extended range designed to function during accident conditions as well as during normal operating conditions; multiple monitors are considered to be necessary to cover the ranges of interest.

    a. Noble gas effluent monitors with an upper range of 10^5 µCi/cc (Xe-133) are considered to be practical and should be installed in all operating plants.

    b. Noble gas effluent monitoring shall be provided for the total range of concentration extending from a minimum of 10^{-7} µCi/cc (Xe-133). Multiple monitors are considered to be necessary to cover the ranges of interest. The range capacity of individual monitors shall overlap by a factor of ten.

2. Since iodine gaseous effluent monitors for the accident condition are not considered to be practical at this time, capability for effluent monitoring of radioiodines for the accident condition shall be provided with sampling conducted by adsorption on charcoal or other media, followed by onsite laboratory analysis.

3. In containment radiation level monitors with a maximum range of 10^8 rad per hour shall be installed. A minimum of two such monitors that are physically separated shall be provided. Monitors shall be designed and qualified to function in an accident environment.

The NRC position in NUREG-0578 was subsequently modified to permit a maximum range of 10^7 R per hour for gamma – only monitoring.

The NRC position (in NUREG-0578) was subsequently modified to require noble gas effluent monitors with a range from as low as reasonably achievable to 10^5 µC/cc. In addition, the overlap requirement was deleted.

Response

1. Extended range noble gas monitors have been installed to meet this requirement. Post-accident samplers are located in enclosures outside of the Auxiliary Building to reduce radiation exposures to individuals obtaining post-accident plant vent samples. The detection range of extended range monitors is as follows:

Intermediate range monitors:

Unit 1 – 1R41B, Unit 2 – 2R41B = $1x10^{-4}$ to $1x10^{2}$ µCi/cc

High range monitors:

Unit 1 – 1R41C, Unit 2 – 2R41C = 0.1 to $1x10^{5}$ µCi/cc

Readings from these monitors are continuously available in the Control Room.

The existing R41A noble gas monitors provide for low detection ranges ($1x10^{-7}$ to $1x10^{1}$ µCi/cc) during normal plant operations.

Channels R41B,C are assigned the Reg. Guide 1.97 extended range noble gas monitoring functions.

High range monitors have also been installed to measure releases from the atmospheric steam relief and/or safety valves. A separate channel is installed for each of the four main steam lines of each unit. The range of these monitors is as follows:

Unit 1 - 1R46, A,B,C,D	0.1 to $2x10^{3}$ µCi/cc Xe-133
Unit 2 - 2R46, A,B,C,D	0.1 to $2x10^{3}$ µCi/cc Xe-133

2. The Salem R45 design provides for iodine sampling by absorption on charcoal or silver zeolite cartridges. Plant vent iodine grab samples can be obtained in the extended range monitoring enclosure to prevent personnel from receiving excessive radiation exposure.

3. There are two high range containment monitors in Salem Units 1 and 2. The R44A and R44B have a high range gamma radiation response of 10^{7} R/hr. These monitors are in separate locations on different elevations.

A full description of these higher range containment monitors is provided in FSAR Section 11.4.

2.1.8.c In Plant Iodine Instrumentation

<u>Improved In Plant Iodine Instrumentation</u>

<u>NRC Position</u>

Each licensee shall provide equipment and associated training and procedures for accurately determining the airborne iodine concentration throughout the plant under accident conditions.

Response

Sufficient instrumentation for iodine concentration monitoring throughout the plant under accident conditions is being provided.

The capability exists to accurately detect the presence of iodine in areas of the plant that may be occupied during an accident. This capability is provided by the equipment discussed below, which are part of the emergency equipment specified in the Emergency Plan. Operating procedures are available and training is conducted in accordance with the Emergency Plan.

> Portable Air Samplers – A charcoal or silver zeolite cartridge sampler can be used on a portable air sampler which can be carried around and plugged into an electrical receptacle.
>
> Continuous Air Monitors – These units are on carts with wheels and can be moved to sample different areas utilizing local electrical receptacles, can use charcoal or silver zeolite cartridge.
>
> In the event that the count room facility cannot be used to analyze iodine samples, Emergency Plan procedures specify that the Hope Creek count room facility be used.
>
> Emergency Plan procedures contain instructions for performing a field analysis of iodine samples using portable radiation detection instrumentation.

The capability to purge a charcoal cartridge with clean air or nitrogen and to remove the cartridge to a low background area for further analysis will be established.

2.1.9 Analysis of Accidents and Instrumentation to Follow Containment Conditions

Analysis of Design and Off-Normal Transients and Accidents NRC Position

Analysis, procedures, and training addressing the following are required:

1. Small break LOCAs,
2. Inadequate core cooling,
3. Transients and accidents.

Some analysis requirements for small breaks have already been specified by the Bulletins and Orders Task Force. These should be completed. In addition, pretest calculations of some of the Loss of Fluid Test (LOFT) small break tests (scheduled to start in September 1979) shall be performed as means to verify the analyses performed in support of the small break emergency procedures and in support of an eventual long-term verification of compliance with Appendix K of 10CFR50.

In the analysis of inadequate core cooling, the following conditions shall be analyzed using realistic (best-estimate) methods:

1. Low Reactor Coolant System inventory (two examples will be required – LOCA with forced flow, LOCA without forced flow).

2. Loss of natural circulation (due to loss of heat sink).

These calculations shall include the period of time during which inadequate core cooling is approached as well as the period of time during which inadequate core cooling exists. The calculations shall be carried out in real time far enough that all important phenomena and instrument indications are included. Each case should then be repeated taking credit for correct operator action. These additional cases will provide the basis for developing appropriate emergency procedures. These calculations should also provide the analytical basis for the design of any additional instrumentation needed to provide operators with an unambiguous indication of vessel water level and core cooling adequacy (see Section 2.1.3.b).

The analysis of transients and accidents shall include the design basis events specified in Section 15 of each FSAR. The analyses shall include a single active failure for each system called upon to function for a particular event. Consequential failures shall also be considered. Failures of the operators to perform required control manipulations shall be given consideration for permutations of the analyses. Operator actions that could cause the complete loss of function of a safety system shall also be considered. At present, these analyses need not address passive failures or multiple system failures in the short term. In the recent analysis of small break LOCAs, complete loss of auxiliary feedwater was considered. The complete loss of auxiliary feedwater may be added to the failures being considered in the analysis of transients and accidents if it is concluded that more is needed in operator training beyond the short-term actions to upgrade AFWS reliability. Similarly, the long-term, multiple failures and passive failures may be considered depending in part on staff review of the results of the short-term analyses.

The transient and accident analyses shall include event tree analyses, which are supplemented by computer calculations for those cases in which the system response to operator actions is unclear or these calculations could be used to provide important quantitative information not available from an event tree. For example, failure to initiate high-pressure injection could lead core uncovery for some transients, and a computer calculation could provide information on the amount of time available for corrective action. Reactor simulators may provide some information in defining the event trees and would be useful in studying the information available to the operators. The transient and accident analyses are to be performed for the purpose of identifying appropriate and inappropriate operator actions relating to important safety considerations such as natural circulation, prevention of core uncovery, and prevention of more serious accidents.

The information derived from the preceding analyses shall be included in the plant emergency procedures and operator training. It is expected that analyses performed by the Nuclear Steam Supply System (NSSS) vendors will be put in the form of emergency procedure guidelines and that the changes in the procedures will be implemented by each licensee or applicant.

In addition to the analyses performed by the reactor vendors, analyses of selected transients should be performed by the NRC Office of Research, using the best available computer codes to provide the basis for comparisons with the analytical methods being used by the reactor vendors. These comparisons together with comparisons to data, including LOFT small break test data, will constitute the short-term verification effort to assure the adequacy of the analytical methods being used to generate emergency procedures.

Response

Public Service Electric & Gas is a member of the Westinghouse Owners' Group and is actively supporting the generic analysis work described above. This analysis work will be completed on a schedule compatible with the industry effort. Emergency Procedures EI 4.4, "LOCA", and EI 4.17, "Leakage Greater than Charging Flow," (in effect, inadequate core cooling) will incorporate the results of the analysis work performed. Operating personnel have been advised of small break LOCA procedural changes and will receive the appropriate training for inadequate core cooling emergency procedures upon completion of the generic analysis work in early 1980.

Analyses of small break LOCA, symptoms of inadequate core cooling and required actions to restore core cooling, and analysis of transient and accident scenarios including operator actions not previously analyzed are being performed on a generic basis by the Westinghouse Owners' Group, of which PSE&G is a member. The small break analyses have been completed and were reported in WCAP-9600, which was submitted to the Bulletins and Orders Task Force by the Owners' Group on June 29, 1979. Incorporated in that report were guidelines that were developed as a result of small break analyses. These guidelines have been reviewed and approved by the Bulletins and Orders Task Force and have been presented to the Owners' Group utility representatives in a seminar held on October 16-19, 1979. Following this seminar, each utility has developed plant-specific procedures and trained their personnel on the new procedures. Revised procedures and training are in place in accordance with the requirement in Enclosure 6 to Mr. Eisenhut's letter of September 13, 1979, and Enclosure 2 to Mr. Denton's letter of October 30, 1979.

The work required to address the other two areas, inadequate core cooling and other transient and accident scenarios, was performed in conjunction with schedules and requirements established by the Bulletins and Orders Task Force. Analysis related to the definition of inadequate core cooling and guidelines for recognizing the symptoms of inadequate core cooling based on existing plant instrumentation and for restoring core cooling following a small break LOCA were detailed analysis than was originally proposed, and will be followed up with a more extensive and detailed analysis which will be available during the first quarter of 1980. The Salem 2 Emergency Procedures have been revised on an interim basis as described in the response to Section 2.1.3.b.

With respect to other transient accidents contained in the Salem FSAR, the Westinghouse Owners' Group has performed an evaluation of the actions which occur during an event by constructing a sequence of event trees for each of the non-LOCA and LOCA transients. From these event trees, a list of decision points for operator action has been prepared, along with a list of information available to the operator at each decision point. Following this, criteria have

been set for credible misoperation, and time available for operator decisions have been qualitatively assessed. The information developed was then used to test Abnormal and Emergency Operating Procedures against the event sequences and determine if inadequacies exist in them.

The Owners' Group has also provided test predictions of the LOFT L3-1 nuclear small break experiment.

Instrumentation to Monitor Containment Conditions During the Course of an Accident

NRC Position

Consistent with satisfying the requirements set forth in General Design Criterion 13 to provide the capability in the Control Room to ascertain containment conditions during the course of an accident, the following requirements shall be implemented:

1. A continuous indication of containment pressure shall be provided in the Control Room. Measurement and indication capability shall include three times the design pressure of the containment for concrete, four times the design pressure for steel, and minus five psig for all containments.

2. A continuous indication of hydrogen concentration in the containment atmosphere shall be provided in the Control Room. Measurement capability shall be provided over the range of 0 to 10 percent hydrogen concentration under both positive and negative ambient pressure.

3. A continuous indication of containment water level shall be provided in the Control Room for all plants. A narrow range instrument shall be provided for PWRs and cover the range from the bottom to the top of the containment sump. Also for PWRs, a wide range instrument shall be provided and cover the range from the bottom of the containment to the elevation equivalent to a 500,000-gallon capacity. For boiling water reactors (BWRs), a wide range instrument shall be provided and cover the range from the bottom to 5 feet above the normal water level of the suppression pool.

The containment pressure, hydrogen concentration and wide range containment water level measurements shall meet the design and qualification provisions of Regulatory Guide 1.97, including qualification, redundancy, and testability. The narrow range containment water level measurement instrumentation shall be qualified to meet the requirements of Regulatory Guide 1.89 and shall be capable of being periodically tested.

Response

Containment pressure instrumentation, containment water level instrumentation, and containment hydrogen instrumentation have been installed.

Installation of Remotely-Operated High Point Vents in the Reactor Coolant System

NRC Position

Each applicant and licensee shall install Reactor Coolant System and reactor vessel head high point vents remotely operated from the Control Room. Since these vents form a part of the reactor coolant pressure boundary, the design of the vents shall conform to the requirements of Appendix A to 10CFR50, General Design Criteria. In particular, these vents shall be safety grade, and shall satisfy the single failure criterion and the requirements of IEEE Standard 279 to ensure a low probability of inadvertent actuation.

Each applicant and licensee shall provide the following information concerning the design and operation of these high point vents:

1. A description of the construction, location, size, and power supply for the vents along with results of analyses of LOCAs initiated by a break in the vent pipe. The results of the analyses should be demonstrated to be acceptable in accordance with the acceptance criteria of 10CFR50.46.

2. Analyses demonstrating that the direct venting of non-condensable gases with perhaps high hydrogen concentrations does not result in violation of combustible gas concentration limits in containment as described in 10CFRS50.44, Regulatory Guide 1.7 (Rev. 1), and SRP, Section 6.2.5.

3. Procedural guidelines for the operators' use of the vents. The information available to the operator for initiating or terminating vent usage shall be discussed.

Response

The Reactor Vessel Head Venting System is described in Section 5.1.

2.2.1.a Shift Supervisor's Responsibilities

Shift Supervisor's Responsibilities NRC Position

1. The highest level of corporate management of each licensee shall issue and periodically reissue a management directive that emphasizes the primary management responsibility of the shift supervisor for safe operation of the plant under all conditions on his shift and that clearly establishes his command duties.

2. Plant procedures shall be reviewed to assure that the duties, responsibilities, and authority of the shift supervisor and Control Room operators are properly defined to effect the establishment of a definite line of command and clear delineation of the command decision authority of the shift supervisor in the Control Room relative to other plant management personnel. Particular emphasis shall be placed on the following:

a. The responsibility and authority of the shift supervisor shall be to maintain the broadest perspective of operational conditions affecting the safety of the plant as a matter of highest priority at all times when on duty in the Control Room. The idea shall be reinforced that the shift supervisor should not become totally involved in any single operation in times of emergency when multiple operations are required in the Control Room.

b. The shift supervisor, until properly relieved, shall remain in the Control Room at all times during accident situations to direct the activities of Control Room operators. Persons authorized to relieve the shift supervisor shall be specified.

c. If the shift supervisor is temporarily absent from the Control Room during routine operations, a lead Control Room operator shall be designated to assume the Control Room command function. These temporary duties, responsibilities, and authority shall be clearly specified.

3. Training programs for shift supervisors shall emphasize and reinforce the responsibility for safe operation and the management function the shift supervisor is to provide for assuring safety.

4. The administrative duties of the shift supervisor shall be reviewed by the senior officer of each utility responsible for plant operations. Administrative functions that detract from or are subordinate to the management responsibility for assuring the safe operation of the plant shall be delegated to other operations personnel not on duty in the Control Room.

Response

A written directive describing and emphasizing the primary management responsibilities of shift supervisors and establishing their command duties was placed in effect September 12, 1979. Using the guidance of this directive, Administrative Procedure AP-5 and the requalification program have been revised as necessary.

Shift administrative activities have been reviewed to determine which duties should be delegated to personnel who are not on duty in the Control Room. Duties which were found to detract from the shift supervisor's responsibility for safe operation of the plant have been reassigned.

2.2.1.b Shift Technical Advisor

Shift Technical Advisor

NRC Position

Each licensee shall provide an on-shift technical advisor to the shift supervisor. The shift technical advisor may serve more than one unit at a multi-unit site if qualified to perform the advisor function for the various units.

The shift technical advisor shall have a bachelor's degree or equivalent in a scientific or engineering discipline and have received specific training in the response and analysis of the plant for transients and accidents. The shift technical advisor shall also receive training in plant design and layout, including the capabilities of instrumentation and controls in the Control Room. The licensee shall assign normal duties to the shift technical advisors that pertain to the engineering aspects of assuring safe operations of the plant, including the review and evaluation of operating experience.

Response

See UFSAR Section 13 and Technical Specification Section 6.0 for Shift Technical Advisor staffing and qualification requirements.

2.2.1.c Shift Turnover Procedures

Shift and Relief Turnover Procedures

NRC Position

The licensees shall review and revise as necessary the plant procedure for shift and relief turnover to assure the following:

1. A checklist shall be provided for the oncoming and offgoing Control Room operators and the oncoming shift supervisor to complete and sign. The following items, as a minimum, shall be included in the checklist:

    a. Assurance that critical plant parameters are within allowable limits (parameters and allowable limits shall be listed on the checklist).

    b. Assurance of the availability and proper alignment of all systems essential to the prevention and mitigation of operational transients and accidents by a check of the control console (what to check and criteria for acceptable status shall be included on the checklist).

    c. Identification of systems and components that are in a degraded mode of operation permitted by the Technical Specifications. For such systems and components, the length of time in the degraded mode shall be compared with the Technical Specifications action statement (this shall be recorded as a separate entry on the checklist).

2. Checklists or logs shall be provided for completion by the offgoing and oncoming auxiliary operators and technicians. Such checklists or logs shall include any equipment under maintenance of test that by themselves could degrade a system critical to the prevention and mitigation of operational transients and accidents or initiate an operational transients (what to check and criteria for acceptable status shall be included on the checklist); and

3. A system shall be established to evaluate the effectiveness of the shift and relief turnover procedure (for example, periodic independent verification of system alignments).

Response

The Salem operating logs contain checklists which provide oncoming and offgoing shifts with a status of critical plant parameters. In order to provide for a more formalized shift turnover, a program has been established in Administrative Procedure AP-5 and the Operating Department Manual to ensure that the oncoming shift log and plant status review has been properly accomplished. An adequate evaluation system, which provides for a frequent management review of shift supervision logs to determine the quality of shift operations is already in use at Salem. This inspection consists of verification of operator understanding of equipment status and plant alarms, and direct observation of the conduct of operations in the Control Room.

2.2.2.a Control Room Access

Control Room Access

NRC Position

The licensee shall make provisions for limiting access to the Control Room to those individuals responsible for the direct operation of the nuclear power plant (e.g., operations supervisor, shift supervisor, and control room operators), to technical advisors who may be requested or required to support the operation, and to pre-designated NRC personnel. Provisions shall include the following:

Develop and implement an administrative procedure that establishes the authority and responsibility of the person in charge of the Control Room to limit access.

1. Develop and implement an administrative procedure that establishes the authority and responsibility of the person in charge of the Control Room to limit access.

2. Develop and implement procedures that establish a clear line of authority and responsibility in the Control Room in the event of an emergency. The line of succession for the person in charge of the Control Room shall be established and limited to persons possessing a current senior reactor operator's license. The plan shall clearly define the lines of communication and authority for plant management personnel not in direct command of operations, including those who report to stations outside of the Control Room.

Response

The Control Room access points are card key controlled.

Only those personnel who are required in the Control Room have unescorted access to the Control Room area.

Administrative controls adequately restrict access to the Control Rooms to only those personnel who can demonstrate an actual need to be there.

Responsibilities and lines of authority in the Control Room are addressed in the response to Section 2.2.1.a.

2.2.2.b Onsite Technical Support Center

Onsite Technical Support Center

NRC Position

Each operating nuclear power plant shall maintain an onsite technical support center separate from and in close proximity to the Control Room that has the capability to display and transmit plant status to those individuals who are knowledgeable of and responsible for engineering and management support of reactor operations in the event of an accident. The center shall be habitable to the same degree as the control room for postulated accident conditions. The licensee shall revise his emergency plans as necessary to incorporate the role and location of the technical support center.

A complete set of as-built drawings and other records, as described in ANSI N45.2.9-1974, shall be properly stored and filed at the site and accessible to the technical support center under emergency conditions. These documents shall include, but not be limited to, general arrangement drawings, P&IDs, piping system isometrics, electrical schematics, and photographs of components installed without layout specifications (e.g., field-run piping and instrument tubing).

Response

1. An onsite Technical Support Center (TSC) is provided in accordance with 10CFR50.47(b) and 10CFR50 Appendix E.IV.E.8 on the third floor of the Clean Facilities Building and meets the detailed criteria of NUREG-0737, Supplement 1 and NUREG-0696 as described in the Emergency Plan.

   Additional descriptions and details of the TSC and support systems are confirmed in the Emergency Plan provided under separate submittal pursuant to 10CFR50.4(b)(5).

   The TSC is provided with plant parameters and derived variables representative of the safety status of the plant via the Safety Parameter Display System (SPDS) in accordance with NUREG-0737 supplement 1 and NUREG 0696. A description of SPDS is contained in Section 7.10.

2.2.2.c Onsite Operational Support Center

Onsite Operational Support Center (Section 2.2.2.c)

NRC Position

An area to be designated as the onsite operational support center shall be established. It shall be separate from the Control Room and shall be the place to which the operations support personnel will report in an emergency situation. Communications with the Control Room shall be provided. The emergency plan shall be revised to reflect the existence of the center and to establish the methods and lines of communication and management.

Response

The Operational Control Center (OSC) is located on 127 foot elevation, Service Building. This area is separate from each Control Room and is the place to which operations support personnel report in an emergency situation. Communications with each Control Room are provided at this location.

Location Number	Location Description	Zone No. 1 Hour	1 Day	1 Week
01045001	11 RHR Rm	VIII	VII	VII
	Shielded (2)	V	IV	III
01045002	11 RHR Pmp Rm	VIII	VII	VII
	11 RHR Pmp Rm (3)	III	II	I
01045003	11 RHR Hx Area	VIII	VII	VI
01045004	11 Cont. Sump Valve Rm	VIII	VII	VI
01045005	Instrument Rack 102	VIII	VII	VII
	Shielded (2)	V	IV	III
01045006	12 RHR Pmp Rm	VIII	VII	VII
	12 RHR Pmp Rm (3)	III	II	I
01045007	12 RHR Hx Area	VIII	VII	VI
01045008	12 Cont. Sump Valve Rm	VIII	VII	VI
01055002	11 RHR Valve Rm	V	IV	III
01055003	11 RHR Hx Rm	VIII	VII	VI
01055004	11 RHR Valve Rm	VIII	VII	VI
01055005	12 RHR Valve Rm	V	IV	III
01055006	12 RHR Hx Rm	VIII	VII	VI
01055007	12 RHR Valve Rm	VIII	VII	VI
01064001	4KV Vital Bus & Battery RM	IV	III	II
01064003	Laundry & Chem Dr Pump Rm	IV	III	II
01064004	11 Monitor Tk & Pump Rm	IV	III	III
01064005	12 Monitor Tk & Pump Rm	V	IV	III
01064008	Waste Holdup Tk Rm	IV	III	II
01064009	Waste Holdup Tk Rm	IV	III	II
01064011	Waste Monitor Holdup Tank Rm	IV	III	II
01064014	11 Waste Gas Compressor Rm	IV	III	II
01064017	12 Waste Gas Compressor Rm	IV	III	II
01064020	Waste Gas Valve Area	IV	III	II
01064021	11 Gas Decay Tk	IV	III	II
01064022	12 Gas Decay Tk	IV	III	II
01064028	11 HUT Rm	IV	III	II
01064029	12 HUT Rm	IV	III	II
01064031	Pipeway	VIII	VII	VII
01084002	460/230V Vital	IV	III	III
01084003	Aisle Outside SFP Pump	I	I	I
01084004	Spent Fuel Pit Pump Rm	V	III	III
01084005	11 Safety Inj Pmp Rm	VIII	VII	VI
01084006	12 Safety Inj Pmp Rm	VIII	VII	VI
01084007	Pipe Alley	VIII	VII	VII
	West End of Pipeway	VII	VII	VI
01084008	Corridor at CC Hx	V	III	III
01084009	11 CC Pump Rm	V	III	III
01084010	11 Comp Cooling VA Rm	VII	VI	VI
01084012	12 CC Hx Rm	III	II	II

Location Number	Location Description	Zone No. 1 Hour	1 Day	1 Week
01084016	13 Aux FW Pump	III	II	II
01084018	LTD Hx Valve Rm	I	I	I
01084019	LTD Hx Rm	IV	III	II
01084020	Seal Water Hx Valve Rm	I	I	I
01084021	Seal Water Hx Rm	IV	III	II
01084022	Elec. Control Center	I	I	I
01084024	Vent Duct & Equipment Hatch	IV	III	II
01084025	Corridor at 11 Cont Spray	IV	III	II
01084026	Rm Next to 11 Chrg Pmp	VII	VII	VI
01084027	CO2 Fire Equip Rm	III	II	II
01084029	11 Diesel FOST Rm	IV	III	II
01084034	Chrg Pmp Valve Alley (11)	VII/III	VII/N/A	VI/N/A
01084035	11 Chrg Pmp Rm	VIII	VII	VI
01084036	12 Chrg Pmp Rm	VIII	VII	VI
01084037	13 Chrg Pmp Rm	VII	VII	VI
01084038	Cont. Spray Area	IV	III	II
01084041	No.1 Aisle (4)	V	IV	III
01084046	Stairwell	V	III	III
01100013	Waste Evap Rm	V	III	III
01100005	Chemistry Lab (8)	V	IV	III
01100006	Access Aisle	V	IV	III
01100007	Corridor at BAE (5)	III	II	I
01100008	Boric Acid Evap Rm (5)	V	IV	III
01100008	BAE Room Door	I	I	I
01100009	S/G Blowdown Sample Rm	V	IV	III
01100014	Ion Exchanger Rm	V	III	III
01100015	BA Transfer Pumps	V	IV	III
01100019	Drum Storage Area	V	IV	III
01113001	Primary Sampling Room (9)	IV	III	II
02078001	A, B, & C East Valves C.C	V	III	II
02078002	Top of 12 Hx Rm	VIII	VII	VII
02078003	BIT Rm Lower Level	VII	VII	VI
02078004	BIT Rm Upper Level	VIII	VII	VI
02078005	Mech. Pen. Area	VII	VII	VI
02078006	Stm Gen Blowdown Area	VII	VII	VI
02078007	S.W. Piping Room	III	II	II
02078008	Mech. Pen Area	VII	VII	VI
02078009	S.W. Piping Rm	III	II	II
02078011	S.W. Piping Rm	III	II	II
02078012	Elec Pen/Cont Rad Mon. Area (6) (7)	V	III	II
02100002	Main Steam Pen Area	V	IV	III
02100003	Personnel Hatch	V	IV	III

Location Number	Location Description	Zone No. 1 Hour	1 Day	1 Week
12100005	Counting Room (10)	V	IV	III

Notes:

1. Dose rates are due to post accident sources only and do not include normal operating dose rates. Any area not listed is Zone I.
2. This area is behind the partial wall separating this zone from the RHR pump room (Location Code # 01045002, 01045006).
3. This dose rate is due to the adjacent RHR pump room (Location Code # 01045002 or 01045006).
4. This area has the same shielding as the adjacent pump cubicles.
5. The designations shown correspond to conditions prior to initiation of SS.
6. Dose rates are due to shine through containment.
7. Containment atmosphere samples can be obtained approximately 17 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
8. Containment sump boron analysis can be performed approximately 46 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
9. Reactor coolant and containment sump samples can be obtained approximately 19 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
10. Bounding dose rates are those associated with a Design Basis LOCA in Salem Unit 2.
11. The second designation is for a LOCA in Salem Unit 2.

Location Number	Location Description	Zone No. 1 Hour	1 Day	1 Week
01100005	Chemistry Lab (10)	V	IV	III
01113001	Primary Sampling Room (11)	IV	III	II
12045001	21 RHR Rm	VIII	VII	VII
	Shielded (2)	V	IV	III
12045002	21 RHR Pmp Rm	VIII	VII	VII
	21 RHR Pmp Rm (3)	III	II	I
12045002	21 RHR Pmp Rm	VIII	VII	VII
12045003	21 RHR Hx Area	VIII	VII	VI
12045004	21 Cont. Sump Valve Rm	VIII	VII	VI
12045005	Instrument Rack 102	VIII	VII	VII
12045006	22 RHR Pmp Rm	VIII	VII	VII
	22 RHR Pmp RM (3)	III	II	I
12045007	22 RHR Hx Area	VIII	VII	VI
12045008	22 Cont. Sump Valve Rm	VIII	VII	VI
12055002	21 RHR Valve Rm	V	IV	III
12055003	21 RHR Hx Rm	VIII	VII	VI
12055004	21 RHR Valve Rm	VIII	VIII	VI
12055005	22 RHR Valve Rm	V	IV	III
12055006	22 RHR Hx Rm	VIII	VII	VI
12055007	22 RHR Valve Rm	VIII	VII	VI
12064001	4KV Vital Bus & Battery Rm	IV	III	II
12064003	Laundry & Chem Dr Pump Rm	IV	III	II
12064004	21 Monitor Tk & Pump Rm	IV	III	III
12064005	22 Monitor Tk & Pump Rm	V	IV	III
12064008	Waste Holdup Tk Rm	IV	III	II
12064009	Waste Holdup Tk Rm	IV	III	II
12064011	Waste Monitor Holdup Tank Rm	IV	III	II
12064014	21 Waste Gas Compressor Rm	IV	III	II
12064017	22 Waste Gas Compressor Rm	IV	III	II
12064020	Waste Gas Valve Area	IV	III	II
12064021	21 Gas Decay Tk	IV	III	II
12064022	22 Gas Decay Tk	IV	III	II
12064028	21 HUT Rm	IV	III	II
12064029	22 HUT Rm	IV	III	II
12064031	Pipeway	VIII	VII	VII
12084002	460/230V Vital	IV	III	III
12084003	Aisle Outside SFP Pump	I	I	I
12084004	Spent Fuel Pit Pump Rm	V	III	III
12084005	21 Safety Inj Pmp Rm	VIII	VII	VI
12084006	22 Safety Inj Pmp Rm	VIII	VII	VI
12084007	Pipe Alley	VIII	VII	VII
	West End of Pipeway	VII	VII	VI
12084008	Corridor at CC Hx	V	III	III
12084009	21 CC Pump Rm	V	III	III
12084010	21 Comp Cooling VA Rm	VII	VI	VI
12084012	22 CC Hx Rm	III	II	II
12084016	23 Aux FW Pump	III	II	II
12084018	LTD Hx Valve Rm	I	I	I
12084019	LTD Hx Rm	IV	III	II

Location Number	Location Description	Zone No. 1 Hour	1 Day	1 Week
12084020	Seal Water Hx Valve Rm	I	I	I
12084021	Seal Water Hx Rm	IV	III	II
12084022	Elec. Control Center	I	I	I
12084024	Vent Duct & Equipment Hatch	IV	III	II
12084025	Corridor at 21 Cont Spray	IV	III	II
12084026	Rm Next to 21 Chrg Pmp	VII	VII	VI
12084027	CO2 Fire Equip Rm	III	II	II
12084029	21 Diesel FOST Rm	IV	III	II
12084034	Chrg Pmp Valve Alley (12)	VII/III	VII/N/A	VI/N/A
12084035	21 Chrg Pmp Rm	VIII	VII	VI
12084036	22 Chrg Pmp Rm	VIII	VII	VI
12084037	23 Chrg Pmp Rm	VII	VII	VI
12084038	Cont. Spray Area	IV	III	II
12084041	No.2 Aisle (4)	V	IV	III
12084046	Stairwell	V	III	III
12100005	Counting Room (6)	V	IV	III
12100006	Access Aisle	V	IV	III
12100007	Corridor at BAE (5)	III	II	I
12100008	Boric Acid Evap Rm (5)	V	IV	III
	BAE Room Door	I	I	I
12100009	PASS Room (8)	V	IV	III
12100013	Waste Evap Rm	V	III	III
12100014	Ion Exchanger Rm	V	III	III
12100015	BA Transfer Pumps	V	IV	III
12100019	Drum Storage Area	V	IV	III
13078001	A, B, & C East Valves C.C	V	III	II
13078002	Top of 22 Hx Rm	VIII	VII	VII
13078003	BIT Rm Lower Level	VII	VII	VI
13078004	BIT Rm Upper Level	VIII	VII	VI
13078005	Mech. Pen. Area	VII	VII	VI
13078006	Stm Gen Blowdown Area	VII	VII	VI
13078007	S.W. Piping Room	III	II	II
13078008	Mech. Pen Area	VII	VII	VI
13078009	S.W. Piping Rm	III	II	II
13078011	S.W. Piping Rm	III	II	II
13078012	Elec Pen/Cont Rad Mon. Area (7) (9)	V	III	II
13100002	Main Steam Pen Area	V	IV	III
13100003	Personnel Hatch	V	IV	III

Notes

1. Dose rates are due to post accident sources only and do not include normal operating dose rates. Any area not listed is Zone I.
2. This area is behind the partial wall separating this zone from the RHR pump room (Location Code # 12045002, 12045006).
3. This dose rate is due to the adjacent RHR pump room (Location Code # 12045002 or 12045006).
4. This area has the same shielding as the adjacent pump cubicles.
5. The designations shown correspond to conditions prior to initiation of SS.
6. The dose rates apply only in the north end of the room. Other areas will be a zone I.

7. Dose rates are due to shine through containment.
8. Dose rates are due to external sources and will allow limited access to obtain samples. Post accident sampling is addressed in the response to item 2.1.8.a.
9. Containment atmosphere samples can be obtained approximately 17 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
10. Containment sump boron analysis can be performed approximately 46 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
11. Reactor coolant and containment sump samples can be obtained approximately 19 hours after accident initiation without exceeding personnel exposure guidelines based on SS.
12. The second designation is for a LOCA in Salem Unit 1.

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A LICENSE RENEWAL FINAL SAFETY ANALYSIS REPORT SUPPLEMENT

A.1 Introduction

The application for a renewed operating license is required by 10 CFR 54.21(d) to include a FSAR Supplement. This appendix, which includes the following sections, comprises the FSAR supplement:

- Section A.1.1 contains a listing of the aging management programs that correspond to NUREG-1801 Chapter XI programs.
- Section A.1.2 contains a listing of the plant-specific aging management programs.
- Section A.1.3 contains a listing of aging management programs that correspond to NUREG-1801 Chapter X programs associated with Time-Limited Aging Analyses.
- Section A.1.4 contains a listing of the Time-Limited Aging Analyses (TLAA).
- Section A.1.5 contains a discussion of the Quality Assurance Program and Administrative Controls.
- Section A.2 contains a summarized description of the aging management programs.
- Section A.2.1 contains a summarized description of the NUREG-1801 Chapter XI programs for managing the effects of aging.
- Section A.2.2 contains a summarized description of the plant-specific programs for managing the effects of aging.
- Section A.3 contains a summarized description of the NUREG-1801 Chapter X programs that support the TLAAs.
- Section A.4 contains a summarized description of the TLAAs applicable to the period of extended operation.
- Section A.5 contains the License Renewal Commitment List.

The integrated plant assessment for license renewal identified new and existing aging management programs necessary to provide reasonable assurance that systems, structures, and components within the scope of license renewal will continue to perform their intended functions consistent with the Current Licensing Basis (CLB) for the period of extended operation. The period of extended operation is defined as 20 years from the unit's original operating license expiration date.

A.1.1 NUREG-1801 Chapter XI Aging Management Programs

The NUREG-1801 Chapter XI Aging Management Programs (AMPs) are described in the following sections. The AMPs are either consistent with generally accepted industry methods as discussed in NUREG-1801 or require enhancements.

Commitments for program additions and enhancements are identified in the Appendix A.5 License Renewal Commitment List.

1. ASME Section XI Inservice Inspection, Subsections IWB, IWC, and IWD (Section A.2.1.1)
2. Water Chemistry (Section A.2.1.2)
3. Reactor Head Closure Studs (Section A.2.1.3)
4. Boric Acid Corrosion (Section A.2.1.4)
5. Nickel-Alloy Penetration Nozzles Welded to the Upper Reactor Vessel Closure Heads of Pressurized Water Reactors (Section A.2.1.5)
6. Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (CASS) (Section A.2.1.6)
7. PWR Vessel Internals (Section A.2.1.7)
8. Flow-Accelerated Corrosion (Section A.2.1.8)
9. Bolting Integrity (Section A.2.1.9)
10. Steam Generator Tube Integrity (Section A.2.1.10)
11. Open-Cycle Cooling Water System (Section A.2.1.11)
12. Closed-Cycle Cooling Water System (Section A.2.1.12)
13. Inspection of Overhead Heavy Load (Related to Refueling) Handling Systems (Section A.2.1.13)
14. Compressed Air Monitoring (Section A.2.1.14)
15. Fire Protection (Section A.2.1.15)
16. Fire Water System (Section A.2.1.16)
17. Aboveground Steel Tanks (Section A.2.1.17)
18. Fuel Oil Chemistry (Section A.2.1.18)

19. Reactor Vessel Surveillance (Section A.2.1.19)
20. One-Time Inspection (Section A.2.1.20)
21. Selective Leaching of Materials (Section A.2.1.21)
22. Buried Piping Inspection (Section A.2.1.22)
23. One-Time Inspection of ASME Code Class 1 Small Bore-Piping (Section A.2.1.23)
24. External Surfaces Monitoring (Section A.2.1.24)
25. Flux Thimble Tube Inspection (Section A.2.1.25)
26. Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components (Section A.2.1.26)
27. Lubricating Oil Analysis (Section A.2.1.27)
28. ASME Section XI, Subsection IWE (Section A.2.1.28)
29. ASME Section XI, Subsection IWL (Section A.2.1.29)
30. ASME Section XI, Subsection IWF (Section A.2.1.30)
31. 10 CFR Part 50, Appendix J (Section A.2.1.31)
32. Masonry Wall Program (Section A.2.1.32)
33. Structures Monitoring Program (Section A.2.1.33)
34. RG 1.127 Inspection of Water-Control Structures (Section A.2.1.34)
35. Protective Coating Monitoring and Maintenance Program (Section A.2.1.35)
36. Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (Section A.2.1.36)
37. Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits (Section A.2.1.37)
38. Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (Section A.2.1.38)
39. Metal Enclosed Bus (Section A.2.1.39)
40. Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (Section A.2.1.40)

A.1.2 Plant-Specific Aging Management Programs

The plant-specific programs are described in the following sections. Commitments for program additions and enhancements are identified in Section A.5 License Renewal Commitment List.

1. High Voltage Insulators (Section A.2.2.1)
2. Periodic Inspection (Section A.2.2.2)
3. Aboveground Non-Steel Tanks (Section A.2.2.3)
4. Buried Non-Steel Piping Inspection (Section A.2.2.4)
5. Boral Monitoring Program (Section A.2.2.5)
6. Nickel Alloy Aging Management Program (Section A.2.2.6)

A.1.3 NUREG-1801 Chapter X Aging Management Programs

The NUREG-1801 Chapter X Aging Management Programs associated with Time-Limited Aging Analyses are described in the following sections. The AMPs are either consistent with generally accepted industry methods as discussed in NUREG-1801 Chapter X or require enhancements. Commitments for program additions and enhancements are identified in Section A.5 License Renewal Commitment List.

1. Metal Fatigue of Reactor Coolant Pressure Boundary (Section A.3.1.1)
2. Environmental Qualification (EQ) of Electric Components (Section A.3.1.2)

A.1.4 Time-Limited Aging Analyses

Summaries of the Time-Limited Aging Analyses applicable to the period of extended operation are included in the following sections:

1. Reactor Vessel Neutron Embrittlement (Section A.4.2)
2. Metal Fatigue of Piping and Components (Section A.4.3)
3. Other Plant-Specific Analyses (Section A.4.4)
4. Fuel Transfer Tube Bellows Design Cycles (Section A.4.5)
5. Crane Load Cycle Limits (Section A.4.6)
6. Environmental Qualification of Electric Equipment (Section A.4.7)

A.1.5 Quality Assurance Program and Administrative Controls

The Quality Assurance Program implements the requirements of 10 CFR 50, Appendix B, and is consistent with the summary in Appendix A.2, “Quality Assurance for Aging Management Programs (Branch Technical Position IQMB-1)” of NUREG-1800. The Quality Assurance Program includes the elements of corrective action, confirmation process, and administrative controls, and these elements are applicable to the safety-related and non-safety-related systems, structures, and components (SSCs) that are subject to Aging Management Review (AMR). In many cases, existing activities were found adequate for managing aging effects during the period of extended operation.

A.2 Aging Management Programs

A.2.1 NUREG-1801 Chapter XI Aging Management Programs

This section provides summaries of the NUREG-1801 programs credited for managing the effects of aging.

A.2.1.1 ASME Section XI Inservice Inspection, Subsections IWB, IWC, and IWD

The ASME Section XI Inservice Inspection, Subsections IWB, IWC, and IWD aging management program is an existing program that consists of periodic volumetric and visual examinations of components for assessment, identification of signs of degradation, and establishment of corrective actions. The program includes inspections performed to manage cracking, loss of fracture toughness and loss of material in Class 1, 2, and 3 piping and components exposed to reactor coolant, steam and treated water environments. The inspections will be implemented in accordance with 10 CFR 50.55(a). These activities include inspections, and monitoring and trending of results to confirm that aging effects are managed.

A.2.1.2 Water Chemistry

The Salem Water Chemistry aging management program is an existing program that provides activities for monitoring and controlling the chemical environments of the Salem primary cycle and secondary cycle systems such that aging effects of system components are minimized. Aging effects include cracking, loss of material, reduction of neutron-absorbing capacity and reduction of heat transfer. The primary cycle scope of this program consists of the reactor coolant system and related auxiliary systems containing treated water, reactor coolant, treated borated water and steam, including the primary side of the steam generators that contain treated water and steam. The secondary cycle portion of the program consists of the various secondary side systems and the secondary side of the steam generators. Major component types include reactor vessel, reactor internals, piping, piping elements and piping components, heat exchangers and tanks. The Water Chemistry aging management program is consistent with EPRI, Pressurized Water Reactor Primary Chemistry Guidelines, and Plant UFSAR limits for fluorides, chlorides, and dissolved oxygen. The Water Chemistry program is consistent with EPRI, Pressurized Water Secondary Water Chemistry Guidelines.

The One-Time Inspection program will be used to verify the effectiveness of the Water Chemistry program in managing the aging effects for stainless steel components in a treated borated water environment where dissolved oxygen may not be controlled to less than 100 ppb.

A.2.1.3 Reactor Head Closure Studs

The Reactor Head Closure Studs program is an existing program that provides for condition monitoring and preventive activities to manage reactor head closure stud cracking caused by stress corrosion cracking. The program is implemented through station procedures based on the examination and inspection requirements specified in ASME Section XI, Table IWB-2500-1 and preventive measures described in NRC Regulatory Guide 1.65, “Materials and Inspection for Reactor Vessel Closure Studs.”

A.2.1.4 Boric Acid Corrosion

The Boric Acid Corrosion aging management program is an existing program that manages loss of; material on piping, piping elements, heat exchangers, bolting, panels, racks, cabinets and enclosures, insulation jacketing, cable trays, concrete anchors, concrete embedments, manhole covers, conduit, miscellaneous steel, and other structural components, component supports, cracking, blistering, flaking, peeling, and delamination of coatings, and corrosion of electrical connector contact surfaces. The program includes provisions to identify, inspect, examine and evaluate leakage, and initiate corrective action. The program relies in part on implementation of recommendations contained in NRC Generic Letter 88-05, “Boric Acid Corrosion of Carbon Steel Reactor Components in PWR plants”.

A.2.1.5 Nickel-Alloy Penetration Nozzles Welded to the Upper Reactor Vessel Closure Heads of Pressurized Water Reactors

The Nickel-Alloy Penetration Nozzles Welded to the Upper Reactor Vessel Closure Heads of Pressurized Water Reactors aging management program is an existing program that consists of a combination of periodic bare metal visual inspections of the outer surface of the upper reactor vessel closure head (closure head) and periodic non-destructive examinations (surface and volumetric) of the upper vessel head penetration (VHP) nozzles and associated J-groove welds, for assessment, identification of signs of degradation, and establishment of corrective actions. The program monitors the condition of nickel-alloy components and J-groove welds for the effects of cracking in a reactor coolant environment. The inspections will be implemented in accordance with 10 CFR 50.55(a), which endorses ASME Code Case N-729-1. The program ensures the structural integrity of the closure head, VHP nozzles, and associated J-groove welds, and the detection of cracking and any loss of material/wastage prior to a loss of intended function. These activities include examinations, and monitoring and trending of results to confirm that aging effects are managed.

A.2.1.6 Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (CASS)

The Thermal Aging Embrittlement of Cast Austenitic Stainless Steel aging management program is a new program that includes condition monitoring activities to provide assurance that reactor coolant system CASS components susceptible to thermal aging embrittlement meet the

intended functions. The reactor coolant system CASS components are maintained by inspecting and evaluating the extent of thermal aging embrittlement in accordance with the requirements of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Section XI, 1998 Edition, 2000 Addenda. The Salem ASME Section XI Inservice Inspection program is augmented by the implementation of the Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (CASS) aging management program which monitors the aging effect of loss of fracture toughness due to thermal aging embrittlement of CASS components.

The Thermal Aging Embrittlement of CASS program will include a screening for components susceptible to thermal aging embrittlement based on casting method, molybdenum content, and percent ferrite. For “potentially susceptible” components, thermal aging embrittlement management will be accomplished through either a component-specific flaw tolerance evaluation or enhanced volumetric examinations. Inspections or evaluations are not required for components that are determined not to be susceptible to thermal aging embrittlement. Screening for CASS components susceptible to thermal aging embrittlement is not required for pump casings and valve bodies. The existing ASME Section XI inspection requirements are adequate for managing the aging effects of Class 1 pump casings and valve bodies. This new program will be implemented prior to the period of extended operation.

A.2.1.7 PWR Vessel Internals

Salem Units 1 and 2 commit to the following activities for the new PWR Vessel Internals program:

1. Participate in the industry programs for investigating and managing aging effects on reactor internals.

2. Evaluate and implement the results of the industry programs as applicable to the reactor internals.

3. Upon completion of these programs, but not less than 24 months before entering the period of extended operation, submit an inspection plan for reactor internals to the NRC for review and approval.

This new aging management program will be implemented prior to the period of extended operation.

A.2.1.8 Flow-Accelerated Corrosion

The Flow-Accelerated Corrosion (FAC) aging management program at Salem is an existing program based on EPRI guidelines in NSAC-202L, “Recommendations for an Effective Flow Accelerated Corrosion Program.” The program provides for predicting, detecting, and monitoring wall thinning in piping, fittings, valve bodies, and heat exchangers due to FAC. Analytical evaluations and periodic examinations of locations that are most susceptible to wall thinning due to FAC are used to predict the amount of wall thinning in pipes, fittings, and feedwater heater shells. Program activities include analyses to determine critical locations, baseline inspections

to determine the extent of thinning at these critical locations, and follow-up inspections to confirm the predictions. Repairs and replacements are performed as necessary.

A.2.1.9 Bolting Integrity

The Bolting Integrity aging management program is an existing program that provides for aging management of pressure retaining bolted joints, component support bolting and structural bolting within the scope of license renewal. The Bolting Integrity program incorporates NRC and industry recommendations delineated in NUREG-1339, “Resolution of Generic Safety Issue 29: Bolting Degradation or Failure in Nuclear Power Plants,” EPRI TR-104213, “Bolted Joint Maintenance & Applications Guide,” and EPRI NP 5769, “Degradation and Failure of Bolting in Nuclear Power Plants,” as part of the comprehensive corporate component bolting program. The program provides for managing cracking, loss of material and loss of preload of bolted joints in the following environments: air, groundwater/soil, raw water, treated borated water, and soil. Included in the aging management activities directed by this program are visual inspections for pressure retaining bolted joint leakage and preventive measures.

The Bolting Integrity aging management program will be enhanced to include:

1. In the following cases, bolting material should not be reused:

    a. Galvanized bolts and nuts,

    b. ASTM A490 bolts; and

    c. Any bolt and nut tightened by the turn of nut method.

This enhancement will be implemented prior to the period of extended operation.

A.2.1.10 Steam Generator Tube Integrity

The Steam Generator Tube Integrity aging management program is an existing program that establishes the operation, maintenance, testing, inspection and repair of the steam generators to ensure that Technical Specification surveillance requirements, ASME Code requirements and the Maintenance Rule performance criteria are met, thereby adequately managing the aging effects of the steam generator tubes, plugs, and tube support plates. Aging effects include cracking, loss of material, loss of preload, reduction of heat transfer and wall thinning. The program identifies and maintains the steam generator design and licensing bases and implements NEI 97-06. NEI 97-06 establishes a framework for prevention, inspection, evaluation, repair and leakage monitoring measures.

A.2.1.11 Open-Cycle Cooling Water System

The Salem Open-Cycle Cooling Water System aging management program is an existing program that includes mitigative, performance-monitoring, and condition-monitoring activities to manage the internal corrosion of piping to minimize susceptibility of corrosion and to verify that corrosion has not exceeded acceptance limits. More than one type of aging management

program is necessary to ultimately ensure that the aging effects are adequately managed and the intended function(s) are maintained for the extended period of operation. These activities provide assurance that cracking, material loss, and heat transfer reduction aging effects are maintained at acceptable levels for systems and components within the scope of license renewal. The GL 89-13 activities provide for management of aging effects in raw water cooling systems through tests and inspections per the guidelines of NRC Generic Letter 89-13. System and component testing, visual inspections, non-destructive examination (e.g., RT-Radiographic Testing, UT-Ultrasonic Testing, and/or ECT-Eddy Current Testing), and sodium hypochlorite injection are conducted to ensure that aging effects are managed such that system and component intended functions and integrity are maintained. Major component types include pumps, piping, piping elements, piping components, heat exchangers and tanks.

The Salem Open-Cycle Cooling Water System (OCCWS) aging management program primarily consists of station GL 89-13 activities that include sodium hypochlorite injection, system testing, periodic inspections and non-destructive examination. The program includes surveillance and control techniques to manage aging effects caused by bio-fouling, corrosion, erosion, protective coating failures, and silting in the Service Water System components and on the systems, structures, and components supported by the Service Water System. Other activities include station maintenance inspections, component preventive maintenance, plant surveillance testing, and inspections. These activities provide for management of loss of material (without credit for protective coatings) and heat transfer reduction (including fouling from biological, corrosion product, and external sources) aging effects where applicable in system components exposed to a raw water environment.

A.2.1.12 Closed-Cycle Cooling Water System

The Closed-Cycle Cooling Water System aging management program is an existing program that manages aging of piping, piping components, piping elements, tanks, and heat exchangers that are included in the scope of license renewal for loss of material, stress corrosion cracking, and reduction of heat transfer and are exposed to a closed cooling water environment at Salem. The Closed-Cycle Cooling Water System aging management program relies on mitigation measures to minimize corrosion by maintaining inhibitors and by performing non-chemistry monitoring consisting of inspection and nondestructive examinations based on industry-recognized guidelines of EPRI 1007820 for closed-cycle cooling water systems. Station maintenance inspections and nondestructive examinations provide condition monitoring of heat exchangers exposed to closed-cycle cooling water environments.

The following enhancements will be incorporated to the Closed-Cycle Cooling Water System Program.

1. The Component Cooling System is not currently analyzed for sulfates, which is not consistent with the EPRI standard. The program will be enhanced to include monitoring this parameter as part of the Closed-Cycle Cooling Water program.

2. The emergency diesel generator jacket water system is not currently analyzed for azole or ammonia, chlorides, fluorides, and microbiologically-influenced corrosion in

accordance with the current EPRI standard. The program will be enhanced to include monitoring these parameters as part of the Closed-Cycle Cooling Water program.

3. The Closed-Cycle Cooling Water program for the Chilled Water System will have a program or hardware change to bring the system chemistry parameters into compliance with EPRI 1007820, prior to the period of extended operation.

4. New recurring tasks will be established to enhance the performance monitoring of selected heat exchangers cooled by the Component Cooling System.

5. New recurring tasks will be established for enhancing the performance monitoring of selected Chilled Water System components.

6. A one-time inspection of selected components will be established for Chilled Water System piping to confirm the effectiveness of the Closed-Cycle Cooling Water program.

7. A one-time inspection of selected closed-cycle cooling water components in stagnant flow areas will be conducted to confirm the effectiveness of the Closed-Cycle Cooling Water program.

8. A one-time inspection of selected closed-cycle cooling water chemical mixing tanks and associated piping will be conducted to confirm the effectiveness of the Closed-Cycle Cooling Water program on the interior surfaces of the tanks and associated piping.

9. The program will be enhanced such that the Heating Water and Heating Steam System will have a pure water control program instituted, in accordance with EPRI 1007820, prior to the period of extended operation.

10. New recurring tasks will be established for enhancing the performance monitoring of selected Heating Water and Heating Steam System components.

11. A one-time inspection of selected Heating Water and Heating Steam System piping will be conducted to confirm the effectiveness of the Closed-Cycle Cooling Water program.

These enhancements will be implemented prior to entering the period of extended operation. In addition, the one-time inspections will be performed prior to the period of extended operation.

A.2.1.13 Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems

The Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems aging management program is an existing program that is credited for managing aging effects of cranes and hoists in the scope of license renewal. Administrative controls ensure that only allowable loads are handled. Cranes and hoists structural components, including the bridge, the trolley, bolting, lifting devices, and the rail system are visually inspected periodically

for loss of material. Bolting is also monitored for loss of preload by inspecting for missing, detached, or loosened bolts. The program relies on procurement controls and installation practices, defined in plant procedures, to ensure that only approved lubricants and proper torque are applied to bolting.

The Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems Program will be enhanced to include:

1. The program will be enhanced to include visual inspection of structural components and structural bolts for loss of material due to general, pitting, and crevice corrosion and structural bolting for loss of preload due to self-loosening.

2. The program will be enhanced to require visual inspection of the rails in the rail system for loss of material due to wear.

3. The acceptance criteria will be enhanced to require evaluation of significant loss of material due to corrosion for structural components and structural bolts, and significant loss of material due to wear of rail in the rail system.

These enhancements will be implemented prior to the period of extended operation.

A.2.1.14 Compressed Air Monitoring

The Compressed Air Monitoring aging management program is an existing program that manages piping, piping components, and piping elements, compressor housings, and tanks for loss of material due to general, pitting, and crevice corrosion in the Compressed Air System. The Compressed Air Monitoring aging management activities consist of preventive maintenance and condition monitoring measures to manage the aging effects.

A.2.1.15 Fire Protection

The Fire Protection aging management program is an existing program that includes a fire barrier inspection, diesel-driven fire pump inspection and Halon and Carbon Dioxide systems inspections and functional tests. These inspections and functional tests provide assurance that the fire protection components within the scope of license renewal are maintained operational. The fire protection components are comprised of piping, piping elements, piping components, doors, dampers and fire barriers. The Fire Protection program provides for visual inspections of fire barriers and penetration seals for signs of degradation such as cracking, loss of material and hardening, through periodic inspection and functional testing. These components within the scope of license renewal are maintained in accordance to the guidance contained within NFPA Codes and Standards. The fire barrier inspections require periodic visual inspections of fire barrier penetration seals, fire barrier walls, ceilings, and floors, and periodic visual inspection. Functional testing and inspections of the fire rated doors and dampers is performed to ensure that their operability is maintained. The program includes surveillance tests of fuel oil systems for the diesel-driven fire pumps to ensure that the fuel supply lines can perform their intended functions. The program also includes visual inspections and periodic operability tests

of Halon and Carbon Dioxide fire suppression systems using NFPA Codes and Standards for guidance.

The Fire Protection aging management program will be enhanced to include:

1. The Salem routine inspection procedures will be enhanced to provide additional inspection guidance to identify degradation of fire barrier walls, ceilings, and floors for aging effects such as cracking, spalling and loss of material caused by freeze-thaw, chemical attack, and reaction with aggregates.

2. The Salem fire pump supply line functional tests will be enhanced to provide specific guidance for examining exposed external surfaces of the fire pump diesel fuel oil supply line for corrosion during pump tests.

3. The Halon and Carbon Dioxide fire suppression system functional test procedures will be enhanced to include visual inspection of system piping and component external surfaces for signs of corrosion or other age related degradation, and for mechanical damage. The system functional test procedures will also be enhanced to include acceptance criteria stating that identified corrosion or mechanical damage will be evaluated with corrective action taken as appropriate.

These enhancements will be implemented prior to the period of extended operation.

A.2.1.16 Fire Water System

The Fire Water System aging management program is an existing program that provides for system pressure monitoring, fire system header flushing and flow testing, pump performance testing, hydrant flushing, and visual inspection activities. Testing, inspection and flushing activates are performed periodically assure that the aging effect of loss of material due to corrosion, microbiologically influenced corrosion (MIC), or biofouling are managed such that the system intended functions are maintained. System flow tests measure hydraulic resistance and compare results with previous testing, as a means of evaluating the internal piping conditions. Major component types include piping and fittings, heat exchangers, tanks and pumps. Monitoring system piping flow characteristics ensures that signs of loss of material will be detected in a timely manner. Pump performance tests, hydrant flushing and system inspections are based on guidance from the applicable NFPA standards.

The Fire Water System program will be enhanced as follows:

1. The Fire Water System aging management program will be enhanced to inspect selected portions of the water based fire protection system piping located aboveground and exposed to the fire water internal environment by non-intrusive volumetric examinations. These inspections shall be performed prior to the period of extended operation and will be performed every 10 years thereafter.

2. The Fire Water System aging management program will be enhanced to replace or perform 50-year sprinkler head inspections and testing using the guidance of NFPA-25

"Standard for the Inspection, Testing and Maintenance of Water-Based Fire Protection Systems" (2002 Edition), Section 5-3.1.1. These inspections will be performed by the 50-year in-service date and every 10 years thereafter.

These enhancements will be implemented prior to the period of extended operation, with the inspections and testing performed in accordance with the schedule described above.

A.2.1.17 Aboveground Steel Tanks

The Aboveground Steel Tanks aging management program is an existing program that will manage loss of material aging effects of outdoor carbon steel tanks (Fire Protection Water Storage Tanks). Paint is a corrosion preventive measure, and periodic visual inspections will monitor degradation of the paint and any resulting metal degradation of carbon steel tanks. Inspection of the grout or sealant at the tank-foundation interface is included for tanks that are located on a sand bed on top of concrete foundations.

The Aboveground Steel Tanks program will be enhanced as follows:

1. The program will be enhanced to include UT measurements of the bottom of the tanks that are supported on concrete foundations (Fire Protection Water Storage Tanks). Measured wall thickness will be monitored and trended if significant material loss is detected. These thickness measurements of the tank bottom will be taken and evaluated against design thickness and corrosion allowance to ensure that significant degradation is not occurring and the component intended function would be maintained during the extended period of operation.

2. The program will be enhanced to provide routine visual inspections of the Fire Protection Water Storage Tanks external surfaces. The visual inspection activities will include inspection of the grout or sealant between the tank bottom and the concrete foundation for signs of degradation.

These enhancements will be implemented prior to the period of extended operation. Tank bottom UT inspections will also be performed prior to the period of extended operation.

A.2.1.18 Fuel Oil Chemistry

The Fuel Oil Chemistry aging management program is an existing program that includes preventive activities to provide assurance that contaminants are maintained at acceptable levels in fuel oil for systems and components within the scope of License Renewal. The fuel oil tanks within the scope of License Renewal are maintained by monitoring and controlling fuel oil contaminants in accordance with the guidelines of the American Society for Testing and Materials (ASTM). Fuel oil sampling and analysis is performed in accordance with approved procedures for new fuel oil and stored fuel oil. Fuel oil tanks are periodically drained of accumulated water and sediment, cleaned, and internally inspected. These activities effectively manage the effects of aging by providing reasonable assurance that potentially harmful contaminants are maintained at low concentrations.

The Fuel Oil Chemistry aging management program will be enhanced to include:

1. Equivalent requirements for fuel oil purity and fuel oil testing as described by the Standard Technical Specifications.

2. Analysis for particulate contamination in new and stored fuel oil.

3. Addition of biocides, stabilizers and inhibitors as determined by fuel oil sampling or inspection activities.

4. Quarterly analysis for bacteria in new and stored fuel oil.

5. Internal inspection of 350-gallon Fire Pump Day Tanks (S1DF-1DFE21 and S1DF-1DFE23) using visual inspections and ultrasonic thickness examination of tank bottoms.

6. Sampling of new fuel oil deliveries for API gravity and flash point prior to off load.

7. Internal inspection of the 30,000-gallon Fuel Oil Storage Tanks (S1DF-1DFE1, S1DF-1DFE2, S2DF-2DFE1 and S2DF-2DFE2) using visual inspections and ultrasonic thickness examination of tank bottoms.

8. To confirm the absence of any significant aging effects, a one-time inspection of each of the 550-gallon Diesel Fuel Oil Day Tanks will be performed.

These enhancements will be implemented prior to the period of extended operation. In addition, the one-time inspections will be performed prior to the period of extended operation.

A.2.1.19 Reactor Vessel Surveillance

The Reactor Vessel Surveillance Program is an existing program that manages the loss of fracture toughness due to neutron irradiation embrittlement of the reactor vessel beltline materials. The program fulfills the intent and scope of 10 CFR 50, Appendix H. This program evaluates neutron embrittlement by projecting Upper Shelf Energy (USE) for all reactor materials with projected neutron exposure greater than 10^{17} n/cm2 (E > 1.0 MeV) after 60 years of operation and with the development of pressure-temperature limit curves. Embrittlement information is obtained in accordance with Regulatory Guide 1.99, Rev. 2. In accordance with 10 CFR Part 50, Appendix H, Salem Units 1 and 2 will submit their proposed capsule withdrawal schedules for approval prior to implementation.

The Reactor Vessel Surveillance program will be enhanced as follows:

1. The Reactor Vessel Surveillance program will be enhanced to state the bounding vessel inlet temperature (cold leg) limits and fluence projections, and to provide instructions for changes.

    a. Inlet Temperature Range Limitation: 525°F (min) to 590°F (max)

b. Fluence Limitation (max.): 1.00 x 10^{20} n/cm^2 (E > 1.0 MeV)

2. The Reactor Vessel Surveillance program will be enhanced to describe the capsule storage requirements and the need to retain future pulled capsules.

3. The Reactor Vessel Surveillance program will be enhanced to specify a scheduled date for withdrawal of capsules, including pulling one of the remaining four capsules during the period of extended operation to monitor the effects of long-term exposure to neutron embrittlement for each Salem Unit. Those dates shall be approved by the NRC prior to withdrawal of the capsules, in accordance with 10 CFR Part 50, Appendix H.

4. The Reactor Vessel Surveillance Program will be enhanced to incorporate the requirements for (1) withdrawing the remaining capsules when the monitor capsule is withdrawn during the period of extended operation and placing them in storage for the purpose of reinstituting the Reactor Vessel Surveillance Program if required, i.e. if the reactor vessel exposure conditions (neutron flux, spectrum, irradiation temperature, etc.) are altered, and subsequently the basis for the projection to 60 years warrant the reinstitution, and (2) changes to the reactor vessel exposure conditions and the potential need to re-institute a vessel surveillance program will be discussed with the NRC staff prior to changing the plant's licensing basis.

5. Enhancements to the current Reactor Vessel Surveillance Program will be made to require that if future plant operations exceed the limitations or bounds specified for cold leg temperatures (vessel inlet) or higher fluence projections, then the impact of plant operation changes on the extent of reactor vessel embrittlement will be evaluated and the NRC shall be notified.

   a. Inlet Temperature Range Limitation: 525°F (min) to 590°F (max)

   b. Fluence Limitation (max.): 1.00 x 10^{20} n/cm^2 (E > 1.0 MeV)

These enhancements will be implemented prior to the period of extended operation.

A.2.1.20 <u>One-Time Inspection</u>

The One-Time Inspection aging management program is a new program that will provide reasonable assurance that an aging effect is not occurring, or that the aging effect is occurring slowly enough to not affect a component intended function during the period of extended operation, and therefore will not require additional aging management. The program will be credited for cases where either (a) an aging effect is not expected to occur but there is insufficient data to completely rule it out, (b) an aging effect is expected to progress very slowly in the specified environment, but the local environment may be more adverse than that generally expected, or (c) the characteristics of the aging effect include a long incubation period. Major component types covered by the program include piping, piping elements and piping components, steam generators, heat exchangers and tanks.

The One-Time Inspection aging management program will be used for the following:

1. To confirm the effectiveness of the Water Chemistry program to manage the loss of material, cracking, and the reduction of heat transfer aging effects for aluminum, copper alloy, steel, stainless steel, and cast austenitic stainless steel in treated water, treated borated water where dissolved oxygen may not be controlled to less than 100 ppb, steam, and reactor coolant environments.

2. To confirm the effectiveness of the Fuel Oil Chemistry program to manage the loss of material aging effect for aluminum, copper alloy, gray cast iron, steel and stainless steel in a fuel oil environment.

3. To confirm the effectiveness of the Lubricating Oil Analysis program to manage the loss of material and the reduction of heat transfer aging effects for aluminum, copper alloy, ductile cast iron, gray cast iron, steel, stainless steel, cast austenitic stainless steel and titanium alloy in a lubricating oil environment.

The sample plan for inspections associated with the One-Time Inspection program will be developed to ensure there are adequate inspections to address each of the material, environment, and aging effect combinations. A sample size of 20% of the population (up to a maximum of 25 inspections) will be established for each of the sample groups. Inspection methods will include visual examination or volumetric examinations. Acceptance criteria are in accordance with industry guidelines, codes, and standards, including the applicable edition of ASME Boiler and Pressure Vessel Code, Section XI. The One-Time Inspection program provides for the evaluation of the need for follow-up examinations to monitor the progression of aging if age-related degradation is found that could jeopardize an intended function before the end of the period of extended operation. Should aging effects be detected, the program triggers actions to characterize the nature and extent of the aging effect and determines what subsequent monitoring is needed to ensure intended functions are maintained during the period of extended operation.

The new program, including performance of physical inspections and evaluation of results, will be implemented prior to the period of extended operation.

A.2.1.21 Selective Leaching of Materials

The Selective Leaching of Materials aging program is a new program that will include one-time inspections of a representative sample of susceptible components to determine where loss of material due to selective leaching is occurring in susceptible material and environment combinations. The program will also include aging management activities, for material and environment combinations where selective leaching is identified, to manage loss of material due to selective leaching. Components include valve bodies, filter housing, heat exchanger components, pump casings, strainer bodies, piping and fittings, drain traps, and tanks. One-time inspections will include visual examinations, supplemented by hardness tests, and other examinations, as required. If selective leaching is found, the condition will be evaluated to determine the need to expand inspection scope.

One-time inspections of susceptible material and environment combinations, where selective leaching has not previously been confirmed, will be performed in the last 10 years of the current term, prior to entering the period of extended operation. A sample size of 20% of susceptible components will be subjected to a one-time inspection with a maximum of 25 inspections for each of the susceptible material groups. For material and environment combinations where selective leaching is identified, aging management activities, such as periodic inspections, will be implemented to manage aging such that the component intended function is maintained consistent with the current licensing basis through the period of extended operation.

A.2.1.22 Buried Piping Inspection

The Buried Piping Inspection aging management program is an existing program that manages the external surface aging effects of loss of material for piping and components in a soil or groundwater (external) environment. The Salem buried component activities consist of preventive and condition-monitoring measures to manage, detect and monitor the loss of material due to external corrosion for piping and components in the scope of license renewal that are in a soil environment.

External inspections of buried components will occur opportunistically when they are excavated during maintenance.

The Buried Piping Inspection aging management program will be enhanced to include:

1. A study will be performed prior to entering the period of extended operation to assess the possibility and benefits of installing a cathodic protection system, versus other mitigative and preventive actions.

2. A soil characterization study will be performed prior to entering the period of extended operation to determine soil corrosivity in the vicinity of buried piping. The results of the study will be used as an input to the program so that inspections will be performed at the locations of highest risk.

3. At least one (1) opportunistic or focused excavation and inspection will be performed on each of the Fire Protection System material groupings, which include carbon steel, ductile cast iron, and gray cast iron piping and components during each ten (10) year period, beginning ten (10) years prior to entry into the period of extended operation.

4. The following inspections apply to buried, carbon steel, safety-related portions of the specified systems. A different segment for each system will be inspected for each ten year period.

    a) At least one (1) opportunistic or focused excavation and inspection on each of the Auxiliary Feedwater and Compressed Air systems during the ten (10) years prior to entering the period of extended operation.

b) At least three (3) opportunistic or focused excavations and inspection of the Service Water System during the ten (10) years prior to entering the period of extended operation.

c) If, as a result of the soil characterization study, it is determined that the soil is not corrosive in the vicinity of all of the Auxiliary Feedwater, Service Water, and Compressed Air systems, Salem will perform at least one (1) opportunistic or focused excavation and inspection on each of the respective systems every ten (10) years during the period of extended operation.

d) If, as a result of the soil characterization study, it is determined that the soil is corrosive in the vicinity of the Auxiliary Feedwater, Service Water, or Compressed Air systems, Salem will perform at least two (2) opportunistic or focused excavations and inspections on the respective susceptible system(s) every ten (10) years during the period of extended operation.

5. If, based on the results of the initial soil characterization study, it is determined that the soil is not corrosive in the vicinity of the Auxiliary Feedwater, Service Water, or Compressed Air systems, Salem will perform a second Soil Characterization Study within approximately fifteen (15) years of the original study. The results of the second soil study will be entered into the Corrective Action Program for evaluation.

6. The buried Auxiliary Feedwater System piping located inside the Unit 2 Fuel Transfer Tube Area (approximately 125 feet) will be replaced and rerouted above ground prior to entering the period of extended operation.

These enhancements will be implemented prior to the period of extended operation, with the inspections performed in accordance with the schedule described above.

A.2.1.23 <u>One-Time Inspection of ASME Code Class 1 Small Bore-Piping</u>

The One-Time Inspection of ASME Code Class 1 Small-Bore Piping program is a new program that will manage the aging effect of cracking in stainless steel small-bore, less than nominal pipe size (NPS) 4 inches and greater than or equal to NPS 1 Class 1 piping through the use of a combination of volumetric examinations and visual inspections. The components include piping, piping elements and piping components. This program is a part of the Salem Risk Informed Inservice Inspection program.

The program will manage the aging effect through the identification and evaluation of cracking in small-bore Class 1 piping. The program will include one-time volumetric examination of a sample of class 1 butt welds for pipe size less than 4 inches NPS and greater than or equal to NPS 1. In addition, Salem Units 1 and 2 will perform four volumetric examinations, two per unit, from a population of 36 susceptible Class 1 small-bore socket welds on Unit 1 and 34 susceptible Class 1 small-bore socket welds on Unit 2. Provided the technology is available, these inspections shall be performed prior to entering the period of extended operation. More specifically, the volumetric examinations will analyze Class 1 small-bore socket welds as follows:

- Two Class 1 small-bore socket welds (one per unit) for intergranular stress corrosion cracking
- Two Class 1 small bore socket welds (one per unit) for cracking caused by thermal fatigue (thermal and mechanical loading)

The current examination method and frequency, VT-2, as allowed per the code, each refueling outage as defined in Code Case N-578-1, Table 1 will continue to be conducted. Any cracking identified in small-bore Class 1 piping resulting from stress corrosion or thermal and mechanical loading will result in periodic inspections. The program will effectively manage the aging effect by identifying and evaluating cracking in small-bore Class 1 piping prior to loss of intended function.

The One-Time Inspection of ASME Code Class 1 Small-Bore Piping Program will be implemented and one-time inspections completed and evaluated prior to the period of extended operation.

A.2.1.24 External Surfaces Monitoring

The External Surfaces Monitoring aging management program is a new program that directs visual inspections that are performed during system walkdowns. The program consists of periodic visual inspection of components such as piping, piping components, ducting, and other components within the scope of license renewal. The program manages aging effects through visual inspection of external surfaces for evidence of loss of material. Loss of material due to boric acid corrosion is managed by the Boric Acid Corrosion program. The external surfaces of components that are buried are inspected via the Buried Piping Inspection and Buried Non-Steel Piping Inspection programs. The external surfaces of above ground tanks are inspected via the Aboveground Steel Tanks and Aboveground Non-Steel Tanks programs.

This new aging management program will be implemented prior to the period of extended operation.

A.2.1.25 Flux Thimble Tube Inspection

The Flux Thimble Tube Inspection Program is a new program that manages the loss of material of the flux thimble tube materials by use of inspection methods such as eddy current testing. The reason it is a new program is that in Salem's response and supplements to NRC Bulletin 88-09, Salem implemented a flux thimble tube inspection program in 1988, then discontinued the program in 1993 after replacement of the flux thimble tubes with those of an improved design and satisfactory follow-up inspections of the improved materials for Unit 1. This new program implements the recommendations of NRC Bulletin 88-09, "Thimble Tube Thinning in Westinghouse Reactors". Acceptance criteria were established from industry guidance.

This new aging management program will be implemented prior to the period of extended operation.

A.2.1.26 Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components

The Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components aging management program is a new program that manages the aging of the internal surfaces of steel piping, piping components and piping elements, ducting components, tanks and heat exchanger components. This program will manage the aging effect of loss of material. The program includes provisions for visual inspections of the internal surfaces of components not managed under other aging management programs. Identified deficiencies due to age related degradation are evaluated under the Corrective Action Program.

This new aging management program will be implemented prior to the period of extended operation.

A.2.1.27 Lubricating Oil Analysis

The Lubricating Oil Analysis aging management program is an existing program that provides oil condition monitoring activities to manage the loss of material and the reduction of heat transfer in piping, piping components, piping elements, heat exchangers, and tanks within the scope of license renewal exposed to a lubricating oil environment. Sampling, analysis, and condition monitoring activities identify specific wear products and contamination and determine the physical properties of lubricating oil within operating machinery. These activities are used to verify that the wear product and contamination levels and the physical properties of the lubricating oil are maintained within acceptable limits to ensure that intended functions are maintained.

A.2.1.28 ASME Section XI, Subsection IWE

The ASME Section XI, Subsection IWE aging management program is an existing program based on ASME Code and complies with the provisions of 10 CFR 50.55a. The program consists of periodic inspection of the containment structure liner plate, including its integral attachments, penetration sleeves, pressure retaining bolting, personnel airlock and equipment hatches, moisture barrier, and other pressure retaining components for cracking, loss of material, loss of preload, and loss of sealing (of the moisture barrier). The moisture barrier is a sealant installed at the junction of the Containment concrete floor and the carbon steel Containment liner.

Examination methods include visual and volumetric testing as required by ASME Section XI, Subsection IWE. Observed conditions that have the potential for impacting an intended function are evaluated for acceptability in accordance with ASME requirements or corrected in accordance with corrective action process.

The ASME Section XI, Subsection IWE aging management program will be enhanced to include:

1. Inspection of a sample of the inaccessible liner covered by insulation and lagging prior to the period of extended operation and every 10 years thereafter. Should

unacceptable degradation be found, additional insulation will be removed as necessary to determine extent of condition in accordance with the corrective action process.

Prior to the period of extended operation

- The samples shall include 57 randomly selected containment liner insulation panels per unit.
- The randomly selected containment liner insulation panels will not include containment liner insulation panels previously removed to allow for inspection
- The examination will be performed by either removing the containment liner insulation panels and performing a visual inspection, or by using a pulsed eddy current (PEC) remote inspection, with the containment liner insulation left in place, to detect evidence of loss of material. If evidence of loss of material is detected using PEC, the containment liner insulation panel will be subsequently removed to allow for visual and UT examinations.
- All inspections will be completed by August 2016 for both Salem Units. Approximately one third of the 57 inspections will be completed during each refueling outage (Salem Unit 1 involves the following refuel outages: Spring 2013, Fall 2014, and Spring 2016. Salem Unit 2 involves the following refuel outages: Fall 2012, Spring 2014, and Fall 2015.). It is acceptable to perform greater than one third of the inspections in any refuel outage to accelerate the inspection schedule.

During the period of extended operation

- One containment liner insulation panel will be selected, at random, for removal from each quadrant, during each of the three Periods in an Inspection Interval. Therefore, a total of 12 containment liner insulation panels will be selected, in each unit, during each ten-year Inspection Interval, to allow for examination of the containment liner behind the containment liner insulation.
- The randomly selected containment liner insulation panels in each quadrant will not include containment liner insulation panels previously selected.

2. Visual inspection of 100 % of the moisture barrier, at the junction between the containment concrete floor and the containment liner, will be performed in accordance with ASME Section XI, Subsection IWE program requirements, to the extent practical within the limitation of design, geometry, and materials of construction of the components. The bottom edge of the stainless steel insulation lagging will be trimmed, if necessary, to perform the moisture barrier inspections. This inspection will be performed prior to the period of extended operation, and on a frequency consistent with IWE inspection requirements thereafter. Should unacceptable degradation be found, corrective actions, including extent of condition, will be addressed in accordance with the corrective action process.

As a follow up to inspections performed during the 2009 refueling outage, the following specific corrective actions will be performed on Unit 2 prior to entry into the period of extended operation:

- Examine the accessible 3/4" knuckle plate. If corrosion is observed to extend below the surface of the moisture barrier, excavate the moisture barrier to sound metal below the floor level and perform examinations as required by IWE.
- Perform remote visual inspections, of the six capped vertical leak chase channels, below the containment floor to determine extent of condition.
- Remove the concrete floor and expose 1/4" containment liner plate (floor) for a minimum of two of the vertical leak chase channels with holes. Perform examination of exposed 1/4" containment liner plate (floor) as required by IWE. Additional excavations will be performed, if necessary, depending upon conditions found at the first two channels.
- Remove 1/2" containment liner insulation panels, adjacent to accessible areas where there are indications of corrosion, to determine the extent of condition of the existing corroded areas of the containment liner plate.
- Perform augmented examinations of the area of the 1/2" containment liner plate behind insulation panels, where loss of material was previously identified, in accordance with IWE-2420.
- Examine 100% of the moisture barrier in accordance with IWE-2310 and replace or repair the moisture barrier to meet the acceptance standard in IWE-3510.

As a follow-up to inspections performed during the 2010 refueling outage, the following specific corrective actions will be performed on Unit 1 prior to entry into the period of extended operation:

- Perform augmented examinations of the 3/4" containment liner (knuckle plate) at 78’ elevation in accordance with IWE-2420.
- Perform augmented examinations of the areas of the 1/2" containment liner plate behind insulation panels, where loss of material was previously identified, in accordance with IWE-2420.
- Remove 1/2" containment liner insulation panels, adjacent to accessible areas where there are indications of corrosion, to determine the extent of condition of the existing corroded areas of the containment liner plate.

3. ASME Section XI, Subsection IWE program scope will be revised to include the following welds that are currently exempted from Subsection IWE and governed under ASME Section XI, Subsection IWB or IWC. The scope of the revision will include the cap plate to penetrating pipe pressure boundary welds, for penetrating pipe

constructed of stainless steel for those penetrations with a normal operating temperature greater than 140 degrees F.

4. Owner augmented inspections will be performed at the Salem Unit 1 and Unit 2 area of the Containment liner, under the fuel transfer canal and behind the Containment liner insulation, which are subjected to leaks from the reactor cavity. These owner augmented inspections will be performed on a frequency of once per Containment Inservice Inspection Period, starting with the current Period. These owner augmented inspections will continue, under the IWE program, as long as leakage from the reactor cavity or fuel transfer canal is observed between the Containment liner and the Containment liner insulation, including during the PEO.

These enhancements will be implemented prior to the period of extended operation, with the inspections performed in accordance with the schedule described above.

A.2.1.29 ASME Section XI, Subsection IWL

The ASME Section XI, Subsection IWL aging management program is an existing program based on ASME Code and complies with the provisions of 10 CFR 50.55a. The program requires periodic inspection of Containment Structure concrete surfaces to identify areas of deterioration and distress such as defined in ACI 201.1, including loss of material, cracks and distortion, and loss of bond.

Inspection methods, inspected parameters, and acceptance criteria are in accordance with ASME Section XI, Subsection IWL as approved by 10 CFR 50.55a. Observed conditions that have the potential for impacting an intended function are evaluated for acceptability in accordance with ASME Section XI, Subsection IWL requirements or corrected in accordance with the corrective action process.

The ASME Section XI, Subsection IWL, aging management program will be enhanced to include:

1. Examination and acceptance criteria in accordance with the guidance contained in ACI 349.3R.

A.2.1.30 ASME Section XI, Subsection IWF

The ASME Section XI, Subsection IWF aging management program is an existing program that consists of periodic visual examinations of ASME Class 1, 2, and 3 piping and component supports for identification of signs of degradation such as loss of material, loss of mechanical function and loss of pre-load. The inspections are in accordance with American Society of Mechanical Engineers (ASME), Boiler and Pressure Vessel Code, Section XI, Subsection IWF as approved in 10 CFR 50.55(a). The program activities are relied upon to detect and confirm that aging effects of ASME Class 1, 2, and 3 piping and component supports are adequately managed.

A.2.1.31 10 CFR 50, Appendix J

The 10 CFR 50, Appendix J aging management program is an existing program that monitors leakage rates through the containment pressure boundary, including penetrations, fittings and other access openings, in order to detect age related degradation of the containment pressure boundary. Corrective actions are taken if leakage rates exceed acceptance criteria. The Primary Containment Leakage Rate Testing Program (LRT) provides for aging management of pressure boundary degradation due to aging effects from cracking, loss of leakage tightness, loss of sealing, loss of material, or loss of preload in various systems penetrating containment. The 10 CFR 50 Appendix J program also detects age related degradation in material properties of gaskets, o-rings and packing materials for the containment pressure boundary access points. Consistent with the current licensing basis, the containment leakage rate tests are performed in accordance with the regulations and guidance provided in 10 CFR 50 Appendix J Option B, NEI 94-01 “Industry Guideline for Implementing Performance-Based Option of 10 CFR Part 50 Appendix J”, and ANSI/ANS 56.8, “Containment System Leakage Testing Requirements.”

A.2.1.32 Masonry Wall Program

The Masonry Wall Program is an existing program implemented as part of the Structures Monitoring Program. The Masonry wall condition monitoring is based on guidance provided in IE Bulletin 80-11, "Masonry Wall Design", and IN 87-67, "Lessons Learned from Regional Inspections of Licensee Actions in Response to IE Bulletin 80-11". The Masonry Wall aging management program addresses loss of material, and cracking due to age-related degradation of concrete for masonry walls during the period of extended operation. The program relies on periodic visual inspections to monitor and maintain the condition of masonry walls within the scope of license renewal.

The Masonry Wall Program will be enhanced to include:

1. Add buildings, and masonry walls that have been determined to be in the scope of License Renewal.

    a. Fire Pump House

    b. Masonry Wall Fire Barriers

    c. Office Buildings (The clean and controlled facilities buildings)

    d. SBO Yard Buildings

    e. Service Building

    f. Turbine Building

2. Add an Examination Checklist for masonry wall inspection requirements.

3. Specify an inspection frequency of not greater than 5 years for masonry walls.

These enhancements will be implemented prior to the period of extended operation.

A.2.1.33 Structures Monitoring Program

The Structures Monitoring Program is an existing program that was developed to implement the requirements of 10 CFR 50.65 and is based on NUMARC 93 01, “Industry Guideline for Monitoring the Effectiveness of Maintenance at Nuclear Power Plants,” and Regulatory Guide 1.160, “Monitoring the Effectiveness of Maintenance at Nuclear Power Plants,”. The program includes the Masonry Wall Program and the RG 1.127, Inspection of Water-Control Structures Associated With Nuclear Power Plants aging management program.

The program relies on periodic visual inspections to monitor the condition of structures and structural components, structural bolting, component supports, masonry block walls, and water control structures. The inspections are conducted on a frequency not greater than 5 years.

The Structures Monitoring Program will be enhanced to include:

1. The scope of the program will be enhanced to include the following structures and components:

    a. Fire Pump House

    b. Office Buildings (The clean and controlled facilities buildings)

    c. SBO Yard Buildings

    d. Service Building

    e. Switchyard

    f. Turbine Building

    g. Transmission towers

    h. Yard Structures (Foundations for fire water and demineralized water tanks, the plant vent radiation monitoring enclosures, the turbine crane runway extensions, and manholes)

    i. Building penetrations and pipe encapsulations that perform flood barrier, pressure boundary, shelter and protection intended functions

    j. Pipe whip restraints and jet impingement/spray shields

    k. Trench covers and sump liners

    l. Masonry walls, including Fire Barriers

m. Miscellaneous steel (catwalks, vents, louvers, platforms, etc.)

n. Vortex Suppressor, Ice Barrier, Marine Dock Bumper (Service Water Intake Structure)

o. Panels, Racks, Cabinets, and Other Enclosures

p. Metal-enclosed Bus

q. Components supports including, electrical cable trays, electrical conduit, tubing, HVAC ducts, instrument racks, battery racks, and supports for piping and components that are not within the scope of ASME Section XI, Subsection IWF

r. Duct banks that contain safety-related cables, and cables credited for SBO or ATWS

2. Concrete structures will be observed for a reduction in equipment anchor capacity due to local concrete degradation. This will be accomplished by visual inspection of concrete surfaces around anchors for cracking and spalling.

3. Clarify that inspections are performed for loss of material due to corrosion and pitting of additional steel components, such as embedments, panels and enclosures, doors, siding, metal deck, and anchors.

4. Require inspection of penetration seals, structural seals, and elastomers, for degradations that will lead to a loss of sealing by visual inspection of the seal for hardening, shrinkage and loss of strength.

5. Require the following actions related to the spent fuel pool liner:

   a. Perform periodic structural examination of the Fuel Handling Building per ACI 349.3R to ensure structural condition is in agreement with the analysis.

   b. Monitor telltale leakage and inspect the leak chase system to ensure no blockage.

   c. Test water drained from the telltales and seismic gap for boron, chloride, iron, and sulfate concentrations; and pH. Acceptance criteria will assess any degradation from the borated water. Sample readings outside the acceptance criteria will be entered into and evaluated in the corrective action program.

   d. Perform one shallow core sample in each of the Unit 1 Spent Fuel Pool walls (east and west) that have shown ingress of borated water through the concrete. The core samples will be examined for degradation from borated water. Also the core samples (east and west walls) will expose rebar, which will be examined for signs of corrosion. The core sample from the west wall will be taken by the end of 2013 and the core sample from the east wall will be taken by the end of 2015.

e. Perform a structural examination per ACI 349.3R every 18 months of the Unit 1 Spent Fuel Pool wall in the sump room where previous inspections have shown ingress of borated water through the concrete.

6. Require monitoring of vibration isolators, associated with component supports other than those covered by ASME XI, Subsection IWF.

7. Add an Examination Checklist for masonry wall inspection requirements.

8. Parameters monitored for wooden components will be enhanced to include: Change in Material Properties, Loss of Material due to Insect Damage and Moisture Damage.

9. Specify an inspection frequency of not greater than 5 years for structures including submerged portions of the service water intake structure.

10. Require individuals responsible for inspections and assessments for structures to have a B.S. Engineering degree and/or Professional Engineer license, and a minimum of four years experience working on building structures.

11. Perform periodic sampling, testing, and analysis of ground water chemistry for pH, chlorides, and sulfates on a frequency of 5 years. Groundwater samples in the areas adjacent to Unit 1 containment structure and Unit 1 auxiliary building will also be tested for boron concentration.

12. Require supplemental inspections of the affected in scope structures within 30 days following extreme environmental or natural phenomena (large floods, significant earthquakes, hurricanes, and tornadoes).

13. Perform a chemical analysis of ground or surface water in-leakage when there is significant in-leakage or there is reason to believe that the in-leakage may be damaging concrete elements or reinforcing steel.

14. Implementing procedures will be enhanced to include additional acceptance criteria details specified in ACI 349.3R-96.

15. When the reactor cavity is flooded up, Salem will periodically monitor the telltales associated with the reactor cavity and refueling canal for leakage. If telltale leakage is observed, then the pH of the leakage will be measured to ensure that concrete reinforcement steel is not experiencing a corrosive environment. In addition, Salem will periodically inspect the leak chase system associated with the reactor cavity and refueling canal to ensure the telltales are free of significant blockage. Salem will also inspect concrete surfaces for degradation where leakage has been observed, in accordance with this Program.

These enhancements will be implemented prior to entering the period of extended operation.

The following table is provided to tabulate the acceptance criteria from the Structures Monitoring Program Enhancement 5 c. associated with testing the water drained from the Salem Unit 1 SFP telltales and seismic gap drain.

Acceptance Criteria- Salem Unit 1 SFP Telltales and Seismic Gap Drain

Chemical Analysis	Acceptance Criteria		Frequency for monitoring
	SFP Telltales (West Wall)	Seismic Gap Drain (East Wall)	
pH	> 6.0 and < 9.0	> 6.5 and < 10.0	Samples taken monthly
Chloride	≤ 500 ppm	≤ 500 ppm	Samples taken every 6 months
Sulfate	≤ 1,500 ppm	≤ 1,500 ppm	Samples taken every 6 months
Boron	For Information Only	For Information Only	Samples taken monthly
Iron	For Information Only	For Information Only	Samples taken every 6 months

Chemistry results that do not meet one of the criteria will be entered into the corrective action program for an investigation and evaluation.

A.2.1.34 RG 1.127, Inspection of Water-Control Structures Associated With Nuclear Power Plants

The RG 1.127, Inspection of Water-Control Structures Associated With Nuclear Power Plants is implemented through the Structures Monitoring Program. The RG 1.127, Inspection of Water-Control Structures Associated with Nuclear Power Plants program is an existing program that will be enhanced to require inspection of water control structures and components that are in scope for license renewal. These structures include the Service Water Intake structure and Shoreline Protection and Dike structures (including outer walls of the Circulating Water Intake Structure). The RG 1.127, Inspection of Water-Control Structures Associated with Nuclear Power Plants aging management program addresses age-related deterioration, degradation due to extreme environmental conditions, and the effects of natural phenomena that may affect the safety function of the water control structures. The program manages loss of material, cracking, and change in material properties for concrete components, loss of material and loss of preload for steel and metal components, loss of material and change in material properties for wooden components, hardening and loss of strength for elastomers, and loss of material and loss of form for earthen water control structures. Elements of the program are designed to detect degradations and take corrective actions to prevent a loss of an intended function.

The RG 1.127, Inspection of Water-Control Structures Associated With Nuclear Power Plants Program will be enhanced to include:

1. Parameters monitored for wooden components will be enhanced to include change in material properties and loss of material due to insect damage and moisture damage.

2. Parameters monitored for elastomers will be enhanced to include hardening, shrinkage and loss of strength due to weathering and elastomer degradation.

3. The inspection requirement for submerged concrete structural components will be enhanced to require that inspections be performed by dewatering a pump bay or by a diver if the pump bay is not dewatered.

4. Specify an inspection frequency of not greater than 5 years for structures including submerged portions of the Service Water Intake Structure.

5. Require supplemental inspections of the in scope structures within 30 days following extreme environmental or natural phenomena (large floods, significant earthquakes, hurricanes, and tornadoes).

These enhancements will be implemented prior to the period of extended operation.

A.2.1.35 Protective Coating Monitoring and Maintenance Program

The Protective Coating Monitoring and Maintenance Program is an existing program that provides for aging management of Service Level I coatings inside the containment structure. Service Level I coatings are used in areas where coating failure could adversely affect the operation of post-accident fluid systems and thereby impair safe shutdown. The Protective Coating Monitoring and Maintenance Program provides for inspections, assessments, and repairs for any condition that adversely affects the ability of Service Level I coatings to function as intended.

A.2.1.36 Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements

The Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements aging management program is a new program that will be used to manage aging of non-EQ cables and connections during the period of extended operation. A representative sample of accessible cables and connections located in adverse localized environments will be visually inspected at least once every 10 years for indications of accelerated insulation aging such as embrittlement, discoloration, cracking, swelling, or surface contamination. An adverse localized environment is a condition in a limited plant area that is significantly more severe than the specified service environment for the cable or connection.

This new aging management program, including performance of initial inspections, will be implemented prior to the period of extended operation.

A.2.1.37 Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits

The Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits aging management program is a new program that will be implemented to manage the aging of the cable and connection insulation of the in scope portions of the Radiation Monitoring System and the Reactor Protection System (i.e., the nuclear instrumentation system). This program applies to sensitive instrumentation cable and connection circuits with low-level signals that are in scope for license renewal and are located in areas where the cables and connections could be exposed to adverse localized environments caused by heat, radiation, or moisture. These adverse localized environments can result in reduced insulation resistance causing increases in leakage currents.

Calibration results and findings of surveillance programs for the in-scope portions of the Radiation Monitoring System and the Reactor Protection System will be assessed for cable aging degradation prior to the period of extended operation and at least once every 10 years afterwards.

This new program will be implemented prior to the period of extended operation.

A.2.1.38 Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements

The Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements aging management program is a new program that will be used to manage the aging effects and mechanisms of non-EQ, in scope inaccessible medium voltage cables (4,160 volts and 13,800 volts). These cables may at times be exposed to significant moisture simultaneously with significant voltage. Significant moisture is defined as periodic exposures that last more than a few days (e.g., cable in standing water). Periodic exposures that last less than a few days (e.g., normal rain and drain) are not significant. Significant voltage exposure is defined as being subject to system voltage for more than twenty-five percent of the time. Note that no inaccessible medium voltage cable exposed to significant moisture was excluded from the program due to the “significant voltage” criterion. The Salem cables in the scope of this aging management program will be tested using a proven test for detecting deterioration of the insulation system due to wetting that is state-of-the-art at the time the test is performed. The cable test frequency will be established based on test results and industry operating experience. The maximum time between tests will be no longer than 6 years. The first tests will be completed prior to the period of the extended operation.

Prior to the period of extended operation, manholes and cable vaults associated with the cables included in this aging management program will be inspected for water collection (with water removal as necessary). The objective of the inspections, as a preventive action, is to minimize the exposure of medium voltage cables to significant moisture. The frequency of inspections for accumulated water will be established based on inspection results. This also recognizes that a recurring inspection, set at the optimum frequency, would result in the cables being submerged only as a result of event driven, rain and drain, type occurrences. Station procedures will direct the assessment of the cable condition as a result of rain or other event driven occurrences. As a

limit on the amount of time between inspections, the maximum time between inspections will be no more than 1 year.

The Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements aging management program will be enhanced as follows:

1. Change cable testing maximum frequency from 10 years to 6 years. Change manhole and cable vault inspection maximum frequency from 2 years to 1 year.

This new program, including the enhancement, will be implemented prior to the period of extended operation. In addition, initial cable tests will be implemented prior to the period of extended operation and sufficient manhole and cable vault inspections will be performed prior to the period of extended operation so that proper inspection frequencies are established minimize the exposure of medium voltage cables to significant moisture during the period of extended operation.

A.2.1.39 Metal Enclosed Bus

The Metal Enclosed Bus aging management program is a new condition monitoring program that will manage the aging of in-scope metal enclosed busses at Salem.

The internal portions of the in-scope metal enclosed bus enclosures will be visually inspected for cracks, corrosion, foreign debris, excessive dust build-up and evidence of moisture intrusion. The bus insulation will be visually inspected for signs of embrittlement, cracking, melting, swelling, or discoloration, which may indicate overheating or aging degradation. The internal bus supports will be visually inspected for structural integrity and signs of cracks. Bolted connections are not accessible, but will be checked (sampled) for loose connection using thermography from outside the metal enclosed bus.

Metal enclosed busses are to be free from unacceptable visual indications of surface anomalies, which suggest that conductor insulation degradation exists. In addition no unacceptable indication of corrosion, cracks, foreign debris, excessive dust buildup or evidence of moisture intrusion is to exist. An unacceptable indication is defined as a noted condition or situation that, if left unmanaged, could lead to a loss of intended function. Thermography results will be confirmed to be within the acceptance criteria of administrative program procedures.

This new aging management program will be implemented prior to the period of extended operation. In addition, the first inspections will be completed prior to the period of extended operation and every 10 years thereafter.

A.2.1.40 Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements

The Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements program is a new program that will be used to confirm the absence of an aging effect with respect to electrical cable connection stressors. A representative sample of non-EQ electrical cable connections will be selected for one-time testing considering application

(medium and low voltage), circuit loading (high loading) and location, with respect to connection stressors. The technical basis for the sample selected will be documented. The specific type of test performed will be a proven test for detecting loose connections, such as thermography or contact resistance measurement, as appropriate to the application.

This new aging management program will be implemented and the one-time tests will be completed prior to the period of extended operation.

A.2.2 Plant-Specific Aging Management Programs

This section provides summaries of the plant-specific programs credited for managing the effects of aging.

A.2.2.1 High Voltage Insulators

The High Voltage Insulators program is a new program that manages the degradation of insulator quality due to the presence of salt deposits or surface contamination. This aging effect will be identified through visual inspections of the external surfaces of the high voltage insulators. The visual inspections will be performed on a twice per year frequency.

This new aging management program will be implemented prior to the period of extended operation.

A.2.2.2 Periodic Inspection

The Periodic Inspection aging management program is a new condition-monitoring program that manages the aging of piping, piping components, piping elements, ducting components, tanks and heat exchanger components. This program will manage the aging effects of loss of material, cracking, reduction of heat transfer and hardening and loss of strength. The program includes provisions for visual inspections of stainless steel, aluminum, copper alloy and elastomer components not managed under other aging management programs. The program also includes provisions for ultrasonic wall thickness measurements to detect loss of material. Identified deficiencies due to age related degradation are evaluated under the corrective action program.

This new aging management program will be implemented prior to the period of extended operation.

A.2.2.3 Aboveground Non-Steel Tanks

The Aboveground Non-Steel Tanks aging management program is a new program that will manage loss of material of outdoor non-steel tanks in the scope of license renewal. Periodic visual inspections will monitor for degradation of the non-steel tanks external surfaces. Periodic visual inspections will also monitor for degradation of the seal at the interface between the tank bottom and the concrete foundation. Tanks within the scope of this program are the Auxiliary Feedwater Storage Tanks, Primary Water Storage Tanks, Refueling Water Storage Tanks and Demineralized Water Storage Tanks.

The Aboveground Non-Steel Tanks program will include a UT wall thickness inspection of the bottom of the tanks. The UT measurements will be taken to ensure that significant degradation is not occurring and that the component intended function will be maintained during the extended period of operation.

This new program, including tank bottom UT inspections, will be implemented prior to the period of extended operation.

A.2.2.4 Buried Non-Steel Piping Inspection

The Salem Buried Non-Steel Piping Inspection program is an existing condition monitoring program that manages buried reinforced concrete piping and components in the Service Water System and Circulating Water System that are exposed to an external soil or groundwater environment for cracking, loss of bond, increase in porosity and permeability, and loss of material. The Salem Buried Non-Steel Piping Inspection aging management program also inspects the buried stainless steel penetration bellows between the Containment Structure and the Fuel Handling Building, including the penetration sleeves, for loss of material. These aging effects will be identified through visual inspections of the external surfaces of the buried piping and components.

The Buried Non-Steel Piping Inspection aging management program will be enhanced to include:

1. At least one (1) opportunistic or focused excavation and inspection will be performed on buried reinforced concrete piping and components during each ten (10) year period beginning ten (10) years prior to entry into the period of extended operation.

2. At least one (1) opportunistic or focused excavation and inspection will be performed on buried stainless steel penetration bellows between the Containment Structure and the Fuel Handling Building, including the penetration sleeves, during each ten (10) year period beginning ten years prior to entry into the period of extended operation.

3. Guidance for inspection of concrete aging effects.

These enhancements will be implemented prior to the period of extended operation, with the inspections performed in accordance with the schedule described above.

A.2.2.5 Boral Monitoring Program

The Boral Monitoring Program is an existing program that manages the aging effects of the Boral neutron-absorbing material used in the Exxon and Holtec spent fuel storage rack assemblies in the units 1 and 2 spent fuel pools at Salem. The aging affects that need managing for the Boral neutron-absorbing material during the period of extended operation are reduction of neutron-absorbing capacity and loss of material.

The Boral Monitoring Program performs inspections and/testing on Boral test specimens or coupons. The program monitors changes in physical properties of the Boral by performing

measurements on representative Boral test coupons. The Boral test coupons simulate as nearly as possible the actual in-service geometry, physical mounting, materials, and flow conditions of the Boral panels in the spent fuel storage rack assemblies. Monitoring of the Boral neutron-absorbing material is accomplished through periodic examination of the Boral test coupons, consisting of visual observations (which may include photography), and may consist of dimensional measurements (length, width, and thickness), weight and density determinations, and neutron attenuation measurements (for B-10 areal density). The results are evaluated against acceptance criteria for determination of any follow-up activities as appropriate (e.g., removal and examination of additional Boral test coupons, wet chemical analyses, radiography, etc.).

The Boral Monitoring Program will be enhanced to include:

1. The program will be enhanced to perform a neutron attenuation measurement on one each of the three (no vent holes, one vent holes and two vent holes) flat plate sandwich Boral test coupons during the first three two-year inspection frequency periods and every six years thereafter for the Exxon spent fuel storage rack assemblies.

2. The program will be enhanced to include acceptance criteria of the neutron attenuation measurement on the Boral test coupons for the Exxon spent fuel storage rack assemblies: ·A decrease of no more than 5% in Boron-10 content as determined by neutron attenuation measurements. The benchmark Boron-10 content used for comparison will be based on the nominal B-10 areal density in the design basis specification.

These enhancements will be implemented prior to the period of extended operation.

A.2.2.6 Nickel Alloy Aging Management

The Salem Nickel Alloy Aging Management program is an existing program that manages cracking for nickel alloy components in the reactor vessel and steam generators. The Nickel Alloy Aging Management program implements mitigative and condition monitoring activities. Mitigative actions include replacement of components whose materials are susceptible to cracking with materials with improved susceptibility to cracking, and Mechanical Stress Improvement Process on the reactor vessel primary nozzle to safe end welds. The condition monitoring portion of the program uses a number of inspection techniques to detect cracking due, including surface examinations, volumetric examinations and bare metal visual examinations. The Nickel Alloy Aging Management program implements the inspection of components through an augmented Inservice Inspection program. The augmented program administers component evaluations, examination methods, scheduling, and site documentation as required to comply with regulatory, code, and industry commitments related to nickel alloy issues. The Nickel Alloy Aging Management program implements applicable NRC Bulletins, Generic Letters and staff-accepted industry guidelines.

A.3 NUREG-1801 Chapter X Aging Management Programs

A.3.1 Evaluation of Chapter X Aging Management Programs

Aging Management Programs evaluated in Chapter X of NUREG-1801 are associated with Time-Limited Aging Analyses for metal fatigue of the reactor coolant pressure boundary and environmental qualification (EQ) of electrical components. These programs are evaluated in this section.

A.3.1.1 Metal Fatigue of Reactor Coolant Pressure Boundary

The Metal Fatigue of Reactor Pressure Boundary program is an existing program that manages cumulative fatigue usage for the reactor vessel, the pressurizer, the steam generators, Class 1 and non-Class 1 piping, and Class 1 components subject to the reactor coolant, treated borated water, and treated water environments.

The Metal Fatigue of Reactor Pressure Boundary program is a preventive program that monitors and tracks the number of critical thermal and pressure transients to ensure that the cumulative usage factors for selected reactor coolant system (RCS) components remain less than 1.00 through the period of extended operation. The program determines the number of transients that occur and updates the 60-year projections as required on an annual basis. A software program, WESTEMS, computes cumulative usage factors for select locations.

The effect of the reactor coolant environment on fatigue usage, known as environmental fatigue, has been evaluated for the period of extended operation using the formulae contained in NUREG/CR-6583 for carbon and low-alloy steels and NUREG/CR-5704 for austenitic stainless steels. The fatigue usage associated with the effects of the reactor coolant environment will be included into the ongoing monitoring program.

The program requires the generation of a periodic fatigue monitoring report, including a listing of transient events, cycle summary event details, cumulative usage factors, a detailed fatigue analysis report, and a cycle projection report. If the fatigue usage for any location has had an unanticipated increase based on cycle accumulation trends or if the number of cycles is approaching their limit, the corrective action program is used to evaluate the condition and determine the corrective action. Acceptable corrective actions include repair of the component, replacement of the component, and a more rigorous analysis of the component to demonstrate that the design code limit will not be exceeded during the period of extended operation. Corrective actions include a review of additional affected reactor coolant pressure boundary locations.

There are several enhancements identified for this existing program as follows.

1. The Metal Fatigue of Reactor Coolant Pressure Boundary program will be enhanced to include additional transients beyond those defined in the Technical Specifications and the UFSAR, and expanding the fatigue monitoring program to encompass other components identified to have fatigue as an analyzed aging effect, which require monitoring.

2. The Metal Fatigue of Reactor Coolant Pressure Boundary program will be enhanced to use a software program to automatically count transients and calculate cumulative usage on select components.

3. The Metal Fatigue of Reactor Coolant Pressure Boundary program will be enhanced to address the effects of the reactor coolant environment on component fatigue life by assessing the impact of the reactor coolant environment on a sample of critical components for the plant identified in NUREG/CR-6260.

4. The Metal Fatigue of Reactor Coolant Pressure Boundary program will be enhanced to require a review of additional reactor coolant pressure boundary locations if the usage factor for one of the environmental fatigue sample locations approaches its design limit.

These enhancements will be implemented prior to the period of extended operation.

A.3.1.2 Environmental Qualification (EQ) of Electric Components

The Environmental Qualification (EQ) of Electric Components is an existing program that manages the aging of electrical equipment within the scope of 10 CFR 50.49, “Environmental Qualification of Electric Equipment Important to Safety for Nuclear Power Plants.” The program establishes, demonstrates, and documents the level of qualification, qualified configurations, maintenance, surveillance and replacements necessary to meet 10 CFR 50.49. A qualified life is determined for components within the scope of the program and appropriate actions such as replacement or refurbishment, or reanalysis, are taken prior to or at the end of the qualified life of the components so that the aging limit is not exceeded. The aging effects are adequately managed so that the intended functions of components within the scope of 10 CFR 50.49 are maintained consistent with the current licensing basis during the period of extended operation.

A.4 Time-Limited Aging Analyses

A.4.1 Introduction

As part of the application for a renewed license, 10 CFR 54.21(c) requires that an evaluation of Time-Limited Aging Analyses (TLAAs) for the period of extended operation be provided. The following TLAAs have been identified and evaluated to meet this requirement.

A.4.2 Reactor Vessel Neutron Embrittlement

The reactor vessel embrittlement calculations for Salem that evaluated reduction of fracture toughness of the Salem reactor vessel beltline materials for 40 years are based upon a predicted End of License fluence of 32 Effective Full Power Years (EFPY). These analyses are considered Time-Limited Aging Analyses (TLAAs) as defined in 10 CFR 54.21(c) and they must be evaluated for the increased neutron fluence associated with 60 years of operation (50 EFPY).

A.4.2.1 Neutron Fluence Analyses

The current reactor vessel embrittlement calculations that evaluated reduction of fracture toughness of the Salem reactor vessel beltline materials for 40 years are based on predicted 40-year EOL fluence values (32 EFPY). These analyses also incorporate the effect of 1.4% power uprate. Therefore, they are Time-Limited Aging Analyses as defined by 10 CFR 54.21(c) and must be evaluated for the increased neutron fluence associated with 60 years of operation.

A value of 50 EFPY for both Salem units was selected to provide conservatism in the fluence projections.

The reactor vessel beltline neutron fluence values applicable to a postulated 20-year license renewal period were calculated for each reactor pressure vessel beltline materials on Salem. The analysis methods used to calculate the Salem vessel fluences satisfy the requirements set forth in Regulator Guide 1.190, “Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence”.

The analyses are projected for the period of extended operation in accordance with 10 CFR54.21(c)(1)(ii).

A.4.2.2 Upper Shelf Energy Analyses

The current Charpy Upper Shelf Energy (USE) calculations were prepared for each reactor vessel beltline material for Salem based upon projected neutron fluence values for 40 years of service (32 EFPY). These are TLAAs requiring evaluation using 60-year fluence values (50 EFPY).

Appendix G of 10 CFR 50 contains screening criteria that establish limits on how far the USE value for a reactor pressure vessel material may be allowed to decrease due to neutron irradiation exposure. The regulations requires the initial USE value to be greater than 75 ft-lbs in the non-irradiated condition and that the value be greater than 50 ft-lbs in the fully irradiated conditions as determined by Charpy V-notch specimen testing throughout the licensed life of the plant. USE values of less than 50 ft-lbs may be acceptable to the NRC if it can be demonstrated that these lower values will provide margins of safety against brittle fracture equivalent to those required by ASME Section XI, Appendix G.

Per Regulatory Guide 1.99, Revision 2, the Charpy USE should be assumed to decrease as a function of fluence according to Figure 2 of the Regulatory Guide when surveillance data is not used. If surveillance data is used, the decrease in USE may be obtained by plotting the reduced plant surveillance data on Figure 2 of the Regulatory Guide and fitting the data with a line drawn parallel to the existing lines as the upper bound of all of the data. Charpy USE for the beltline forgings and welds were determined using surveillance data (Position 2.2 of the Regulatory Guide), and the Charpy USE for the extended beltline materials was determined without the use of surveillance data (Position 1.2 of the Regulatory Guide).

Predictions of the Charpy USE for EOL (50 EFPY) for Salem have been made using the corresponding 1/4T fluence projection, the copper and nickel content of the beltline materials,

and the results of the capsule specimens tested to date, where applicable, using Figure 2 in Regulatory Guide 1.99.

The USE values for the beltline and extended beltline materials are projected to remain above the 50 ft-lb requirement through the period of extended operation.

The analyses are projected for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

A.4.2.3 Pressurized Thermal Shock Analyses

10 CFR 50.61(b)(1) provides rules for the protection of pressurized water reactors against pressurized thermal shock. Licensees are required to assess the projected values of nil ductility reference temperature whenever a significant change occurs in the projected values of Reference Temperature Pressurized Thermal Shock (RT_{PTS}), or upon request for a change in the expiration date for the facility operating license. For the current 40-year period, the RT_{PTS} was analyzed for 32 EFPY, which is considered a TLAA.

Reactor vessel beltline fluence is one of the factors used in determining the margin of acceptability of the reactor vessel to pressurized thermal shock as a result of neutron embrittlement. The margin is the difference between the maximum nil ductility reference temperature in the limiting beltline material and the screening criteria established in accordance with 10 CFR 50.61(b)(2). The screening criteria for the limiting reactor vessel materials are 270°F for beltline plates, forgings, and axial weld materials, and 300°F for beltline circumferential weld materials.

The limiting RT_{PTS} value for the Salem Unit 1 axially oriented welds and plates is 258°F, which corresponds to the Lower Shell Longitudinal Weld Seam 3-042C. The limiting RT_{PTS} value for the Salem Unit 1 circumferentially oriented welds and plates is 229°F, which corresponds to the Intermediate-to-Lower Shell Circumferential Weld Seam 9-042.

The limiting RT_{PTS} value for the Salem Unit 2 axially oriented welds and plates is 239°F which corresponds to the Lower Shell Longitudinal Weld Seams 3-442 A&C. The limiting RT_{PTS} value for the Salem Unit 2 circumferentially oriented welds is 118°F, which corresponds to the Intermediate Shell-to-Lower Shell Circumferential Weld Seam 9-442.

Therefore, all of the Salem Units 1 and 2 reactor vessel materials that exceed a surface fluence of 1.0E+17 n/cm2 (E > 1.0 MeV) at 50 EFPY are below the RT_{PTS} screening criteria values of 270°F, for axially oriented welds and plates/forgings, and 300°F, for circumferentially oriented welds, at 50 EFPY.

The analyses are projected for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

A.4.2.4 Reactor Vessel Pressure–Temperature Limits, Including Low Temperature Overpressure Protection Limits

Appendix G of 10 CFR 50 requires that the reactor pressure vessel be maintained within established pressure-temperature (P-T) limits, including heatup and cooldown operations. These limits specify the maximum allowable pressure as a function of reactor coolant temperature. As the reactor pressure vessel is exposed to increased neutron irradiation, its fracture toughness is reduced. The P-T limits must account for the anticipated reactor vessel fluence.

The current Low temperature Overpressure Protection (LTOP) setpoint for Salem Units 1 and 2 is 375 psig, and is effective through 32 EFPY for both Units. These calculations are associated with the generation of the P-T limit curves that satisfy the criteria of 10 CFR 54.3(a) and are, therefore, TLAAs.

Updated P-T limits were calculated using fluence values valid for 50 EFPY for Salem reactor vessel beltline region, inlet and outlet nozzles, and closure head flange locations for normal heatup, normal cooldown, and in-service leak and hydrostatic test conditions. In addition, minimum bolt up temperatures, minimum temperature of core criticality, and LTOP system limits were determined. These P-T limits are expressed in the form of a curve of allowable pressure versus temperature.

Salem Units 1 and 2 will submit updates to the P-T and LTOP limits to the NRC at the appropriate time to comply with 10 CFR 50 Appendix G.

The P-T and LTOPs limit analyses will be managed through the period of extended operation in accordance with 10 CFR 54.21(c)(1)(iii).

A.4.3 Metal Fatigue of Piping and Components

Metal fatigue was evaluated in the design process for Salem pressure boundary components, including the reactor vessel, reactor coolant pumps, steam generators, pressurizer, piping, valves, and components of primary, secondary, auxiliary, steam, and other systems. The current design analyses for these components have been determined to be Time-Limited Aging Analyses (TLAAs) requiring evaluation for the period of extended operation. Fatigue TLAAs for Salem pressure boundary components are characterized by determining the applicable design code and design specifications that specify the fatigue design requirements.

NUREG-1801 provides a listing of components that are likely to have TLAAs in place that require evaluation for License Renewal. Each of these has been reviewed and the applicable TLAAs are evaluated in the following sections, as appropriate.

In addition, for license renewal, fatigue calculations have been prepared to evaluate the effects of the reactor water environment on the sample of high-fatigue locations applicable to Older-Vintage Westinghouse Plants, identified in Section 5.5 of NUREG/CR-6260. Since several of these components are located within systems currently analyzed to ASA/USAS B31.1 rules, new explicit analyses were prepared in accordance with ASME Section III, Class 1 rules for

each of these components. For these locations environmental fatigue correction factors were computed and applied to the CUF values developed in the Class 1 fatigue analyses.

A.4.3.1 Nuclear Steam Supply System (NSSS) Pressure Vessel and Component Fatigue Analyses

Nuclear Steam Supply System (NSSS) pressure vessels and components for Salem were designed in accordance with ASME Section III, Class A or Class 1 requirements, and were required to have explicit analyses of cumulative fatigue usage.

ASME Section III, Class A and Class 1 fatigue analyses are based upon explicit numbers and amplitudes of thermal and pressure transients described in the design specifications. The intent of the design basis transient definitions is to bound not just specific operations but a wide range of possible events with varying ranges of severity in temperature, pressure, and flow. The most limiting numbers of transients used in these NSSS component analyses are considered design limits. Those that are significant contributors to fatigue usage are monitored to assure the limits are not exceeded.

Each Salem Unit 1 and 2 component designed in accordance with ASME Section III, Class A and Class 1 rules was analyzed and shown to have a CUF less than the design limit of 1.0. Since each of the Class A and Class 1 fatigue analyses described above are based upon a number of cycles postulated to bound 40 years of service, they have been identified as TLAAs that require evaluation for 60 years.

The evaluation method used to determine the adequacy of the existing design transients for 60 years operations for the Salem license renewal was as follows:

- An investigation of the actual number of cumulative cycles of each transient type was performed, along with predictions of future cycles, and the results were compared to the original design number of cycles for each transient type to demonstrate that the original design cycles were bounding.

- An investigation of the actual severity of the plant transients in comparison with the severity of the equivalent design basis transients was performed to demonstrate the original design transient severities are bounding.

- The administrative and operating procedures were reviewed in order to assess the effectiveness of the design transient cycle counting program and to validate the cyclic assumptions.

The overall conclusion of these investigations is that the existing design transients are sufficiently conservative for encompassing 60 years of plant operations.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.3.2 Pressurizer Safety Valve and Pilot-Operated Relief Valve Fatigue Analyses

This section is deleted.

A.4.3.3 ASA / USAS B31.1 Piping Fatigue Analyses

Piping designed in accordance with USAS B31.1 Piping Code is not required to have an analysis of cumulative fatigue usage, but cyclic loading is considered in a simplified manner in the design process. When the Salem B31.1 components were designed, the overall number of thermal and pressure cycles expected during the 40-year lifetime of these components was determined. The total number of cycles was compared to cycle ranges specified in USAS B31.1 for consideration of allowable stress reduction. If the total number of cycles exceeded 7,000 cycles, a stress range reduction factor had to be applied to the allowable stress range for secondary stresses (expansion and displacement) to account for thermal cycling. This is considered to be an implicit fatigue analysis since it is based upon a total number of cycles projected to occur in 40 years, but does not have an explicit Cumulative Usage Factor (CUF) value associated with it. Since the overall number of cycles could potentially increase during the period of extended operation, which could potentially result in further reduction of the allowable stress, these implicit fatigue analyses are also considered to be TLAAs requiring evaluation for the period of extended operation.

Note that after originally being designed in accordance with USAS B31.1 requirements, portions of some components of these systems were reevaluated to ASME Section III, Class 1 requirements. The remaining components within those systems that were not reanalyzed are included within the evaluation provided within this section.

In order to evaluate these TLAAs for 60 years, the numbers of cycles expected to occur within the 60-year operational period should be compared to the numbers of cycles that were originally considered in the design of these components. If this number does not exceed 7,000 cycles, the minimum number of cycles required that would result in application of an allowable stress reduction factor, then there is no impact from the added years of service and the original analyses remain valid. If the total number of cycles exceeds 7,000 cycles, then additional evaluation is required.

The 60-year transient projections show that even if all of the projected operational transients are added together, the total number of cycles projected for 60 years will not exceed 7,000 cycles. Therefore, there is no impact upon the implicit fatigue analyses used in the component design for any system that is only affected by operational transients.

The analyses are valid for the period of extended operation in accordance with 10CFR 54.21(c)(1)(i).

A.4.3.4 Supplementary ASME Section III, Class 1 Piping and Component Fatigue Analyses

Each of the Salem piping systems, including the Reactor Coolant System main loop piping, were originally designed in accordance with ASA B31.1-1955 design requirements. Piping systems for Unit 2 were designed in accordance with USAS B31.1.0-1967 requirements. Since

then, a number of updated fatigue analyses have been prepared for piping systems and components to address transients that have been identified in the industry that were not originally considered. These analyses have been performed in accordance with ASME Section III, Class 1, rules to enable these transients to be thoroughly evaluated.

Each of these analyses resulted in a Cumulative Usage Factor (CUF) value less than 1.0 based upon a number of transients postulated to bound 40 years of plant operations. Therefore, each of these analyses has been identified as a TLAA requiring evaluation for 60 years.

These analyses are separated from those evaluated in the previous sections because the transient definitions have been modified, or additional transients have been postulated for these components, in addition to those previously described. Therefore, the cycle projections for these components must address these revised transients or additional transient types to determine if they also remain bounded for 60 years of service. Each of these analyses is dispositioned separately within this section for clarity.

A.4.3.4.1 NRC Bulletin 88-08, Thermal Stresses in Piping Connected to Reactor Coolant Systems

NRC Bulletin 88-08 was issued June 22, 1988 with supplements in 1988 and 1989 because of observed pipe cracking due to valve leakage in unisolable lines. The Bulletin required that licensees identify potential locations that might be subject to high stresses due to leaking valves, inspect the potential locations, and to assure that susceptible locations will not fail for the remaining life of the unit.

The NRC Safety Evaluation Report (SER) approved Salem's response to NRC Bulletin 88-08, which included the evaluation of the fatigue analyses of the Normal and Alternate Charging Lines and the Auxiliary Spray Lines. The analyses were based on the requirements of ASME Section III, 1986 edition, Subsection NB-3653 and the fatigue curves of I-9.2.1 and I-9.2.2 and concluded that the cumulative usage factor would remain less than 1.0 for the Normal and Alternate Charging Lines. The Auxiliary Spray Line results for the same transients would remain less than 1.0 for twenty-four (24) calendar years since initial plant start-up. A follow-up analysis was performed and concluded that the fatigue usage for the Auxiliary Spray Line was calculated to be less than 1.0 for a life of 40 calendar years. These analyses are considered TLAAs for evaluation through the period of extended operation.

<u>Normal and Alternate Charging Lines</u>

The fatigue evaluation of the charging lines to address potential thermal cycling transients included typical charging line transients from similar Westinghouse plants designed to ASME Section III, plus additional transients assumed to be induced by valve leakage over a 40 year period, based on operation of either charging line for 60% of the 40-year period, or an equivalent 24-year total leakage period. For the analyses to remain valid for the 60-year period, the cycles of the design transients and those due to leakage must be shown to be less than analyzed. For the design cycles, the fatigue monitoring program can be used to show the charging transient cycles in either the normal or alternate charging line to be less than those analyzed.

A re-evaluation has been performed for the charging nozzles to account for the 60-year projected charging transient cycles, and the resulting fatigue usage is less than 1.0. Although the charging nozzles are not subject to thermal stratification and cycling transients, the results of the re-evaluation demonstrate the relative impact of the 60-year projected charging design transients, and the ability of the design to accommodate these cycles.

The Metal Fatigue of Reactor Coolant Pressure Boundary program will manage the effects of aging due to fatigue in accordance with 10 CFR 54.21(c)(1)(iii).

Auxiliary Spray Line

The Pressurizer Auxiliary Spray line was reanalyzed to ASME Section III, Class 1 design rules in order to evaluate postulated thermal events described in Generic Letter 88-08. The potential thermal events would result from cold water leaking past a closed valve seat into the hot Pressurizer Auxiliary Spray line, leading to thermal cycling along the bottom of the pipe. Subsequent valve maintenance and monitoring has minimized the likelihood of this type of thermal cycling, but the analysis remains in effect, and was identified as a TLAA requiring evaluation for 60 years.

The 1999 analysis for 40 years of operation concluded that the cumulative usage factor was less than 1.0 for the limiting location of the Auxiliary Spray Line. The basis was five (5) Inadvertent Auxiliary Spray transients. The analysis also stated that the fatigue usage for thermal cycling and striping due to valve leakage was negligible. The 60-yr projected Inadvertent Auxiliary Spray to Pressurizer events are 2 and 3, respectively for Salem Units 1 and 2.

Therefore, the number of projected transients is bounded by the basis for the ASME III fatigue analysis.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.3.4.2 NRC Bulletin 88-11, Pressurizer Surge Line Thermal Stratification

NRC Bulletin 88-11, issued in December 1988, requested utilities to establish and implement a program to confirm the integrity of the pressurizer surge line. The program required both visual inspection of the surge line and demonstration that the design requirements of the surge line are satisfied, including the consideration of stratification effects.

The Pressurizer Surge Line piping and nozzles were previously evaluated for the effects of thermal stratification and plant-specific transients. The controlling fatigue location was the surge line weld to the pressurizer surge nozzle. In a later evaluation, a plant-specific WESTEMS™ model was developed for the pressurizer and surge line to evaluate the effects of pressurizer insurge/outsurge transients and surge line stratification on the pressurizer surge nozzle safe end to pipe weld and the surge line hot leg nozzle, which is the analysis of record and needs to be evaluated as a TLAA for 60 years.

The 60-year analyses demonstrated compliance with design requirements considering ASME Code requirements and utilized the design set of NSSS transients. The Pressurizer Surge Line stratification sub-transients were developed based on plant operating procedures, surge line monitoring data from similar units and historical records for each Salem Unit, and projected 60-year cycles of surge line stratification and insurge/outsurge transients where these were greater than previously evaluated design cycles. These evaluations resulted in CUF less than 1.0 at the pressurizer surge nozzle safe end to pipe weld and at the surge line hot leg nozzle.

The analyses are projected for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(ii).

A.4.3.4.3 Salem Unit 1 Steam Generator Feedwater Nozzle Transition Piece

As part of the Salem Unit 1 Steam Generators replacement, a new feedwater nozzle transition piece forging was designed as an ASME III, Class 1 component. The specific requirements were ASME III, Subsection NB, 1989. Additionally, the requirements for cyclic operation of Article NB-3200 were incorporated into the design of the transition piece. The remainder of the feedwater piping was designed in accordance with ANSI B31.1.0.

All transients considered in the original design of the feedwater nozzles remain the same, except that the hot standby case was replaced with thermal stratification loadings. This portion of the design assumed 800 hours of auxiliary feedwater flow per cycle, over the course of the fifteen (15) remaining cycles (22 years of balance of life of the plant), resulting in 12,000 hours of auxiliary feedwater operation. This analysis is considered a TLAA for evaluation through the period of extended operation.

The cumulative usage for the Unit 1 Steam Generator Feedwater Nozzle Transition Piece, is based on design basis transients and thermal stratification loads. The cumulative usage will be managed by the Metal Fatigue of Reactor Coolant Pressure Boundary aging management program, where the program will monitor cumulative usage and prevent exceeding its design limit of 1.00.

The Metal Fatigue of Reactor Coolant Pressure Boundary program will manage the effects of aging due to fatigue in accordance with 10 CFR 54.21(c)(1)(iii).

A.4.3.4.4 Salem Unit 1 Steam Generator Primary Manway Studs

The Salem Unit 1 Steam Generator primary manway studs were originally planned for replacement every five (5) years. However, Westinghouse conducted a series of tests to qualify their life for forty (40) years. The analysis conducted for Salem Unit 1 compared the studs installed in the Model F Salem to those studs qualified for extended fatigue life at another Model F plant. The qualification tests were performed in accordance with ASME Code, Section III, Appendix II. This analysis is considered a TLAA for evaluation through the period of extended operation.

Transients were inputs to the fatigue analyses of the studs. The 60-year projected cycles applicable to the Salem Unit 1 Model F Steam Generator primary manway studs are bounded by the cycles in the Westinghouse fatigue analysis.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.3.5 Reactor Vessel Internals Fatigue Analyses

The Salem Reactor Vessel Internals were designed and constructed prior to the development of ASME Code requirements for core support structures, but the reactor coolant system functional design requirements were considered in the design. The Reactor Vessel Internals were implicitly designed for low cycle fatigue based upon the reactor coolant system design transient projections for 40 years, which has been identified as a TLAA.

Post-design analyses consist of two Westinghouse calculation notes; (1) a lower core plate evaluation based on the 1.4% uprate and (2) qualification of the Salem domed lower core support plate, also part of the 1.4% uprate project at Salem.

The two post-design Westinghouse calculation notes are evaluations that analyzed the 1.4% uprate in terms of the effect of changes in significant thermal design transients, which were deemed negligible. Therefore, the cumulative fatigue usage attributable to the significant thermal design transients did not change.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.3.6 Spent Fuel Pool Bottom Plates Fatigue Analyses

Salem responded to the NRC request for additional information (RAI) dated 2/26/96 where the NRC requested an analysis to show that the spent fuel pool (SFP) liner and anchors will not experience significant deformations as a result of thermal loadings. In their RAI, the NRC provided acceptance criteria contained in Tables CC-3720-1 and CC-3730-1 of ASME Section III, Div. 2, 1995. Using a conservative fatigue strength reduction factor of five (5), the Salem analysis determined the maximum alternating stresses and compared them to the austenitic stainless steel curves in the ASME code. The resulting allowable cycles for the bottom liner plates were 1,638 cycles.

A separate analysis evaluated the liner bottom plate for the fuel rack pedestal loads under upset conditions, specifically operating basis earthquake (OBE) loadings. The cumulative usage factor was determined to be 0.00063 for one (1) design basis earthquake (DBE) and twenty (20) OBEs.

The events that would cause full temperature thermal cycles in the SFP are refueling outages which can conservatively be correlated to plant heatups and cooldowns. Since the number of cycles projected to occur in 60 years is well below 1,638 cycles analyzed for the SFP bottom liner, this design analysis remains valid for the period of extended operation.

Since the number of DBEs and OBEs projected to occur in 60 years are below the combination of 1 DBE and 20 OBE cycles analyzed for the SFP bottom liner, this design analysis remains valid for the period of extended operation.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.3.7 Environmentally-Assisted Fatigue Analyses

NUREG-1801, Revision 1, Generic Aging Lessons Learned, contains recommendations on specific areas for which existing programs should be augmented for license renewal. The program description for Aging Management Program X.M1, Metal Fatigue of Reactor Coolant Pressure Boundary Program, provides guidance for addressing environmental fatigue for license renewal. It states that an acceptable program addresses the effects of the reactor coolant environment on component fatigue life by assessing the impact of the reactor coolant environment on a sample of critical components for the plant. Examples of critical components are identified in NUREG/CR-6260, “Application of NUREG/CR-5999 Interim Fatigue Curves to Selected Nuclear Power Plant Components”.

This sample of components can be evaluated by applying environmental life correction factors to the existing ASME Code fatigue analyses using formulae contained in NUREG/CR-6583, “Effects of LWR Coolant Environments on Fatigue Design Curves of Carbon and Low Alloy Steels”, and in NUREG/CR-5704, “Effects of LWR Coolant Environments on Fatigue Design Curves of Austenitic Stainless Steels”. Demonstrating that these components have an environmentally adjusted cumulative usage factor less than or equal to the design limit of 1.0 is an acceptable option for managing metal fatigue for the reactor coolant pressure boundary.

NUREG/CR-6260 provided environmental fatigue calculations for an Older Vintage Westinghouse plant using the interim fatigue curves from NUREG/CR-5999 for the locations of highest design CUF for the components listed below:

1. Reactor Vessel Shell and Lower Head
2. Reactor Vessel Inlet and Outlet Nozzles
3. Pressurizer Surge Line (including hot leg and pressurizer nozzles)
4. RCS Piping Charging System Nozzles
5. RCS Piping System Safety Injection Nozzles
6. RHR System Class 1 Piping

For the NUREG/CR-6260 locations identified above, the plant-specific components were identified. ASME fatigue usage factors were calculated for each plant-specific component, and the environmental fatigue (Fen) penalties were applied to obtain the updated fatigue results. The reactor vessel was designed to the requirements of the ASME Code, Section III, explicit

fatigue usage factors were available from the design evaluations. The plant specific design fatigue results were used to determine the specific locations to be evaluated and apply the applicable Fen penalties to determine updated fatigue usage with environmentally assisted fatigue (EAF).

Salem piping was designed to ANSI B31.1 Code and there is no original explicit fatigue design. To identify Salem specific piping components to be evaluated, design fatigue calculations for similar components were reviewed to determine limiting component locations with respect to the factors influencing fatigue and considering reactor water environmental effects. For the pressurizer surge line, an ASME fatigue evaluation had been previously performed in response to NRC Bulletin 88-11. However, more detailed evaluations were required to accommodate the Fen penalty factors.

Since there was no explicit fatigue design for Salem, no design or transient specifications to the piping exist. Standard transient descriptions for a 4-loop Westinghouse plant were used as the starting point for the fatigue evaluation. Where Salem specific transient information was available, this was incorporated when applicable.

The evaluations showed that no cumulative usage factors with environmental penalties exceeded 1.0 for 60 years of service for the identified plant-specific locations. Future fatigue evaluations using WESTEMS™ “Design CUF” (NB-3200 module) will include written explanation and justification of any user intervention. Future fatigue design calculations will not use or implement the NB-3600 option (module) of the WESTEMS™ program. The Metal Fatigue of Reactor Coolant Pressure Boundary aging management program (B.3.1.1) will be used to manage the aging effects of environmentally assisted fatigue for the components in Salem LRA Tables 4.3.7-1 and 4.3.7-2.

The Metal Fatigue of Reactor Coolant Pressure Boundary program will manage the effects of aging due to environmentally assisted fatigue in accordance with 10 CFR 54.21(c)(1)(iii).

A.4.4 Other Plant-Specific Analyses

A.4.4.1 Reactor Vessel Underclad Cracking Analyses

Intergranular separations (underclad cracking) in low alloy steel heat-affected zones under austenitic stainless steel weld claddings were first detected in SA-508, Class 2, reactor vessel forgings in 1970. They have been reported to exist in SA-508, Class 2, reactor vessel forgings manufactured to a coarse grain practice and clad by high-heat-input submerged arc processes. The regulatory position regarding this issue is found in Regulatory Guide 1.43, “Control of Stainless Steel Weld Cladding of Low-Alloy Steel Components.”

A detailed analysis of underclad cracks is provided in a topical report in which Westinghouse presented a fracture mechanics analysis to justify the continue operation of all Westinghouse Units for 32 EFPY with underclad cracks in the reactor pressure vessels. The Westinghouse Owners’ Group and the NRC identified this topical report as a TLAA that required evaluation for License Renewal.

By letter dated March 1, 2001, as supplemented by letters dated June 15 and July 31, 2001, the Westinghouse Owners' Group submitted another analysis for NRC review. It evaluated the impact of cracks in SA-508 Class 2 and SA-508 Class 3 forgings beneath austenitic stainless steel weld cladding on reactor pressure vessel integrity. The initial NRC SER issued on October 15, 2001 applied only to 3-loop plants. The Westinghouse Owners' Group provided clarification to the NRC in a letter dated June 19, 2002, to include all Westinghouse plants. The NRC issued a revised SER in September 2002 to include all Westinghouse plants and the evaluation was reissued.

The reissued evaluation was used to demonstrate that fatigue growth of the subject flaw is insignificant over 60 years and the presence of the underclad cracks is of no concern relative to the structural integrity of the vessels, since (1) Salem verified that it was bounded by the report, and (2) Salem provided a FSAR summary description of the programs and activities for managing the effects of aging and evaluation of TLAAs for the period of extended operation.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.4.2 Reactor Coolant Pump Flywheel Fatigue Crack Growth Analyses

A Westinghouse report includes a fatigue crack growth analysis that has been identified as a TLAA. The report was submitted for NRC review and the NRC issued a Safety Evaluation Report in September 1996. The purpose of the report was to provide an engineering basis for elimination of flywheel in-service inspection requirements for all operating Westinghouse plants and certain Babcock and Wilcox plants.

The analysis addresses crack growth of a postulated flaw and compares this growth to a critical flaw size to determine whether or not a failure would occur under maximum overspeed conditions. To estimate the magnitude of fatigue crack growth during plant life, an initial radial crack length of 10 percent of the through the flywheel was assumed. The maximum stress intensity range occurs during reactor coolant pump startup. The number of cycles on the flywheel corresponds to the number of reactor coolant pump starts and stops. The number of cycles (pump starts and stops) for a 60-year plant life was assumed to be 6,000 for this analysis. Crack growth was shown to be negligible from exposure to these 6,000 cycles.

The projected number of start/stop cycles for the Salem Units 1 and 2 RCP flywheels are much less than the analyzed 6,000 cycles.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.4.3 Leak-Before-Break Analyses

Appendix A, Criterion 4, of 10 CFR 50 allows for the use of leak-before-break (LBB) methodology for excluding the dynamic effects of postulated ruptures in reactor coolant system piping. The fundamental premise of the LBB methodology is that the materials used in nuclear power plant piping are sufficiently tough that even a large through-wall crack would remain

stable and would not result in a double-ended pipe rupture. Application of the LBB methodology is limited to those high-energy fluid systems not considered to be overly susceptible to failure from such mechanisms as corrosion, water hammer, fatigue, thermal aging or indirectly from such causes as missile damage or the failure of nearby components. The analyses involved with LBB are considered TLAAs.

The 60-year LBB analysis demonstrates that the previous LBB conclusions still remain valid, and the dynamic effects of the pipe rupture resulting from postulated breaks in the reactor coolant primary loop piping need not be considered in the Salem structural design basis for the period of extended operation. This 60-year analysis used input from steam generator snubber elimination program, steam generator replacement design change packages, 1.4% power uprate evaluation, the Tavg operating window, and the Mechanical Stress Improvement Process (MSIP) application at the reactor vessel primary nozzle locations.

In addition to the large primary loop RCS piping, the NRC approved Amendment Nos. 336 and 317 to renewed facility operating License Nos. DPR-70 and DPR-75 to apply Leak-Before-Break (LBB) methodology to piping systems six-inches and larger connected to the Reactor Coolant Loop in both Units 1 and 2, and to the pressurizer surge line. This approval allowed excluding from the Licensing Basis the dynamic effects of postulated ruptures of various piping segments (See Section 3.6.4.2.1 for updated piping segments).

Based on the fatigue crack growth (FCG) analysis of the piping evaluated in Amendment Nos. 336 and 317, the FCG analysis for the subject piping bounds the transient projections for the 60-year operation.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.4.4 Applicability of ASME Code Case N-481 to the Salem Units 1 and 2 Reactor Coolant Pump Casings

Periodic volumetric inspections of the welds of the primary loop pump casings of commercial nuclear power plants are require by Section XI of the ASME Boiler and Pressure Vessel Code. These inspections require a large amount of time and resources to complete. They also result in large radiation exposure (man-rem). Since the pump casings are inspected prior to being placed in service, and no significant mechanisms exist for crack initiation and propagation, it has been concluded that the in-service volumetric inspection can be replaced with an acceptable alternate inspection.

In recognition of this, ASME Code Case N-481, Alternative Examination Requirements for Cast Austenitic Pump Casings, provides an alternative to the volumetric inspection requirement. The code case allows the replacement of volumetric examinations of primary loop pump casings with fracture mechanics-based integrity evaluation (Item (d) of the code case) supplemented by specific visual examinations. Westinghouse demonstrated compliance with ASME Code Case N-481 on a generic basis that is documented in WCAP-13045. In this evaluation, stress analyses were performed to support fracture mechanics analyses for postulated flaws. Salem applied WCAP-13045 to the Salem reactor coolant pump casings for their 40-year plant life.

The TLAA related to Code Case N-481 is thermal aging of cast austenitic stainless steel and its consequence on fatigue crack growth. The 60-year analysis provided a comparison of the Salem pump casing nozzle loadings with the screening loads reported in WCAP-13045. The screening loads in WCAP-13045 bounded the Salem loads anticipated for 60 years of operation. The stability of the flaws postulated in the primary loop pump casings has been established by evaluating the necessary materials properties against the saturated (fully aged) fracture toughness values. The results of the 60-year analysis show that Code Case N-481 is satisfied for the license renewal period when supplemented with the visual inspections specified in the code case (Items a, b, and c).

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.4.5 Salem Unit 1 Volume Control Tank Flaw Growth Analysis

Flaws were identified in the shell to lower head weld of the Salem Unit 1 Volume Control Tank (VCT) during 1RF13 (1999). The flaws found during the inspection were subsurface and not in contact with the environment, therefore, only fatigue would be the contributing mechanism to flaw growth. The analyses concluded that an initial flaw would grow an insignificant amount of only $1.1 \times 10^{(-5)}$ inches, based on 1,000 pressurization cycles.

The VCT is an operating surge volume compensating in part for reactor coolant releases from the reactor coolant system as a result of level changes. The major pressurization cycles (transients) experienced by the VCT would be Inadvertent Safety Injection events and Operating Basis Earthquake cycles, and to a lesser extent, Plant Heatups and Cooldowns.

Therefore, since the number of cycles projected to occur in 60 years for Inadvertent Safety Injection events and Operating Basis Earthquake cycles, Plant Heatups, and Plant Cooldowns is well below 1,000 pressurization cycles analyzed for the Unit 1 VCT flaw, this design analysis remains valid for the period of extended operation.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.5 Fuel Transfer Tube Bellows Design Cycles

The fuel transfer tube connects the fuel transfer canal (inside the Containment Structure) to the transfer pool (inside the Fuel Handling Building). The fuel transfer tube passes through the containment wall and through the exterior wall of the Fuel Handling Building.

The fuel handling building fuel transfer tube is comprised of a 24-inch diameter penetration sleeve penetrating through the containment and fuel handling building walls and three (3) sets of expansion joints (bellows). The penetration sleeve and the three bellows perform a water retaining intended function, and are within the scope of license renewal.

Each of these three bellows was designed for a minimum of 50 cycles of seismic movement, therefore, this design analysis is a TLAA requiring evaluation for the period of extended operation.

In order to determine if the design analyses remain valid for 60 years of operation, the number of seismic cycles for 60 years has been conservatively projected. As of January 2009, the Salem transfer tube bellows have been exposed to zero (0) Operating Basis Earthquake cycles. It is anticipated that 2 and 3 Operating Basis Earthquake cycles would occur in 60 years of operation for Salem Units 1 and 2, respectively.

Therefore, since the number of cycles in 60 years is well below the 50 seismic movement cycles analyzed for these bellows, these design analyses remain valid for the period of extended operation.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.6 Crane Load Cycle Limits

A.4.6.1 Polar Gantry Crane

The purchasing specification for the 230/35-ton Polar Crane in the containment structure at Salem required the crane conform to the design requirements of EOCI-61, “Specifications for Electric Overhead Traveling Cranes – 1961. Issuance of the Crane Manufacturers Association of America (CMAA) Specification 70 was meant to supersede EOCI-61. An engineering study reviewed the design of this crane and determined that it complied with CMAA 70, Rev. 75. As such, the design of this crane corresponds to the cyclic loading requirements of CMAA 70, Class A. This evaluation of cycles over the 40-year life is the basis of a safety determination and is therefore a TLAA Analysis.

The Polar Crane was designed for a minimum of 20,000 load cycles, corresponding to the criteria of CMAA Specification 70 for service Class A. The number of anticipated lifts for the Polar Crane is estimated to be 1,720 through the period of extended operation, which is less than the minimum allowable design value of 20,000 cycles. The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.6.2 Fuel Handling Crane

The purchasing specification for the 5-ton Fuel Handling Crane in the Fuel Handling Building at Salem required the crane conform to the design requirements of EOCI-61, “Specifications for Electric Overhead Traveling Cranes – 1961. Issuance of the Crane Manufacturers Association of America (CMAA) Specification 70 was meant to supersede EOCI-61. An engineering study reviewed the design of this crane and determined that it complied with CMAA 70, Rev. 75. As such, the design of this crane corresponds to the cyclic loading requirements of CMAA 70, Class A. This evaluation of cycles over the 40-year life is the basis of a safety determination and is therefore a TLAA Analysis.

The Fuel Handling Crane was designed for a minimum of 20,000 load cycles, corresponding to the criteria of CMAA Specification 70 for service Class A. The number of anticipated lifts for the Fuel Handling Crane is estimated at 12,000, which is less than the minimum allowable design value of 20,000 cycles.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.6.3 Cask Handling Crane

The existing Salem Cask Handling Cranes were replaced (Unit 1 in Fall 2008; Unit 2 in Fall 2009) by single failure proof Cask Handling Cranes rated for 115 tons (main hoist) and 10 tons (auxiliary hoist). Each of these two cranes was designed to ASME NOG-1-2004, NUREG-0554, and NUREG-0612 criteria in order to be certified as an NRC-approved single failure proof design. The cranes were also designed to CMAA 70-04 standards for Class A service. This evaluation of cycles over the 40-year life is the basis of a safety determination and is therefore a TLAA Analysis.

The Cask Handling Crane was designed for a minimum of 20,000 load cycles, corresponding to the criteria of CMAA Specification 70 for service Class A. The number of anticipated lifts for the Cask Handling Crane is estimated at 1,560, which is less than the minimum allowable design value of 20,000 cycles.

The analyses are valid for the period of extended operation in accordance with 10 CFR 54.21(c)(1)(i).

A.4.7 Environmental Qualification of Electrical Equipment

Thermal, radiation, and cyclical aging analyses of plant electrical and I&C components, developed to meet 10 CFR 50.49 requirements, have been identified as time-limited aging analyses (TLAAs) for Salem. The NRC has established nuclear station environmental qualification (EQ) requirements in 10 CFR 50.49 and 10 CFR 50, Appendix A, Criterion 4. 10 CFR 50.49 specifically requires that an EQ program be established to demonstrate that certain electrical components located in harsh plant environments are qualified to perform their safety function in those harsh environments after the effects of in-service aging. Harsh environments are defined as those areas of the plant that could be subject to the harsh environmental effects of a loss-of-coolant accident (LOCA), high energy line break (HELB), or post-LOCA radiation. 10 CFR 50.49 requires that the effects of significant aging mechanisms be addressed as part of environmental qualification.

Under 10 CFR 54.21(c)(1)(iii), the Salem Environmental Qualification (EQ) of Electric Components program, which implements the requirements of 10 CFR 50.49 (as further defined and clarified by NUREG-0588, and RG 1.89, Rev. 1), is viewed as an aging management program for License Renewal.

Additionally, reanalysis of an aging evaluation to extend the qualifications of components is performed on a routine basis as part of the Salem Environmental Qualification (EQ) of Electric

Components program. Important attributes for the reanalysis of an aging evaluation include analytical methods, data collection and reduction methods, underlying assumptions, acceptance criteria, and corrective actions (if acceptance criteria are not met). TLAA demonstration option (iii), which states that the effects of aging will be adequately managed for the period of extended operation, is chosen and the Salem Environmental Qualification (EQ) of Electric Components program will manage the aging effects of the components associated with the environmental qualification TLAA.

NUREG-1800 states that the staff evaluated the industry's EQ program (10 CFR 50.49) and determined that it is an acceptable aging management program to address environmental qualification according to 10 CFR 54.21(c)(1)(iii).

The Environmental Qualification (EQ) of Electric Components program will manage the effects of aging in accordance with 10 CFR 54.21(c)(1)(iii).

A.5 Salem License Renewal Commitment List

Note: This commitment list represents the commitments in effect at the time of the first UFSAR update after issuance of the renewed plant operating license. As such this list is a “point in time” reference document and no changes should be made to this document. The commitments are controlled per the commitment control process as defined in the “Commitment Management Procedure”.

* License Renewal Application (LRA)

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
1	ASME Section XI Inservice Inspection, Sub Sections IWB, IWC, and IWD	Existing program is credited.	A.2.1.1	Ongoing	LRA Section B.2.1.1
2	Water Chemistry	Existing program is credited.	A.2.1.2	Ongoing	LRA Section B.2.1.2
3	Reactor Head Closure Studs	Existing program is credited.	A.2.1.3	Ongoing	LRA Section B.2.1.3
4	Boric Acid Corrosion	Existing program is credited.	A.2.1.4	Ongoing	LRA Section B.2.1.4
5	Nickel-Alloy Penetration Nozzles Welded to the Upper Reactor Vessel Closure Heads of Pressurized Water Reactors	Existing program is credited.	A.2.1.5	Ongoing	LRA Section B.2.1.5

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
6	Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (CASS)	Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (CASS) is a new program that will provide for aging management of the thermal embrittlement of CASS piping, piping elements and piping components in a reactor coolant environment. The program will include a screening for components susceptible to thermal aging embrittlement based on casting method, molybdenum content, and percent ferrite. For “potentially susceptible” components, thermal aging embrittlement will be managed through either an enhanced volumetric inspection or a component-specific flaw tolerance evaluation.	A.2.1.6	Program to be implemented prior to the period of extended operation.	LRA Section B.2.1.6
7	PWR Vessel Internals	PWR Vessel Internals is a new program that will include the following activities: 1. Participate in the industry programs for investigating and managing aging effects on reactor internals. 2. Evaluate and implement the results of the industry programs as applicable to the reactor internals. 3. Upon completion of these programs, but not less than 24 months before entering the period of extended operation, submit an inspection plan for reactor internals to the NRC for review and approval.	A.2.1.7	Program to be implemented prior to the period of extended operation. Inspection plan to be submitted to NRC not less than 24 months prior to the period of extended operation.	LRA Section B.2.1.7
8	Flow-Accelerated Corrosion	Existing program is credited.	A.2.1.8	Ongoing	LRA Section B.2.1.8

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
9	Bolting Integrity	Bolting Integrity Program is an existing program that will be enhanced to include: 1. In the following cases, bolting material should not be reused: a. Galvanized bolts and nuts, b. ASTM A490 bolts; and c. Any bolt and nut tightened by the turn of nut method.	A.2.1.9	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.9
10	Steam Generator Tube Integrity	Existing program is credited.	A.2.1.10	Ongoing	LRA Section B.2.1.10
11	Open-Cycle Cooling Water System	Existing program is credited.	A.2.1.11	Ongoing	LRA Section B.2.1.11
12	Closed-Cycle Cooling Water System	Closed-Cycle Cooling Water System is an existing program that will be enhanced to include: 1. The Component Cooling System is not currently analyzed for sulfates, which is not consistent with the EPRI standard. The program will be enhanced to include monitoring this parameter as part of the Closed-Cycle Cooling Water program. 2. The emergency diesel generator jacket water system is not currently analyzed for azole or ammonia, chlorides, fluorides, and microbiologically-influenced corrosion in accordance with the current EPRI standard. The program will be enhanced to include these parameters as part of the Closed-Cycle Cooling Water program.	A.2.1.12	Program to be enhanced and one-time inspections to be implemented prior to the period of extended operation.	LRA Section B.2.1.12

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		3. The Closed-Cycle Cooling Water program for the Chilled Water System will have a program or hardware change to bring the system chemistry parameters into compliance with EPRI 1007820, prior to the period of extended operation. 4. New recurring tasks will be established to enhance the performance monitoring of selected heat exchangers cooled by Component Cooling System. 5. New recurring tasks will be established for enhancing the performance monitoring of selected Chilled Water System components. 6. A one-time inspection of selected components will be established for Chilled Water System piping to confirm the effectiveness of the Closed-Cycle Cooling Water program. 7. A one-time inspection of selected closed-cycle cooling water components in stagnant flow areas will be conducted to confirm the effectiveness of the Closed-Cycle Cooling Water program. 8. A one-time inspection of selected closed-cycle cooling water chemical mixing tanks and associated piping will be conducted to confirm the effectiveness of the closed cycle cooling water program on the interior surfaces of the tanks and associated piping.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		9. The program will be enhanced such that the Heating Water and Heating Steam System will have a pure water control program instituted, in accordance with EPRI 1007820, prior to the period of extended operation. 10. New recurring tasks will be established for enhancing the performance monitoring of selected Heating Water and Heating Steam System components. 11. A one-time inspection of selected Heating Water and Heating Steam System piping will be conducted to confirm the effectiveness of the Closed-Cycle Cooling Water program.			
13	Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems	Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems is an existing program that will be enhanced to include: 1. Visual inspection of structural components and structural bolts for loss of material due to general, pitting, and crevice corrosion and structural bolting for loss of preload due to self-loosening. 2. Visual inspection of the rails in the rail system for loss of material due to wear. 3. The acceptance criteria will be enhanced to require evaluation of significant loss of material due to corrosion for structural components and structural bolts, and significant loss of material due to wear of rail in the rail system.	A.2.1.13	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.13

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
14	Compressed Air Monitoring	Existing program is credited.	A.2.1.14	Ongoing	LRA Section B.2.1.14
15	Fire Protection	Fire Protection is an existing program that will be enhanced to include: 1. The routine inspection procedures will be enhanced to provide additional inspection guidance to identify degradation of fire barrier walls, ceilings, and floors for aging effects such as cracking, spalling and loss of material caused by freeze-thaw, chemical attack, and reaction with aggregates. 2. The fire pump supply line functional tests will be enhanced to provide specific guidance for examining exposed external surfaces of the fire pump diesel fuel oil supply line for corrosion during pump tests. 3. The Halon and Carbon Dioxide fire suppression system functional test procedures will be enhanced to include visual inspection of system piping and component external surfaces for signs of corrosion or other age related degradation, and for mechanical damage. The system functional test procedures will also be enhanced to include acceptance criteria stating that identified corrosion or mechanical damage will be evaluated with corrective action taken as appropriate.	A.2.1.15	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.15 Salem Letter LR-N10-0225 RAI B.2.1.15-02

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
16	Fire Water System	Fire Water System is an existing program that will be enhanced to include: 1. The Fire Water System aging management program will be enhanced to inspect selected portions of the water based fire protection system piping located aboveground and exposed to the fire water internal environment by non-intrusive volumetric examinations. These inspections shall be performed prior to the period of extended operation and will be performed every 10 years thereafter. 2. The Fire Water System aging management program will be enhanced to replace or perform 50-year sprinkler head inspections and testing using the guidance of NFPA-25 “Standard for the Inspection, Testing and Maintenance of Water-Based Fire Protection Systems” (2002 Edition), Section 5-3.1.1. These inspections will be performed by the 50-year in-service date and every 10-years thereafter.	A.2.1.16	Program to be enhanced prior to the period of extended operation. Inspection schedule identified in Commitment	LRA Section B.2.1.16

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
17	Aboveground Steel Tanks	Aboveground Steel Tanks is an existing program that will be enhanced to include: 1. The program will be enhanced to include UT measurements of the bottom of the tanks that are supported on concrete foundations (Fire Protection Water Storage Tanks). Measured wall thickness will be monitored and trended if significant material loss is detected. These thickness measurements of the tank bottom will be taken and evaluated against design thickness and corrosion allowance to ensure that significant degradation is not occurring and the component intended function would be maintained during the extended period of operation. 2. The program will be enhanced to provide routine visual inspections of the Fire Protection Water Storage Tanks external surfaces. The visual inspection activities will include inspection of the grout or sealant between the tank bottom and the concrete foundation for signs of degradation.	A.2.1.17	Program to be enhanced prior to the period of extended operation. Tank bottom UT inspections will also be performed prior to the period of extended operation	LRA Section B.2.1.17

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
18	Fuel Oil Chemistry	Fuel Oil Chemistry is an existing program that will be enhanced to include: 1. Equivalent requirements for fuel oil purity and fuel oil testing as described by the Standard Technical Specifications. 2. Analysis for particulate contamination in new and stored fuel oil. 3. Addition of biocides, stabilizers and corrosion inhibitors as determined by fuel oil sampling or inspection activities. 4. Quarterly analysis for bacteria in new and stored fuel oil. 5. Internal inspection of 350-gallon Fire Pump Day Tanks (S1DF-1DFE21 and S1DF-1DFE23) using visual inspections and ultrasonic thickness examination of tank bottoms. 6. Sampling of new fuel oil deliveries for API gravity and flash point prior to off load. 7. Internal inspection of the 30,000-gallon Fuel Oil Storage Tanks (S1DF-1DFE1, S1DF-1DFE2, S2DF-2DFE1 and S2DF-2DFE2) using visual inspections and ultrasonic thickness examination of tank bottoms. 8. To confirm the absence of any significant aging effects, a one-time inspection of each of the 550-gallon Diesel Fuel Oil Day Tanks will be performed.	A.2.1.18	Program to be enhanced and one-time inspections to be implemented prior to the period of extended operation.	LRA Section B.2.1.18

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
19	Reactor Vessel Surveillance	Reactor Vessel Surveillance is an existing program that will be enhanced to include: 1. The Reactor Vessel Surveillance program will be enhanced to state the bounding vessel inlet temperature (cold leg) limits and fluence projections, and to provide instructions for changes. a. Inlet Temperature Range Limitation: 525°F (min) to 590°F (max) b. Fluence Limitation (max.): 1.00 x 10^{20} n/cm^2 (E > 1.0 MeV) 2. The Reactor Vessel Surveillance program will be enhanced to describe the capsule storage requirements and the need to retain future pulled capsules. 3. The Reactor Vessel Surveillance program will be enhanced to specify a scheduled date for withdrawal of capsules including pulling one of the remaining four capsules during the period of extended operation to monitor the effects of long-term exposure to neutron embrittlement for each Salem Unit. Those dates shall be approved by the NRC prior to withdrawal of the capsules, in accordance with 10 CFR Part 50, Appendix H.	A.2.1.19	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.19

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		4. The Reactor Vessel Surveillance program will be enhanced to incorporate the requirements for (1) withdrawing the remaining capsules when the monitor capsule is withdrawn during the period of extended operation and placing them in storage for the purpose of reinstituting the Reactor Vessel Surveillance Program if required, i.e. if the reactor vessel exposure conditions (neutron flux, spectrum, irradiation temperature, etc.) are altered, and subsequently the basis for the projection to 60 years warrant the reinstitution, and (2) changes to the reactor vessel exposure conditions and the potential need to re-institute a vessel surveillance program will be discussed with the NRC staff prior to changing the plant's licensing basis. 5. Enhancements to the current Reactor Vessel Surveillance program will be made to require that if future plant operations exceed the limitations or bounds specified for cold leg temperatures (vessel inlet) or higher fluence projections, then the impact of plant operation changes on the extent of reactor vessel embrittlement will be evaluated and the NRC shall be notified. a. Inlet Temperature Range Limitation: 525°F (min) to 590°F (max) b. Fluence Limitation (max.): 1.00 x 10^{20} n/cm^2 (E > 1.0 MeV)			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
20	One-Time Inspection	One-Time Inspection is a new program and will be used for the following: 1. To confirm the effectiveness of the Water Chemistry program to manage the loss of material, cracking, and the reduction of heat transfer aging effects for aluminum, copper alloy, nickel alloy, steel, stainless steel, and cast austenitic stainless steel in treated water, treated borated water where dissolved oxygen may not be controlled to less than 100 ppb, steam, and reactor coolant environments. 2. To confirm the effectiveness of the Fuel Oil Chemistry program to manage the loss of material aging effect for aluminum, copper alloy, gray cast iron, steel and stainless steel in a fuel oil environment. 3. To confirm the effectiveness of the Lubricating Oil Analysis program to manage the loss of material and the reduction of heat transfer aging effects for aluminum, copper alloy, ductile cast iron, gray cast iron, steel, stainless steel, cast austenitic stainless steel and titanium alloy in a lubricating oil environment.	A.2.1.20	Program to be implemented prior to the period of extended operation. One-time inspections to be performed within the ten-year period prior to the period of extended operation.	LRA Section B.2.1.20 Salem Letter LR-N11-0005 RAI B.2.1.20-01 Salem Letter LR-N11-0148 DRAI 3.2.1.48

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		The sample plan for inspections associated with the One-Time Inspection program will be developed to ensure there are adequate inspections to address each of the material, environment, and aging effect combinations. A sample size of 20% of the population (up to a maximum of 25 inspections) will be established for each of the sample groups.			
21	Selective Leaching of Materials	Selective Leaching of Materials is a new program that will include one-time inspections of a representative sample of susceptible components to determine where loss of material due to selective leaching is occurring. A sample size of 20% of susceptible components will be subjected to a one-time inspection with a maximum of 25 inspections for each of the susceptible material groups. Where selective leaching is identified, further aging management activities will be implemented such that the component intended function is maintained consistent with the current licensing basis through the period of extended operation.	A.2.1.21	Program to be implemented prior to the period of extended operation. One-time inspections to be performed within the ten-year period prior to the period of extended operation.	LRA Section B.2.1.21 Salem Letter LR-N10-0324 Salem Letter LR-N11-0005 RAI B.2.1.21-01
22	Buried Piping Inspection	Buried Piping Inspection is an existing program that will be enhanced to include: 1. A cathodic protection study will be performed prior to entering the period of extended operation to assess the possibility and benefits of installing a system, versus other mitigative and preventive actions.	A.2.1.22	Program to be enhanced prior to the period of extended operation. Inspection Schedule identified in Commitment	LRA Section B.2.1.22 Salem Letter LR-N10-0322 RAI B.2.1.22 Salem Letter LR-N10-0372

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		2. A soil characterization study will be performed prior to entering the period of extended operation to determine soil corrosivity in the vicinity of buried piping. The results of the study will be used as an input to the program so that inspections will be performed at the locations of highest risk. 3. At least one (1) opportunistic or focused excavation and inspection will be performed on each of the Fire Protection System material groupings, which include carbon steel, ductile cast iron, and gray cast iron piping and components during each ten (10) year period, beginning ten (10) years prior to entry into the period of extended operation. 4. The following inspections apply to buried, carbon steel, safety-related portions of the specified systems. A different segment for each system will be inspected in each ten year period. a. At least one (1) opportunistic or focused excavation and inspection on each of the Auxiliary Feedwater and Compressed Air systems during the ten (10) years prior to entering the period of extended operation. b. At least three (3) opportunistic or focused excavations and inspections of the Service Water System during the ten (10) years prior to entering the period of extended operation.			RAI B.2.1.22-02 Salem Letter LR-N10-0444 RAI B.2.1.22-03

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		c. If, as a result of the soil characterization study, it is determined that the soil is not corrosive in the vicinity of all of the Auxiliary Feedwater, Service Water, and Compressed Air systems, Salem will perform at least one (1) opportunistic or focused excavation and inspection on each of the respective systems every ten (10) years during the period of extended operation. d. If, as a result of the soil characterization study, it is determined that the soil is corrosive in the vicinity of the Auxiliary Feedwater, Service Water, or Compressed Air systems, Salem will perform at least two (2) opportunistic or focused excavations and inspections on the respective susceptible system(s) every ten (10) years during the period of extended operation.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		5. If, based on the results of the initial soil characterization study, it is determined that the soil is not corrosive in the vicinity of the Auxiliary Feedwater, Service Water, or Compressed Air systems, Salem will perform a second Soil Characterization Study within approximately fifteen (15) years of the original study. The results of the second soil study will be entered into the Corrective Action Program for evaluation. 6. The buried Auxiliary Feedwater System piping located inside the Unit 2 Fuel Transfer Tube Area (approximately 125 feet) will be replaced and rerouted above ground prior to entering the period of extended operation.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
23	One-Time Inspection of ASME Code Class 1 Small Bore-Piping	One-Time Inspection of ASME Code Class 1 Small-Bore Piping is a new program that will manage the aging effect of cracking in stainless steel small-bore, less than nominal pipe size (NPS) 4 inches and greater than or equal to NPS 1 Class 1 piping through the use of a combination of volumetric examinations and visual inspections. The One-Time Inspection of ASME Code Class 1 Small Bore-Piping is a new program that will be enhanced to include the following activity: Salem Units 1 and 2 will perform four volumetric examinations, two per unit, from a population of 36 susceptible Class 1 small-bore socket welds on Unit 1 and 34 susceptible Class 1 small-bore socket welds on Unit 2. Provided the technology is available, these inspections shall be performed prior to entering the period of extended operation. More specifically, the volumetric examinations will analyze Class 1 small-bore socket welds as follows: • Two Class 1 small-bore socket welds (one per unit) for intergranular stress corrosion cracking; and • Two Class 1 small-bore socket welds (one per unit) for cracking caused by thermal fatigue (thermal and mechanical loading)	A.2.1.23	Program to be implemented prior to the period of extended operation. One-time inspections to be performed within the ten- year period prior to the period of extended operation. Program to be enhanced prior to the period of extended operation. The inspection schedule will be consistent with the Salem ISI Program requirements.	LRA Section B.2.1.23 Salem Letter LR-N10-0247 RAI B.2.1.23-01

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
24	External Surfaces Monitoring	External Surfaces Monitoring is a new program that directs visual inspections of components such as piping, piping components, ducting and other components in the scope of license renewal, exposed to an air environment, to manage aging effects.	A.2.1.24	Program to be implemented prior to the period of extended operation.	LRA Section B.2.1.24
25	Flux Thimble Tube Inspection	Flux Thimble Tube Inspection is a new program that manages the loss of material due to wear of the flux thimble tube materials using inspection methods such as eddy current testing.	A.2.1.25	Program to be implemented prior to the period of extended operation.	LRA Section B.2.1.25
26	Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components	Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components is a new program that manages the aging of the internal surfaces of piping, piping components, piping elements, ducting components, tanks and heat exchanger components.	A.2.1.26	Program to be implemented prior to the period of extended operation.	LRA Section B.2.1.26
27	Lubricating Oil Analysis	Existing program is credited.	A.2.1.27	Ongoing	LRA Section B.2.1.27
28	ASME Section XI, Sub Section IWE	ASME Section XI, Sub Section IWE is an existing program that will be enhanced to include: 1. Inspection of a sample of the inaccessible liner covered by insulation and lagging once prior to the period of extended operation and every 10 years thereafter. Should unacceptable degradation be found additional insulation will be removed as necessary to determine extent of condition in accordance with the corrective action process.	A.2.1.28	Program to be enhanced prior to the period of extended operation. Inspection Schedule identified in Commitment	LRA Section B.2.1.28 Salem Letter LR-N10-0165 RAI B.2.1.28-2 Salem Letter LR-N10-0244 RAI 3.5.2.2.1.7-01

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		Prior to the period of extended operation • The samples shall include 57 randomly selected containment liner insulation panels per unit. • The randomly selected containment liner insulation panels will not include containment liner insulation panels previously removed to allow for inspection • The examination will be performed by either removing the containment liner insulation panels and performing a visual inspection, or by using a pulsed eddy current (PEC) remote inspection, with the containment liner insulation left in place, to detect evidence of loss of material. If evidence of loss of material is detected using PEC, the containment liner insulation panel will be subsequently removed to allow for visual and UT examinations.			Salem Letter LR-N10-0321 RAI B.2.1.28-04 RAI B.2.1.33-06 Salem Letter LR-N10-0382

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		• All inspections will be completed by August 2016 for both Salem Units. Approximately one third of the 57 inspections will be completed during each refuel outage (Salem Unit 1 involves the following refuel outages: Spring 2013, Fall 2014, and Spring 2016. Salem Unit 2 involves the following refuel outages: Fall 2012, Spring 2014, and Fall 2015). It is acceptable to perform greater than one third of the inspections in any refuel outage to accelerate the inspection schedule. During the period of extended operation • One containment liner insulation panel will be selected, at random, for removal from each quadrant, during each of the three Periods in an Inspection Interval. Therefore, a total of 12 containment liner insulation panels will be selected, in each unit, during each ten year Inspection Interval, to allow for examination of the containment liner behind the containment liner insulation. • The randomly selected containment liner insulation panels in each quadrant will not include containment liner insulation panels previously selected.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		2. Visual inspection of 100 % of the moisture barrier, at the junction between the containment concrete floor and the containment liner, will be performed in accordance with ASME Section XI, Section IWE program requirements, to the extent practical within the limitation of design, geometry, and materials of construction of the components. The bottom edge of the stainless steel insulation lagging will be trimmed, if necessary, to perform the moisture barrier inspections. This inspection will be performed prior to the period of extended operation, and on a frequency consistent with IWE inspection requirements thereafter. Should unacceptable degradation be found, corrective actions, including extent of condition, will be addressed in accordance with the corrective action process. As a follow-up to inspections performed during the 2009 refueling outage, the following specific corrective actions will be performed on Unit 2 prior to entry into the period of extended operation:			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		• Examine the accessible 3/4" knuckle plate. If corrosion is observed to extend below the surface of the moisture barrier, excavate the moisture barrier to sound metal below the floor level and perform examinations as required by IWE. • Perform remote visual inspections, of the six capped vertical leak chase channels, below the containment floor to determine extent of condition. • Remove the concrete floor and expose the 1/4" containment liner plate (floor) for a minimum of two of the vertical leak chase channels with holes. Perform examination of exposed 1/4" containment liner plate (floor) as required by IWE. Additional excavations will be performed, if necessary, depending upon conditions found at the first two channels. • Remove 1/2" containment liner insulation panels, adjacent to accessible areas where there are indications of corrosion, to determine the extent of condition of the existing corroded areas of the containment liner plate. • Perform augmented examinations of the areas of the 1/2" containment liner plate behind insulation panels, where loss of material was previously identified, in accordance with IWE-2420.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		• Examine 100% of the moisture barrier in accordance with IWE-2310 and replace or repair the moisture barrier to meet the acceptance standard in IWE-3510. As a follow-up to inspections performed during the 2010 refueling outage, the following specific corrective actions will be performed on Unit 1 prior to entry into the period of extended operation: • Perform augmented examinations of the 3/4" containment liner (knuckle plate) at 78' elevation in accordance with IWE-2420. • Perform augmented examinations of the areas of the 1/2" containment liner plate behind insulation panels, where loss of material was previously identified, in accordance with IWE-2420. • Remove 1/2" containment liner insulation panels, adjacent to accessible areas where there are indications of corrosion, to determine the extent of condition of the existing corroded areas of the containment liner plate.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		3. ASME Section XI, Sub Section IWE program scope will be revised to include the following welds that are currently exempted from Sub Section IWE and governed under ASME Section XI, Sub Section IWB or IWC. The scope of the revision will include the cap plate to penetrating pipe pressure boundary welds, for penetrating pipe constructed of stainless steel for those penetrations with a normal operating temperature greater than 140 degrees F. 4. Owner augmented inspections will be performed at the Salem Unit 1 and Unit 2 area of the Containment liner, under the fuel transfer canal and behind the Containment liner insulation, which are subjected to leaks from the reactor cavity. These owner augmented inspections will be performed on a frequency of once per Containment Inservice Inspection Period, starting with the current Period. These owner augmented inspections will continue, under the IWE program, as long as leakage from the reactor cavity or fuel transfer canal is observed between the Containment liner and the Containment liner insulation, including during the PEO.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
29	ASME Section XI, Sub Section IWL	ASME Section XI, Section IWL, is an existing program that will be enhanced to include: 1. Examination and acceptance criteria in accordance with the guidance contained in ACI 349.3R.	A.2.1.29	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.29 Salem Letter LR-N10-0165 RAI B.2.1.29-1
30	ASME Section XI, Section IWF	Existing program is credited.	A.2.1.30	Ongoing	LRA Section B.2.1.30
31	10 CFR Part 50, Appendix J	Existing program is credited.	A.2.1.31	Ongoing	LRA Section B.2.1.31
32	Masonry Wall Program	Masonry Wall is an existing program that will be enhanced to include: 1. Additional buildings and masonry walls as described in A.2.1.32. 2. Add an Examination Checklist for masonry wall inspection requirements. 3. Specify an inspection frequency of not greater than 5 years for masonry walls.	A.2.1.32	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.32
33	Structures Monitoring Program	Structures Monitoring is an existing program that will be enhanced to include: 1. Additional structures and components as described in A.2.1.33. 2. Concrete structures will be observed for a reduction in equipment anchor capacity due to local concrete degradation. This will be accomplished by visual inspection of concrete surfaces around anchors for cracking, and spalling.	A.2.1.33	Program to be enhanced prior to the period of extended operation. Core sample Inspection schedule identified in commitment.	LRA Section B.2.1.33 Salem Letter LR-N10-0165 RAI B.2.1.33-1 Salem Letter LR-N10-0165 RAI B.2.1.33-2

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		3. Clarify that inspections are performed for loss of material due to corrosion and pitting of additional steel components, such as embedments, panels and enclosures, doors, siding, metal deck, and anchors. 4. Require inspection of penetration seals, structural seals, and elastomers, for degradations that will lead to a loss of sealing by visual inspection of the seal for hardening, shrinkage and loss of strength. 5. Require the following actions related to the spent fuel pool liner: a. Perform periodic structural examination of the Fuel Handling Building per ACI 349.3R to ensure structural condition is in agreement with the analysis. b. Monitor telltale leakage and inspect the leak chase system to ensure no blockage. c. Test water drained from the telltales and seismic gap for boron, chloride, iron, and sulfate concentrations; and pH. Acceptance criteria will assess any degradation from the borated water. Sample readings outside the acceptance criteria will be entered into and evaluated in the corrective action program.			Salem Letter LR-N10-0321 RAI B.2.1.33-05 Salem Letter LR-N10-0414 RAI B.2.1.33-07 Salem letter LR-N11-0041 RAI B.2.1.33-07 update

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		d. Perform one shallow core sample in each of the Unit 1 Spent Fuel Pool walls (east and west) that have shown ingress of borated water through the concrete. The core samples will be examined for degradation from borated water. Also the core samples (east and west walls) will expose rebar, which will be examined for signs of corrosion. The core sample from the west wall will be taken by the end of 2013 and the core sample from the east wall will be taken by the end of 2015. e. Perform a structural examination per ACI 349.3R every 18 months of the Unit 1 Spent Fuel Pool wall in the sump room where previous inspections have shown ingress of borated water through the concrete. 6. Require monitoring of vibration isolators, associated with component supports other than those covered by ASME XI, Sub Section IWF. 7. Add an Examination Checklist for masonry wall inspection requirements. 8. Parameters monitored for wooden components will be enhanced to include: Change in Material Properties, Loss of Material due to Insect Damage and Moisture Damage.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		9. Specify an inspection frequency of not greater than 5 years for structures including submerged portions of the service water intake structure. 10. Require individuals responsible for inspections and assessments for structures to have a B.S. Engineering degree and/or Professional Engineer license, and a minimum of four years experience working on building structures. 11. Perform periodic sampling, testing, and analysis of ground water chemistry for pH, chlorides, and sulfates on a frequency of 5 years. Groundwater samples in the areas adjacent to Unit 1 containment structure and Unit 1 auxiliary building will also be tested for boron concentration. 12. Require supplemental inspections of the affected in scope structures within 30 days following extreme environmental or natural phenomena (large floods, significant earthquakes, hurricanes, and tornadoes). 13. Perform a chemical analysis of ground or surface water in-leakage when there is significant in-leakage or there is reason to believe that the in-leakage may be damaging concrete elements or reinforcing steel.			

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		14. Implementing procedures will be enhanced to include additional acceptance criteria details specified in ACI 349.3R-96. 15. When the reactor cavity is flooded up, Salem will periodically monitor the telltales associated with the reactor cavity and refueling canal for leakage. If telltale leakage is observed, then the pH of the leakage will be measured to ensure that concrete reinforcement steel is not experiencing a corrosive environment. In addition, Salem will periodically inspect the leak chase system associated with the reactor cavity and refueling canal to ensure the telltales are free of significant blockage. Salem will also inspect concrete surfaces for degradation where leakage has been observed, in accordance with this Program.			

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
34	RG 1.127, Inspection of Water-Control Structures	RG 1.127, Inspection of Water-Control Structures Associated With Nuclear Power Plants is an existing program that will be enhanced to include: 1. Parameters monitored for wooden components will be enhanced to include change in material properties and loss of material due to insect damage and moisture damage. 2. Parameters monitored for elastomers will be enhanced to include hardening, shrinkage and loss of strength due to weathering and elastomer degradation. 3. The inspection requirement for submerged concrete structural components will be enhanced to require that inspections be performed by dewatering a pump bay or by a diver if the pump bay is not dewatered. 4. Specify an inspection frequency of not greater than 5 years for structures including submerged portions of the Service Water Intake Structure. 5. Require supplemental inspections of the in scope structures within 30 days following extreme environmental or natural phenomena (large floods, significant earthquakes, hurricanes, and tornadoes).	A.2.1.34	Program to be enhanced prior to the period of extended operation.	LRA Section B.2.1.34

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
35	Protective Coating Monitoring and Maintenance Program	Existing program is credited.	A.2.1.35	Ongoing	LRA Section B.2.1.35
36	Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements	Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements is a new program and will be used to manage aging of non-EQ cables and connections during the period of extended operation.	A.2.1.36	Program and initial inspections to be implemented prior to the period of extended operation.	LRA Section B.2.1.36
37	Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits	Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits is a new program that will be implemented to manage the aging of the cable and connection insulation of the in scope portions of the Radiation Monitoring System and the Reactor Protection System (i.e., the nuclear instrumentation system).	A.2.1.37	Program and initial assessment of testing and calibration results to be implemented prior to the period of extended operation.	LRA Section B.2.1.37
38	Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements	Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements is a new program that will be used to manage the aging effects and mechanisms of non-EQ, in scope inaccessible medium voltage cables (4,160V, 13,800V).	A.2.1.38	Enhanced program, initial cable tests, and initial manhole and cable vault inspections to be implemented prior to the period of extended operation.	LRA Section B.2.1.38 Salem Letter LR-N10-0225 RAI B.2.1.38-01

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		The cable test frequency will be established based on test results and industry operating experience. The maximum time between tests will be no longer than 6 years. Manholes and cable vaults associated with the cables included in this aging management program will be inspected for water collection (with water removal as necessary) with the objective of minimizing the exposure of medium voltage cables to significant moisture. Prior to the period of extended operation, the frequency of inspections for accumulated water will be established based on inspection results to minimize the exposure of medium voltage cables to significant moisture. The maximum time between inspections will be no longer than one year. The Inaccessible Medium Voltage Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements aging management program will be enhanced as follows: 1. Change cable testing maximum frequency from 10 years to 6 years. Change manhole and cable vault inspection maximum frequency from 2 years to 1 year.		Test and inspection schedule identified in commitment.	Salem Letter LR-N10-0348 LRA Supplement

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
39	Metal-Enclosed Bus	Metal Enclosed Bus is a new program that will manage the aging of in-scope metal enclosed busses.	A.2.1.39	Program and initial inspections to be implemented prior to the period of extended operation.	LRA Section B.2.1.39
40	Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements	Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements is a new program that will be used to confirm the slow progression or the absence of an aging effect with respect to electrical cable connection stressors. A representative sample of non-EQ electrical cable connections will be selected, for one-time testing considering application (medium and low voltage), circuit loading (high loading) and location, with respect to connection stressors.	A.2.1.40	Program and one-time testing to be implemented prior to the period of extended operation.	LRA Section B.2.1.40
41	High Voltage Insulators	High Voltage Insulators is a new program that manages the degradation of insulator quality due to the presence of salt deposits or surface contamination.	A.2.2.1	Program to be implemented prior to the period of extended operation.	LRA Section B.2.2.1
42	Periodic Inspection	Periodic Inspection is a new program that manages the aging of piping, piping components, piping elements, ducting components, tanks and heat exchanger components.	A.2.2.2	Program to be implemented prior to the period of extended operation.	LRA Section B.2.2.2

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
43	Aboveground Non-Steel Tanks	Aboveground Non-Steel Tanks is a new program that will manage loss of material of outdoor non-steel tanks. The Aboveground Non-Steel Tanks program will include a UT wall thickness inspection of the bottom of the tanks. The UT measurements will be taken to ensure that significant degradation is not occurring and that the component intended function will be maintained during the extended period of operation.	A.2.2.3	Program to be implemented prior to the period of extended operation. Tank bottom UT inspections will also be performed prior to the period of extended operation.	LRA Section B.2.2.3
44	Buried Non-Steel Piping Inspection	Buried Non-Steel Piping Inspection is an existing program that will be enhanced to include: 1. At least one (1) opportunistic or focused excavation and inspection will be performed on buried reinforced concrete piping and components during each ten (10) year period, beginning ten (10) years prior to entry into the period of extended operation. 2. At least one (1) opportunistic or focused excavation and inspection will be performed on buried stainless steel penetration bellows between the Containment Structure and the Fuel Handling Building, including the penetration sleeves, during each ten (10) year period, beginning ten (10) years prior to entry into the period of extended operation. 3. Guidance for inspection of concrete aging effects.	A.2.2.4	Program to be enhanced prior to the period of extended operation. Inspection Schedule identified in Commitment	LRA Section B.2.2.4 Salem Letter LR-N10-0322 RAI B.2.1.22

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45	Boral Monitoring Program	Boral Monitoring is an existing program that will be enhanced to include: 1. The program will be enhanced to perform a neutron attenuation measurement on one each of the three (no vent holes, one vent holes and two vent holes) flat plate sandwich Boral test coupons during the first three two-year inspection frequency periods and every six years thereafter for the Exxon spent fuel storage rack assemblies. 2. The program will be enhanced to include acceptance criteria of the neutron attenuation measurement on the Boral test coupons for the Exxon spent fuel storage rack assemblies: A decrease of no more than 5% in Boron-10 content as determined by neutron attenuation measurements. The benchmark Boron-10 content used for comparison will be based on the nominal B-10 areal density in the design basis specification.	A.2.2.5	Program to be enhanced prior to the period of extended operation. Inspection Schedule identified in Commitment	LRA Section B.2.2.5
46	Nickel Alloy Aging Management	Existing program is credited.	A.2.2.6	Ongoing	LRA Section B.2.2.6

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
47	Metal Fatigue of the Reactor Coolant Pressure Boundary	Metal Fatigue of the Reactor Coolant Pressure Boundary is an existing program that will be enhanced to include: 1. Adding transients beyond those defined in the Technical Specifications and the UFSAR, and expanding the fatigue monitoring program to encompass other components identified to have fatigue as an analyzed aging effect, which require monitoring. 2. Using a software program to automatically count transients and calculate cumulative usage on select components. 3. Addressing the effects of the reactor coolant environment on component fatigue life by assessing the impact of the reactor coolant environment on a sample of critical components for the plant identified in NUREG/CR-6260. 4. Requiring a review of additional reactor coolant pressure boundary locations if the usage factor for one of the environmental fatigue sample locations approaches its design limit.	A.3.1.1	Program to be enhanced prior to the period of extended operation.	LRA Section B.3.1.1
48	Environmental Qualification of Electric Components (EQ)	Existing program is credited.	A.3.1.2	Ongoing	LRA Section B.3.1.2

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
49	New P-T Curves	Revised Pressure-Temperature (P-T) limits will be submitted to the NRC when necessary to comply with 10 CFR 50 Appendix G.	A.4.2.4	Ongoing	LRA Section 4.2.4
50	Steam Generator Divider Plate Inspection	Salem will perform an inspection of each of the four (4) Unit 1 steam generators to assess the condition of the divider plate assembly. The examination technique(s) used will be capable of detecting primary water stress corrosion cracking (PWSCC) in the steam generator divider plate assemblies and the associated welds. The steam generator divider plate inspections will be completed within the first ten (10) years of the Salem Unit 1 period of extended operation.	Not Applicable	Prior to August 2026	Salem Letter LR-N10-0369 RAI 3.1.1-02
51	Steam Generator Tube to Tubesheet Weld Cracking	Salem will develop a plan for each Unit to address the potential for cracking of the primary to secondary pressure boundary due to PWSCC of tube-to-tubesheet welds. Each plan will consist of two options: <u>Salem Unit 1</u> Option 1 (Analysis): Salem Unit 1 will obtain permanent approval for Alternate Repair Criteria from the NRC, **or**	Not Applicable	Develop a plan prior to the Period of Extended Operation for each Unit. If the analysis option is chosen, implement the requirements of the plan, including obtaining any required NRC approval, by April 2018 for Unit 1, and by April 2028 for Unit 2.	Salem Letter LR-N10-0421 RAI 3.1.1-03 Salem Letter LR-N10-0438 Revised Response to RAI 3.1.1-03

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		Option 2 (Inspection): Salem Unit 1 will perform a One-Time inspection of a representative number of tube-to-tubesheet welds in each of the four (4) steam generators to determine if PWSCC is present. If weld cracking is identified, a) the condition will be resolved through repair or engineering evaluation to justify continued service, as appropriate, and b) a periodic monitoring program will be established to perform routine tube-to-tubesheet inspections for the remaining life of the steam generators. <u>Salem Unit 2</u> Option 1 (Analysis): Salem Unit 2 will perform an analytical evaluation either determining that the tubesheet cladding and welds are not susceptible to PWSCC, or redefining the pressure boundary of the tubes, where the steam generator tube-to-tubesheet welds are not required for the reactor coolant pressure boundary function. The redefinition of the reactor coolant pressure boundary will be submitted as part of a license amendment request requiring approval from the NRC, **or**		If steam generator inspections are to be performed, they will be performed between April 2018 and April 2023 for Unit 1, and April 2028 and April 2033 for Unit 2.	

NO.	PROGRAM OR TOPIC	COMMITMENT	UFSAR SUPPLEMENT LOCATION (LRA* Section)	ENHANCEMENT OR IMPLEMENTATION SCHEDULE	SOURCE
		Option 2 (Inspection): Salem Unit 2 will perform a One-Time inspection of a representative number of tube-to-tubesheet welds in each of the four (4) steam generators to determine if PWSCC is present. If weld cracking is identified, a) the condition will be resolved through repair or engineering evaluation to justify continued service, as appropriate, and b) a periodic monitoring program will be established to perform routine tube-to-tubesheet inspections for the remaining life of the steam generators.			
52	Metal Fatigue of Reactor Coolant Pressure Boundary	Salem will perform a review of design basis ASME Code Class 1 fatigue evaluations to determine whether the NUREG/CR-6260 based locations that have been evaluated for the effects of the reactor coolant environment on fatigue usage are the limiting locations for the Salem plant configuration. If more limiting locations are identified, the most limiting location will be evaluated for the effects of the reactor coolant environment on fatigue usage. If any of the limiting locations consist of nickel alloy, NUREG/CR-6909 methodology for nickel alloy will be used in the evaluation.	Not Applicable	Prior to the period of extended operation.	Salem Letter LR-N10-0445 RAI 4.3-08
53	Salem Fatigue Calculations using WESTEMS™ Program	Salem will include written explanation and justification of any user intervention in future evaluations using the WESTEMS “Design CUF” (NB-3200 module).	A.4.3.7	Within 60 days of issuance of the renewed operating license.	Salem Letter LR-N11-0042 Salem Letter LR-N11-0057

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54	Salem Fatigue Calculations using WESTEMS™ Program	Salem will not use or implement the NB-3600 option (module) of the WESTEMS™ program in future online fatigue monitoring and design calculations.	A.4.3.7	Within 60 days of issuance of the renewed operating license.	Salem Letter LR-N11-0042 Salem Letter LR-N11-0057